TECHNICAL REPORT

ISA-TR84.00.03-2002



Guidance for Testing of Process Sector Safety Instrumented Functions (SIF) Implemented as or Within Safety Instrumented Systems (SIS)

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Guidance for Testing of Process Sector Safety Instrumented Functions (SIF) Implemented as or Within Safety Instrumented Systems (SIS)

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Contents

1 Introduction 11

2 Purpose 12

3 Scope 12

4 Audience 13

5 Definition of terms and acronyms 13

5.1 Definitions 13

5.2 Acronyms 15

6 Off-line testing 16

6.1 When should off-line testing be performed 16

6.2 Deferral of scheduled testing of SIF 20

6.3 How to perform off-line testing of SIF 21

6.4 Component testing 23

6.5 Logic solver test procedures 28

6.6 Testing of final control elements 29

6.7 Testing solenoid valves 30

6.8 Testing of HMI 30

6.9 Testing of communications 30

6.10 Final SIF test procedures 31

7 On-line testing 31

7.1 Preparation 31

7.2 When should on-line tests be performed 32

7.3 Performing on-line testing 34

7.4 Inspection (observation techniques that enhance SIF availability) 38

7.5 Testing documentation 41

8 Inspections 42

9 Auditing 43

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10 References 43

Annex A — Model procedure for approval required for replacing individual components in SIF 45

Annex B — Model procedure for deferring scheduled testing of SIF 47

Annex C — Model procedure for testing turbine thrust position monitors 49

Annex D-1 — Model procedure for electronic over-speed trip testing 57

Annex D-2 — Model procedure for testing turbine overspeed trip 63

Annex E — Model procedure for testing permissive start for turning gear motor 67

Annex F — Model procedure for lube oil pumps autostart test 69

Annex G — Model procedure for testing first-out sequence alarms 71

Annex H — Model procedure for functional testing of TMR-based SIS instrumentation 73

Annex J — Example of a jumper control list 77

Annex K — Model procedure for on-line test of a high level switch 79

Annex L — Model procedure for on-line testing of flow sensors in a 1oo2 configuration (high or low trip) 81

Annex M — Model procedure for on-line testing of pressure sensors in a 2oo3 configuration (high or low trip) 83

Annex N — Model procedure for testing temperature switches 85

Annex O — Example visual inspection form for SIF 87

Annex P — Model procedure for testing a permissive pressure logic point 91

Annex Q — Model procedure for testing a simple SIF 95

Annex R — Model procedure for testing a complex logic system 99

Annex S — Model procedure for testing emergency stop switch 115

Annex T — Model procedure for testing a relay implemented SIF 117

Annex U — Model procedure for testing SIF watchdog timer 123

Annex V-1 — Model procedure for on-line testing of sensor logic 125

Annex V-2 — Model procedure for testing sensor logic 129

Annex V-3 — Model procedure for on-line testing sensor logic 133

Annex W — Model procedure for on-line final control element functional testing 137

Annex X — Model procedure for on-line testing of compressor SIF 141

--`,,,,`,`,,,,,,`,,`,,`

Annex Y — Model procedure for on-line testing of 2oo3 temperature elements 155

Annex Z — Model procedure for testing final control elements when manual bypass valves are provided 169

Annex AA — Example of a testing documentation form for off-line tests 173

Annex BB — Model SIF testing policy statement 175

Annex CC — Possible SIF performance metrics 177

Annex DD — Model technique for testing SIF valves on-line 179

Annex EE — Automated testing of SIF valves on-line 181

Annex FF — Possible audit protocol for safety instrumented functions 185

Annex GG — Example of checklist for auditing an SIF 193

Annex HH — Partial instrument trip test (PITT) 195

Annex JJ — Vendor packages to perform partial stroke testing of SIF valves 201

Annex KK — Possible technique for evaluating benefit of partial stroke testing of SIS valves in PFD_{avg} calculations 203

Annex LL —Example method for partial stroke testing of SIS valves 207

Annex MM — Examples of techniques to perform on-line testing of solenoid valves 211

Annex NN — Model procedure for testing mA pressure transmitters 213

Annex PP — Model procedure for testing mA temperature transmitters 215

Annex QQ — Model procedure for testing mV temperature transmitters 217

Annex RR — Model procedure for testing pressure switches 219

Tables

Table 1 — Calibration work process for SIF components 22

Table 2 — Tests performed to verify operation of SIF components 24

Table 3 — Calibration and testing guidance for repaired or replaced components in SIF 25

Table 4 — Sample documentation for high alarm and trip settings 26

Table 5 — Sample documentation of high temperature alarm and trip settings 27

Table C.1 — Turbine thrust position 50

Table R.1.6A — Thermocouple input, trip, and bypass action validation 101

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Table R.1.7A — Manual trip and reset logic functionality validation..... 110

Table KK.1 — Dangerous failure modes and effects with associated test strategy..... 204

Table NN.1 — Sample documentation for high alarm and trip settings 214

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1 Introduction

The best test of the Safety Instrumented Function (SIF) is the full functional test. Because SIF are designed to act upon an abnormal condition being measured and a corrective action taking place, any test must examine the measurement, logic and final control element activity to be considered a full functional test. This should involve creating an abnormal condition of the measured variable such that the input variable first reaches the alarm state and secondly moves to the interlock point making observations that the rest of the system responds as expected. Any less complete test is necessarily a compromise. Understanding what techniques should be used to ensure that this full functional test is complete is vital.

The sense of well being resulting from this successful test unfortunately deteriorates with time. Therefore, determining when subsequent testing is required to maintain this feeling of comfort is critical. The relative value of the functional test versus the cost of running the test can impact this decision. It is necessary to consider the degree of safety risk caused by a Safety Instrumented Function (SIF) initiated nuisance shutdown and at the same time the safety risk associated with an event not stopped due to a dangerous unrevealed fault in the SIF. Real processes are not ideal. Many systems are at maximum expected risk during startup and shutdown conditions.

NOTE 1 In this document the acronyms SIF and SIS will be used for both singular and plural usage of the term.

NOTE 2 The techniques for testing SIF or SIS described in this document apply to demand mode systems only. Continuous mode systems, which are rare in the process industry, require testing considerations beyond the scope of this document.

SIF applications are normally in a standby mode waiting for an indication of some potentially unsafe condition to occur before taking action. Faults may not become visible until the SIF fails to respond to an unsafe condition in the process. In basic process control loops the sensors and valves are exercised continuously during the Distributed Control System (DCS) and Programmable Logic Controller (PLC) cycles making process or equipment faults visible quickly and rendering them hard to ignore. It is vital that some program of testing and observation of each SIF in the SIS be in place. Any testing scheme, though which is burdensome or difficult has the very real probability of being ignored or bypassed. Where on-line testing techniques are implemented, they should not unnecessarily compromise the process safety integrity during the test. The test equipment and procedure must be carefully evaluated to determine whether the danger of causing an incident due to performing the on-line test is greater than the danger of not discovering the failure. Ill-advised maintenance or troubleshooting might actually increase the process risk.

Effective safety testing is strongly affected by local situations. Hazards differ, resources differ, and even the site conditions differ widely. Rapidly changing technology and ever increasing citizen expectations also impact decisions. Safety incidents can have the political result of closing down entire businesses if the local citizens are sufficiently offended. International competition has put tremendous pressure on manufacturing operations to reduce personnel and costs. Whatever testing schemes are used, they need to be very practical and should minimize maintenance and operating costs while ensuring the integrity of the SIF. The techniques suggested in this document are intended to provide guidance in the development of effective and efficient methods to plan and to manage testing and maintenance of SIF. Users of this document should have a good understanding of the applicable standards or guidelines which apply to SIF and SIS such as ANSI/ISA-84.01-1996, ISA-TR84.00.02-2002, OSHA 1910.119, dIEC 61511, and others.

The records resulting from the testing program should be equally valuable to planned and preventive maintenance and address the requirements of all regulations, as well as quality control and mandated standards.

Another important part of process safety in an operating unit is the knowledge and motivation of the operators and maintenance personnel. It is the responsibility of management to provide training and motivation. Any plan, formula, procedure, or even a standard, which attempts to, or claims to substitute

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procedures and rules for training, motivation, and support is doomed to failure. Therefore, the testing techniques proposed should not be considered just another set of rules, which become burdens to overworked plant personnel, but rather means of improving the work process and reducing frustration.

2 Purpose

Systematic testing of each Safety Instrumented Function (SIF) is required to ensure that dangerous unrevealed failures have not occurred that could render the SIF unable to perform the function for which it was provided. This testing ensures that all operational functions of the SIF are evaluated on a periodic schedule in accordance with the safety integrity requirement of the SIF. Many processes have operating cycles that are longer than the period between testing required achieving the safety integrity. Thus performing the required off-line testing necessitates shutting down the process. This is costly and puts unnecessary strain on equipment and necessitates going through shutdown and startup (which are usually the most dangerous periods of a process lifecycle) again. Therefore, the ability to perform testing while the process remains in operation is desirable.

There are also different ideas on what constitutes an acceptable test for various components of SIF. Whether the test is performed off-line, with the process down, or on-line with the process in operation, there are methods for performing the testing that ensure a high degree of detection of failures that might have occurred. Guidance is needed in the selection of these testing methods for both off-line and on-line situations.

There is also benefit in performing inspection activities on SIS equipment during normal operation of the process to detect any potential problem creating situations that might be developing. Guidance in what to look for, how often to inspect, and what to do when a condition is observed that could lead to a failure will enhance the safety integrity of the SIF.

3 Scope

Testing considerations of SIF should be included in most of the Safety Lifecycle steps described in ANSI/ISA-84.01-1996. Testing frequency is a part of the determination of Safety Integrity Level (SIL) for the SIF. Provision for conducting tests must be included in the selection of equipment and design of the SIF and the Pre-Startup Acceptance Test (PSAT) is an integral part of ensuring the SIF will provide the risk reduction necessary. When modifications are made to SIF, testing can validate that appropriate SIF action will still take place.

This technical report is an informative document providing guidance on performing testing of SIF components and systems that will help achieve full safety benefits of the SIF in the most cost-effective way. Both manual and automated techniques are presented for off-line and on-line testing of SIF and the benefits of each technique described. Existing techniques and proposed new techniques will be described. Utilizing the techniques described in conjunction with an overall safety management program will allow users to meet the testing requirements of ANSI/ISA-84.01-1996 and dIEC 61511. Techniques are described for testing all elements of the SIF including field sensors, final control elements, logic solvers (signal conversion modules included), Human Machine Interface (HMI), communication links with other systems, user application software, and other required auxiliaries such as power. Suggested inspection techniques for regular observation of equipment and components to detect potential problems are also presented.

The techniques described can also be used for testing burner management systems in conjunction with the NFPA 85 code.

These techniques are illustrated by the examples given in Annexes A-MM. **Each Annex is an example of how one company might apply a given technique, and is not intended to represent a consensus solution within the process industry.**

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4 Audience

This document is intended as a guide for those responsible for specifying, designing, constructing, scheduling, implementing, and maintaining SIF applied to the process industries.

It is expected that those persons using this document will have adequate understanding of the ANSI/ISA-84.01-1996 standard and its requirements related to testing of SIS.

5 Definition of terms and acronyms

5.1 Definitions

5.1.1 approved substitution:

a replacement item for a component or system that meets the following requirements:

Is specifically permitted as a substitute or duplicate item in a company standard or practice (i.e., the company standard or practice clearly states that more than one brand and/or model number may be used interchangeably in order for a replacement item other than the exact same brand and model number to be considered for use as an approved substitute)

OR

Is approved as an equivalent substitute by the appropriate plant or company personnel, or his/her designee for approving substitutions; meets process-specific operational safety standards; and is covered by existing training and procedures.

See Annex A for an example of a typical approval procedure for making substitute replacements for SIF components.

5.1.2 automatic testing:

a test which consists of simulated process conditions to a logic solver which cause the logic solver to take specified action and signal a final control element to move to a specified position. The simulated process signal is implemented using another programmable device which controls the sequence and range of testing. Humans may observe the action of the system logic and final control element movement but do not intervene in the testing sequence. All steps of this test are documented by the testing device for validation of system performance to specified conditions.

5.1.3 car seal:

a technique consisting of a restraint placed on a valve actuator in such a manner that it cannot be moved from the “sealed” position without breaking the restraint seal. Operations personnel typically maintain a list of those valves “car sealed” in a fixed position for a process.

5.1.4 communications (external):

data exchange between the SIS and a variety of systems or devices that are outside the SIS. These include operator interfaces, maintenance/engineering interfaces, other SIS, etc.

5.1.5 electrical/electronic/programmable (E/E/PE):

logic technology that is based on electrical (E) and/or electronic (E) and/or programmable electronic (PE) technology. The term is intended to cover any and all devices or systems operating on electrical principles and would include

- electro-mechanical devices (electrical);

- solid state non-programmable electronic devices (electronic); and

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- electronic devices based on computer technology (programmable electronic).

5.1.6 field sensors:
field sensors include the process connections, the sensing device, the transmitter, and the signal connection to the logic solver.

5.1.7 final control elements:
final control elements include the signal connection from the logic solver, the actuation medium supply (typically air), solenoid valves, and the device which effects a process flow change (e.g., valves or pumps).

5.1.8 human machine interface (HMI):
the human machine interface includes the connection between the logic solver and the operator station, the graphical display device, the tools available for operating the system (hand-switches, mouse and keyboard) as well as a printer if supplied.

5.1.9 logic solvers:
in the case of PE devices, the logic solver includes the input module, main processor, and the output module. In the case of electrical or electronic devices, the logic solver may be a single relay or redundant, voting relays.

5.1.10 manual test:
a test which consists of simulating process conditions using the input device (i.e., transmitter) to a logic solver causing the logic solver to take specified action and signal a final control element to move to a specified position. Humans typically generate the simulated process signal using appropriate test equipment. Humans also observe the action of the system logic and final control element movement. All steps of this test are documented for validation of system performance to specified conditions.

5.1.11 off-line testing:
testing performed while the process or equipment being protected is not being operated to carry out its designated function. For example, a compressor is designed to take gas from a low-pressure state to a higher pressure state. If the compressor is not running (compressing gas), it is not performing its designated function. Off-line testing would be performed during the time the compressor is not running.

5.1.12 on-line testing:
testing performed while the process or equipment being protected is operating performing its designated function. For example, a compressor is designed to take gas from a low-pressure state to a higher pressure state. If the compressor is operating (compressing gas) while tests are performed on a transmitter providing an input to the SIF, this is an on-line test of the transmitter. When simplex input devices are used, performing such testing typically requires bypassing of the input function to the SIF. When redundant devices are used, bypassing may not be required, depending on the voting configuration.

5.1.13 permissive:
logic action that requires some condition be met before further actions can be taken. For example, a specific temperature might have to be achieved in the process before some additional chemical can be added; a lubrication system must be in operation before a pump can be started; or certain valves must be closed before others can be opened.

5.1.14 proof test:
test performed to reveal undetected faults in a safety instrumented function so that, if necessary, the system can be restored to its designed functionality.

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5.1.15 replacement in kind:
an exact duplicate of a component or system or an "approved substitution" that does not require other modifications to the SIF as installed. See Annex A for an example of a typical approval procedure required for making substitute replacements for SIF components.

5.1.16 safety instrumented function (SIF):
a safety function with a specified safety integrity level which is necessary to achieve functional safety. A safety instrumented function can be either a safety instrumented protection function or a safety instrumented control function.

5.1.17 safety instrumented control function:
safety instrumented function with a specified SIL operating in continuous mode, which is necessary to prevent a hazardous condition from arising and/or to mitigate the consequences.

5.1.18 safety instrumented protection function:
safety instrumented function with a specified SIL operating in a standby mode to take action should a situation which could lead to a hazardous condition arise and/or to prevent the hazardous condition or to mitigate the consequences.

5.1.19 turnaround:
maintenance activities associated with a process, unit, or total plant which require that the process, unit, or plant be taken out of normal service and all equipment taken to a shutdown or out of service state.

5.2 Acronyms

ANSI/ISA	American National Standards Institute/Instrumentation, Systems, and Automation Society
BPCS	Basic Process Control System
CCF	Common Cause Factor
DCS	Distributed Control System
FMECA	Failure Mode Effect and Criticality Analysis
HMI	Human Machine Interface
ICS	Letters indicating a specific manufacturer of equipment
IEC	International Electrotechnical Commission
MTTF	Mean Time To Failure
PES	Programmable Electronic System
PLC	Programmable Logic Controller
PSAT	Pre-Startup Acceptance Test
RTD	Resistance Temperature Detector
SIF	Safety Instrumented Function

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SIL	Safety Integrity Level
SIS	Safety Instrumented System
SOP	Standard Operating Procedures
SOV	Solenoid Valve
SRS	Safety Requirements Specifications
T/C or TE	Thermocouple
TMR	Triple Modular Redundant
UPS	Uninterruptible Power Supply
WDT	Watch Dog Timer

6 Off-line testing

The most common test of an SIF that uncovers failures or faults that may disable an SIF is the off-line, functional test. This test is performed while the process being protected is not in operation thus allowing all features of the SIF to be validated. The primary purpose of this testing is to detect dangerous unrevealed faults that exist in the SIF. When the SIF is properly designed and maintained, this testing should rarely find faults. The basic requirements of this test are described in ANSI/ISA-84.01-1996 in Clause 9.7 Functional Testing. There are, however, multiple ways that tests can be performed to accomplish the purpose of this functional test. This clause will describe techniques and procedures that are known to be effective in carrying out the functional test to uncover faults or failures, which could result in potentially unsafe conditions in the process.

Each SIF included in the SIS should be identified. All inputs, outputs, and logic associated with each SIF should be identified. A testing procedure should define how each SIF will be validated. All equipment necessary for performing testing should be identified and verified suitable for tests to be performed. This includes calibration equipment with traceable performance.

If any components are shared among multiple SIF, testing should take this into account.

NOTE The procedures identified refer to SIF exclusively. Similar procedures should be available for all systems with limited monitoring such as equipment protection systems. These procedures are outside the scope of this document.

There are two important questions that should be addressed related to off-line testing – (1) when should off-line testing be performed and (2) how should the off-line testing be performed. These questions are addressed in the clauses to follow.

6.1 When should off-line testing be performed

6.1.1 General considerations

Off-line testing of the complete SIS should be performed prior to introduction of hazardous chemicals to the process. This is described as the Pre-Startup Acceptance Test (PSAT) in ANSI/ISA-84.01-1996 Clause 8.4. This test should be a final validation that the system can in fact perform the function(s) for which it was designed. Off-line testing allows each SIF to be completely tested including the application software and any equipment and associated logic provided for on-line testing.

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NOTE After the initial PSAT has been performed, any subsequent tests that validate all SIF in the SIS before placing the system back in service may be referred to as a full functional test.

Follow-up testing of the SIF should be performed at intervals determined by one or more of the following criteria:

- The test interval included in the performance calculations for the SIF. See ANSI/ISA-84.01-1996 Clause 4.2.6.
- When changes are made to logic, impacting the function of the SIF. See ANSI/ISA-84.01-1996 Clause 4.2.14.
- When the process or equipment is taken out of service for scheduled maintenance activities that require work involving components of the SIF. See ANSI/ISA-84.01-1996 Clause 4.2.13.
- Company policy requiring complete testing of the SIF on a predefined schedule. See ANSI/ISA-84.01-1996 Clause 4.2.13.
- After extended down time of the SIS (see deferral of testing section Clause 6.2)

No modification, which could alter any of the following, should be made without first carrying out a review to ensure the change cannot reduce the level of protection and appropriate testing is done to validate correct operation of the modified SIF:

- Performance of a Safety Protection Layer for the original design intent
- Materials of construction
- Mode of operation
- Operating procedures
- Alarm and trip settings
- Speed of response
- Testing intervals or methods
- Device type, other than replacement in kind
- Architecture or voting logic
- Diagnostics

Dependent on the nature of the repair work, which has been completed, functional testing after repair to a SIF component may include the following activities. When the test does not involve a complete functional test of the component, the test does not alter the specified SIF testing frequency.

1) Single input: exercise sensor input and verify alarm and trip setpoints are correct then observe output(s) action. Confirm the process sensor is still connected to the correct input. Use the applicable section of the SIF test procedure and complete the required documentation for the equipment checked.

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2) Single output: exercise all inputs that will actuate desired final control element and observe output action. Confirm final control element is connected to correct output. Use the applicable section of the SIF test procedure and complete the required documentation for the equipment checked.

3) Logic: perform a complete functional test of all SIF affected by the repair using the functional test procedure and complete all documentation. Check for cross contamination in the application software/logic by monitoring for unexpected actions across/between SIFs.

Follow-up testing of individual components in a SIF may be considered at intervals shorter than the complete functional test of the SIF to improve the performance capability of the SIF. Factors, which can impact the frequency of these tests, include

- sensors and final control elements installed in severe environment;
- accuracy of measurements required for safety;
- need for positive isolation of streams by valve action;
- mechanical wear and tear on components; and
- desire for longer test interval between complete functional tests.

In selecting a test interval for an SIF to match the SIL determined during the hazard and risk analysis of the process, the severity of the process characteristics should be considered. For example, a shorter test interval might be used initially for process fluids that are known to be more severe (corrosive, erosive, tending to plug, etc.). The minimum test interval should be determined by the user based on the SIL assigned to the SIF. Typically, annual testing is a reasonable starting point for the determination, which should include the examination of the component failure rate in the operating profile, the voting architecture, and the component diagnostics. The test interval chosen should be re-evaluated periodically and adjusted accordingly, based on the results of several functional tests. Based on user experience, shortening the test interval will not correct a faulty design or equipment problem. Instead, shortening the test interval will at most only allow earlier detection of an equipment problem.

It may also be appropriate to establish a maximum period of time between full functional tests of SIF that does not exceed 3-5 years. Few processes can operate for longer periods of time without some maintenance activity requiring process shutdown, and test schedules should not range beyond these shutdown schedules. There may also be some questions concerning the applicability of the failure rate data used in the SIL verification calculations and subsequent test interval determination that would point toward setting maximum test intervals for the SIF.

The incorporation of internal or external diagnostics in the SIF design often results in the reduction of the required test interval due to the ability to detect faults on-line. Diagnostics may not be able to detect all faults of the component. For example, a plugged tap may not be detected by internal diagnostics within the transmitter, but may be detected using external diagnostics (i.e. comparison of redundant transmitter analog signals using a PE logic solver). Consequently, any diagnostic should be carefully evaluated to determine which faults could be detected by the diagnostic prior to using the diagnostic as justification for reduction of the testing interval.

6.1.2 Sensors (transmitters, switches)

Whether switches or transmitters are used for input signals impacts testing requirements. Transmitters provide signals which indicate the current status of the variable being measured. This gives an indication that the input device is functioning. A switch on the other hand gives no indication of its status until the process variable passes through the trip point of the switch. Therefore, it may be necessary to test switches more often than transmitters used as input devices to SIF.

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Transmitters can also provide diagnostics such as out-of-range high/low and out-of control range indications which switches cannot do. Such diagnostics may reduce the frequency of testing required for transmitters.

The calibration stability of an input device may require testing frequencies that are shorter than that for the complete SIF. Devices that are known to drift due to environmental changes in temperature, for instance, may require more frequent testing and calibration to ensure proper process variable input to the SIF. Devices that maintain their calibration stability through wide changes in temperature may not require frequent testing as long as a signal consistent with other process conditions is being transmitted from the device.

Redundancy of components may impact their testing frequency. Where redundant sensors have their outputs monitored and they are compared with each other, agreement usually means viable measurements which do not need frequent testing or calibration. When the outputs drift apart, testing or calibration is indicated for all the redundant components.

Diversity in the detection of the hazardous condition can provide a means to improve the SIF availability without adding redundant components. For instance, a pressure measurement may be used in redundancy with a temperature measurement for some process conditions. A comparison of the temperature and pressure to expected thermodynamic data can provide diagnostics on the validity of the process measurements, reducing the required testing interval.

User experience with specific sensors and service should be used in determining the test frequency of the device to ensure proper performance of a sensor.

Some companies require yearly performance checks of sensor calibration and verification of set points. Other companies have established testing frequencies based on past history with the equipment they use. **Established company policy for testing frequency should take precedence if more frequent than the guidelines of this document.**

6.1.3 Logic solvers (E/E/PE)

When changes are made to the logic solver, the potential effects of these changes must be evaluated to determine how much of the E/E/PE must be tested. If the program changes can be isolated to a particular section, and it can be shown conclusively that the change does not impact other logic implemented in the logic solver, only that section needs to be fully tested (complete functional test). This applies to logic whether it is electromechanical relay based, solid-state relay based, pneumatic, or Programmable Electronic System (PES) based. Where Watch Dog Timers (WDT) are implemented as external diagnostics on PE logic solvers, they should be tested at the same frequency as the logic solver. For guidance in testing WDT see the American Institute of Chemical Engineers, Center for Chemical Process Safety, guideline series book, “Guidelines for Safe Automation of Chemical/Petrochemical Processes.”

Some companies require that functional performance of logic solvers be verified on a schedule that ranges from one year to several years depending on the risk associated with the process, the complexity of the logic, and company experience with the logic solver being used.

6.1.4 Final control elements (valves, motors)

Valves used for final control elements should be tested when full system functional tests are performed. They should be tested at the frequency used in the performance calculations for the SIF. Final control element (valves) should be tested each time the process is taken out of service. This can typically be performed by verifying appropriate operation of all valves when the process is taken out of service (either manually or due to a failure of some nature that caused the process to trip). For batch operations, verification of proper operation during each batch should provide this function.

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Other devices used as final control elements such as motors should be tested at the frequency used in the performance calculations for the SIF.

Frequency of testing valves as final control elements depends on a number of factors:

- Type of valve used as the final control element
- Service in which the valve is applied
- Whether the valve is used during normal operation or as a standby valve for use only when the SIF takes action
- Whether the valve must provide minimal leakage isolation or some leakage can be tolerated
- Whether the valve actuator has a spring to drive it to the safe state or it depends on motive power to drive it in both directions

When testing final control elements, auxiliaries such as valve positioners, position or limit indicators/sensors, air pressure regulators, etc. should be tested at the same frequency as the valve.

6.1.5 HMI

The Human Machine Interface (HMI) should be tested at the same frequency as the full SIF. When changes are made to information displayed in the HMI, the changes should be tested to confirm appropriate status is displayed. If the HMI is used to initiate the SIF logic, all devices associated with the initiation should be tested, including the HMI, output circuit, and final element.

6.1.6 Communications

Communications between the SIF and other control equipment such as the Basic Process Control System (BPCS) should be tested at the same frequency as the SIF. When completing full functional tests of the SIF, the testing should include all communication to auxiliary equipment such as the DCS. When changes are made to the communications links between the SIF and any other equipment, testing should confirm that appropriate information is being communicated.

6.2 Deferral of scheduled testing of SIF

Documented justification for deferral of scheduled inspection and/or testing activities should make use of failure rate data and/or quantitative methods to establish that the design intent and the performance requirements are not compromised. Company or plant-specific failure rate data for the process of concern should be used when available, because this provides the best estimation of component performance. When company or plant specific data is not available, published failure rate data can be used as long as it has been determined that the data agrees with past operational experience and includes the failure modes of interest. The method(s) used for validating the failure rate data should be appropriate to the complexity of the system and the severity of the event consequence.

Scheduled testing of SIF may be deferred if it meets the following criteria:

- The equipment that the SIF is protecting is out of service. An analysis of the impact of such a deferral on the SIF provided should be made prior to the decision to defer. The SIF should be tested prior to the equipment being returned to service in this case.

A plant turnaround is scheduled shortly after the scheduled full functional test of the SIF. This turnaround will allow a complete functional test of the SIF. The time period of this delay should not result in a compromise of the SIF or its safety integrity level. When the SIF is designed with the intent to be full

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functional tested every three to five years, the time delay should not exceed three months unless a safety assessment has determined that the longer delay would not compromise the SIF.

See Annex B for an example of a deferral procedure for SIF testing. The approval process, including levels of management and technical responsibility required for deferring a scheduled test, should be pre-determined, understood, and documented before an SIF is put into service

6.3 How to perform off-line testing of SIF

6.3.1 General guidelines

This clause will outline techniques for performing tests that have been proven and some proposed techniques, including automated techniques that can achieve adequate off-line testing of SIF. The advantages and disadvantages of each technique will be discussed where appropriate.

A key question concerns whether testing of the SIF must be done as an integrated system or whether various parts of the SIF can be tested at different times and credit be taken for the testing required to achieve the SIL specified. The requirement for testing stated in ANSI/ISA-84.01-1996 does not say that all testing of the SIF must take place at the same time. However an integrated test must be performed as the Pre-startup Acceptance Test (PSAT) (ANSI/ISA-84.01-1996, clause 8.4), prior to introduction of hazardous chemicals to the process, to ensure that the SIF can provide the functionality specified in the safety requirement specification. After that, the user is free to structure testing consistent with the integrity requirements of their SIF.

It is highly recommended that a complete functional test of the SIS including all implemented SIF be performed on some prescribed interval to ensure proper functioning of the entire system. Where the dynamics of the entire end-to-end SIF is crucial—i.e., the thermowell, the T/C, the transmitter, the input cycle time, the logic cycle time, the output signal cycle time, as well as all necessary components of the final control elements, such as volume boosters, pneumatic tubing size and length—the complete SIF should be tested together to ensure specification compliance.

Why would a user desire to perform non-integrated testing of the SIF? Testing is looking for dangerous unrevealed or covert failures that have taken place and would prevent a SIF from performing its function. Whether these are uncovered piecemeal or in a total integrated functional test is immaterial. The important factor is that they are discovered and corrected before a demand is placed on the SIF and it cannot perform the specified function.

The properly applied logic solver is generally the most available component of the SIF and thus should require complete tests less frequently than the field devices. Sensors can easily be tested on-line when provisions for testing and/or device redundancy is included in the design. Valve testing may require bypassing in order to perform a full functional test, when a short interruption of the process cannot be tolerated. But, the valve may be partially tested while in operation with a complete functional test performed off-line. Any partial testing should be evaluated to determine which failure modes and components are tested during the partial test, so that this can be considered in the SIL verification calculations. It should be emphasized that provision for this non-integrated testing of SIF components must be factored into the SIF design as required in ANSI/ISA-84.01-1996, Clause 7.9 and into the SIL verification for the SIF.

Many recognized and generally accepted good engineering practices such as NFPA and FM suggest on-line testing of valves using the process chemicals at normal operating pressure to do performance testing. This often provides better validation of the functional performance of the valve and can be a cost-effective alternative to removing the valve and taking it to a calibration facility. This type of testing could be performed as a part of a scheduled shutdown of the process with the appropriate documentation of results.

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6.3.2 SIF component calibration and performance validation

All components of the SIF should be calibrated prior to placing the SIF in service. Calibration test equipment traceable to a recognized standards performance organization should be used to perform a minimum three-point calibration (5%, 50%, 95% to prevent scaling errors) over the full signal range of the loop's sensor/transmitter to the final readout device. Valves should be calibrated to proper stroke length for full open and full closed positions. Any valve that is not required to close or open to full stroke position should be calibrated to the appropriate position prior to placing the SIF in service.

6.3.3 Calibration procedures

Calibration procedures should be available for each type of component in the SIF. In general, calibration procedures recommended by the manufacturer of the component should be used. Where additional requirements (e.g., response time of sensors or valves) are necessary to meet the specified function in the SIF, these should be taken into account in the calibration procedures.

Procedures for calibration of SIF components should include a final step in which Operations verifies the "reasonableness" of the newly calibrated, field sensor(s) actual process readings. This step is very important to minimize the likelihood of a Common Cause Failure (CCF), during calibration of redundant process sensors.

NOTE Common cause calibration failure can arise where redundant sensors are calibrated at the same time by the same person using the same test equipment or standard. Where an instrument technician mis-calibrates one sensor, he/she is very likely to mis-calibrate the other(s). Special concerns for these failures arise in calibration of redundant process analyzers using a single mixed sample and SIL 3 safety controls in batch processes.

Table 1 offers guidelines for calibration tasks and resources for calibration of SIF components:

Table 1 — Calibration work process for SIF components

Devices Being Calibrated	Calibration Tasks and Resources
Most SIF Components	• Trained staff using plant procedures and/or technical data on an "as-needed" basis when performing periodic component calibrations. • Calibration procedures and/or vendor technical data that include step-by-step calibration instructions applicable to each SIF component are available.
Safety instruments not covered in specific Maintenance Staff Training	• Skilled staff using manufacturer's step-by-step calibration instructions to calibrate devices that are not part of the staff maintenance qualification process.
Process Analyzers	• Analyzer calibration may require special considerations in addition to using the manufacturer's step-by-step calibration instructions. **Example:** Limited availability of check-gas may make executing a standard three-point calibration difficult. A calibration procedure that proves operation using one known composition sample that is close to the safety-critical trip point is often adequate.

Many field devices require periodic calibration and checkout to ensure that the process service has not affected the device's ability to respond to process changes. The use of redundancy in process measurements will allow early detection of many device failures, reducing maintenance costs by focusing

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efforts on known problems. An example[1] of what might be achieved in a reasonable process service with instrumentation redundancy is as follows:

- Smart pressure transmitters can go 2 to 4 years between calibrations.
- Coriolis and magnetic flow meters should not be calibrated unless there is evidence of a problem. *(Coriolis and magnetic flow meters should be calibrated using a prover loop at turnaround.)*
- Smart four-wire RTD transmitters should only be calibrated if there is evidence of a problem.
- Smart thermocouple transmitters can go 5 years between calibrations.
- Vortex meters should only be calibrated if the kinematic viscosity permanently changes.
- Radar level gauges should only be calibrated if vessel internals change.
- Smart nuclear level gauges should only be calibrated if process density permanently changes.
- Smart digital positioners on valves should only be calibrated when valves are overhauled.

6.4 Component testing

Both general and specific guidelines are presented in the following clauses for performing off-line testing of SIF components.

6.4.1 General guidelines

Verify permissive values of field sensors and any other devices such as timers used in permissive logic. Note that permissive logic may have manual or logic implemented bypass capability for startup. Both techniques, if provided, should be tested prior to placing the SIF in operation. Verify all alarms and or lights associated with each sensor and switch by observing and documenting correct indication when alarm conditions are reached. See Annex P for a model procedure for testing permissive logic.

Verify all hand trip switch action by observing and documenting observed action when switch is actuated.

An example of a test procedure for a simple SIF is shown in Annex Q.

Table 2 provides general guidance on testing required for verifying proper operation of components typically used in SIF.

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Table 2 — Tests performed to verify operation of SIF components

To verify the operation of …	Test …
sensors	• the operation of the complete field sensor, including - primary sensing element, - switch or transmitter, - wiring, and - logic solver input module.
logic solver	• the operation of the logic solver, including - hardware and software associated with each input device, - combined inputs, - trip setpoints, - operating sequence, - diagnostics, and - computations.
alarm functions	• operation of alarm functions and readout, including the alarms that signal the bypass of automatic trips
final control elements	• the operation of the complete final control element, including - logic solver output module, - wiring, - actuation device (e.g. relay or solenoid), and - final control element affecting the process operation.
safety system functions	• individual SIF and complete system functionality, • speed of response, when a safety parameter must act in a specified period of time, • manual trip function to take the SIF outputs to a safe state, • user-implemented diagnostics, and • SIF operability following testing. NOTE A separate manual trip function, which is not dependent on SIF logic solver, is recommended per ANSI/ISA-84.01-1996 and this function should also be tested.

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Where repair or replacement of SIF components has taken place, the guidance in Table 3 may be used.

Table 3 — Calibration and testing guidance for repaired or replaced components in SIF

Field Device **Examples:** transmitters computational relays switches, and valves.	• Calibrate the transmitter; verify switch setting and valve stroke • Verify correct operation of replacement/repaired component in the SIF; e.g., ❖ Functional testing of all inputs and outputs of the repaired or replaced component. ❖ Functionally verify correct signal flow from replacement transmitter-to-next component in SIF (typically the Logic Solver) ❖ Functionally verify correct signal flow from Logic Solver to replacement valve
Logic Solver and/or I/O module	• Input-to-output functional tests of a replaced Logic Solver component (e.g., a CPU card, and I/O module) is not necessary if the Logic Solver system contains internal self-diagnostics and reporting that verifies component operability.
All	• Document the component calibration and performance verification. NOTE Documentation for replacement of a Logic Solver component includes recording diagnostic information observed that proved component operability.

A test to confirm SIF action on total power supply failure should be carried out and if battery supplied power is provided, it should also be tested to confirm that desired time of backup is available.

Measure the power supply voltage, AC or DC, for the SIF components and verify that the power is within the acceptable range (AC ± 2.5 volts; DC ± 0.4 volts).

Check the power line-to-ground voltage and the phase angle between the current and voltage for each phase line for motors, heaters etc., where applicable.

6.4.2 Component specific guidelines

6.4.2.1 Sensor testing – transmitters

Testing sensors may involve (1) use of process to drive transmitter, (2) simulating the sensor input via appropriate measurement source, or (3) simulating the sensor output via a mA simulation tool. The particular technique used should be specified in the test procedure for the SIF. Using the process to drive the transmitter will provide assurance the transmitter can measure the process conditions but this technique may not always be available if the process is not in operation. Using simulated measurement input to the transmitter is probably the most reliable and available technique. This technique tests the function of the transmitter, the wiring, and the receiving device. Using a current simulation on the output tests the wiring and the receiving device but does not test the transmitter function.

Measure the sensor output conditions; if the output is linear, measure the output level with respect to the current process condition such as temperature, pressure, product level etc.

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Sensor testing will vary depending on the type of sensor used. The guidelines which follow outline proven in use techniques for verifying sensor operation in the SIF.

Root valves on all sensors should be verified open at end of test. Secondary valves, manifolds, vents, etc., on all sensors should also be verified as being in the "in the service" condition at end of test.

Each individual component's off-line condition should be checked and verified based on the expected value with respect to the process off-line conditions.

6.4.2.2 mA pressure transmitter

Refer to Annex NN for example procedure for testing mA pressure transmitters.

Table 4 is an example of a way to document test results for this testing.

Table 4 — Sample documentation for high alarm and trip settings

Pressure Input	Input Range P1234 (0-xxx psi) (0-yyy " H2O)	High Pre-Alarm Setpoint P1234 (xxx psi) (yyy "H_2O) (zzz mA)	High Trip Setpoint P1234 (xxx psi) (yyy " H_2O) (zzz mA)	Pre-Alarm Setpoint (As Found)	Pre-Alarm Setpoint (As Left)	Trip Setpoint (As Found)	Trip Setpoint (As Left)
PT1234							

Note that this same procedure can be used for differential pressure transmitters with the appropriate test equipment.

6.4.2.3 mA temperature transmitters

See Annex PP for example procedure for testing mA temperature transmitters.

6.4.2.4 mV temperature transmitters

See Annex QQ for example procedure for testing mV temperature transmitters.

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Table 5 is an example of how temperature transmitter testing might be documented.

Table 5 — Sample documentation of high temperature alarm and trip settings

T/C Input	T/C Fault (Upscale Burnout) T1234	Input Range T1234 (0-xxxx Deg F)	High Pre-alarm Setpoint T1234 (xxx Deg F)	Pre-alarm Setpoint (As found)	Pre-alarm Setpoint (As Left)	High Trip Setpoint T1234 (xxx Deg F)	Trip Setpoint (As Found)	Trip Setpoint (As Left)
TE1234								

6.4.2.5 Process analyzers

Process analyzers should be calibrated in accordance with manufacturers' specific instructions.

Signals from process analyzers to SIF are typically current signals representing values and ranges of components being measured. Verification of correct setpoints for pre-alarm and trip values should be done using current sources in like manner to that for other current transmitters. (See Annex NN.) As found and as left values for pre-alarm and trip setpoints should be documented.

6.4.3 Sensors – switches

6.4.3.1 Pressure switches

See Annex RR for example procedure for testing pressure switches.

6.4.3.2 Temperature switches

See Annex N for example procedure for testing temperature switches.

6.4.3.3 Level switches

Testing of level switches can be performed using the procedure outlined in Annex K. This procedure was developed for use in on-line testing but is applicable for off-line testing as well.

6.4.4 Miscellaneous sensors

This clause will offer guidance for testing a variety of sensors that might be included in SIF.

6.4.4.1 Vibration monitors

Refer to Annex C for example procedure for testing vibration monitors.

6.4.4.2 Thrust position monitors

Refer to Annex C for example procedure for testing thrust monitors.

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6.4.4.3 Overspeed trip

See Annex D-1 and D-2 for example procedures for testing overspeed trip logic.

6.4.4.4 Permissive start of turning gear motor

See Annex E for example of a turning gear motor permissive start test procedure.

6.4.4.5 Lube oil pump auto start test

See Annex F for example procedure for lube oil pump auto start test.

6.4.4.6 First out alarm tests

See Annex G for example procedure for testing first-out sequence alarms.

6.5 Logic solver test procedures

Use SIF-specific functional test procedures when testing the logic solver. Functional test procedures may include

- written procedures;
- logic diagrams;
- control loop drawings;
- electrical control schematics; and/or
- checklists.

Using HMI, test each SIF manually by creating each fault condition and verifying proper response on the HMI and observation of the final control device(s).

Using PLC programmer for the logic device being tested and HMI screen, test the logic programmed function by function. Thoroughly check and verify the internal scaling factors for calibration and test range limit flags with manual input and output value variation. Test each individual sensor, the measured value with separate certified Test Meter and the value measured in PLC. Verify that the PLC value is scaled to match the Test Meter measured value. Performance should be considered unacceptable if variation between Test Meter measurement and Logic Solver indicated values exceeds ± 2% of measurement range.

Validate logic solver performance by executing the appropriate procedure from the following tests.

6.5.1 Complex application logic systems

For an example functional test procedure for a complex application logic system, refer to Annex H.

6.5.2 PLC logic solvers connected to field devices

An example of a test procedure for complex logic that involves field devices also in included as Annex R.

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6.5.3 PLC logic solvers connected to simulators – Hardwired simulators

Some companies have developed hardwired simulators for use in testing PLC logic. These simulators consist of panels with potentiometers, lights, and switches to represent all input devices and lights to represent output device positions. The simulators may be connected to the input terminals of the PLC directly or an arrangement using plug connection cables may be used. With the simulator connected, a procedure which exercises all possible combinations of logic that the PLC might encounter is conducted to validate that the logic solver will perform as required for each safety function implemented. In some instances the simulation panel is arranged graphically to represent the process being protected. When this is done, the simulator can also be used as an operations training tool for the SIF functionality.

6.5.4 PLC logic solvers connected to simulators – Software based simulators

Some companies have developed software-based simulators to accomplish the testing described in the clause above. In this instance, the test program is developed in application software using another PLC or in some instances a personal computer. Connection to the logic solver for testing is similar to above. However, the use of such a simulation requires complete validation of the embedded, application and utility software in the simulator prior to testing the SIF Logic Solver. The software simulator might also be used in training operators in the functionality of the SIF. In some instances this software simulator might operate in an automated mode in performing the test.

6.5.5 PLC logic solvers not connected to field or simulators

Testing PES based logic solvers that are not yet connected to field devices or a simulator is limited to manual testing of application logic using the PES configuration device. This type of testing primarily takes place during the initial programming and configuration phases of the PES implementation for the SIF application. Since changes are numerous during these phases, formal documentation of this "testing" should not be necessary. The final application logic documentation should reflect the results of this testing.

6.5.6 Electromechanical relay logic solvers

See Annex T for an example of a procedure for testing an electromechanical relay based SIF.

6.6 Testing of final control elements

Manually open or close valves and start or stop motors individually. In some applications, this test might have to be repeated 2 or 3 times to ensure proper functioning of the valves. Failure to properly open or close on the first attempt might be considered a failure by some companies and repeating the test 2 or 3 times to see the valves function would not ensure proper operation when the SIF called for a trip. Others might just want to see the valves operate more than once to obtain a confident feeling of proper functioning.

Manually change the output value for linearly controlled devices such as control valves. Observe the response of the device by watching the feedback value on the HMI and directly at the device. Document response of each valve in field and indication on HMI.

A test of the SIF valve should determine whether the valve can meet the functional requirements provided in the safety requirements specification. In addition to full stroke testing, the valve test may involve leak testing in cases where the valve has been specified with a maximum leak rate. Stroke times may be determined and recorded if valve stroke speed is critical. Stroke time should include the time from output signal change to valve position change, not just from start to finish of valve stroke. It has been shown

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that the pre-stroke dead time as actuators fill or exhaust and achieve breakaway force on the valve is generally the longest time component of the total stroke time.

Leak testing of SIF valves may require installation of bleed valves with pressure gauges downstream of the valve so that the valve can be monitored for positive shutoff. The burner management standard NFPA (8502)[2] gives guidance on this for fuel valves to furnaces and boilers that is also applicable to other process valves requiring positive shutoff.

6.7 Testing solenoid valves

Verify solenoid valve normal and trip condition status. If solenoid is normally energized during process operation, verify that coil is energized and no air is venting through vent port. If solenoid is normally de-energized during process operation, verify that coil is de-energized and vent port is open to vent. De-energize or energize coil as required and verify that air is either vented from valve actuator or applied to valve actuator as required by SIF logic. Verify that solenoid installed position allows gravity assist in taking valve to de-energized position. For examples of testing solenoid valves see example procedures for testing of final control elements (Annexes W, Z, DD, and MM).

6.8 Testing of HMI

All indications of SIF variables that are displayed on a human machine interface whether they be the BPCS operator workstation, a separate operator display station, or lights on a panel should be verified as each variable is tested. The correct range of process variable, the pre-alarm and trip setpoints, and any other variable information that is provided should be verified and documented during the testing. Both as found and as left values should be documented. Where multiple pages (video, CRT, etc.) of SIF information are provided, all displayed pages should be verified for appropriate labeling and access control.

If the HMI is used to initiate output functions for the SIF such as may be the case in batch control applications or a manual shutdown function, this function should also be tested.

6.9 Testing of communications

Where provided all communications with other systems such as the BPCS should be tested to verify correct transfer of information and data from the SIF to the other system(s). All information transferred should be verified by comparing the sent information with the received and displayed information on the system(s) other than the SIF.

Techniques used for blocking communications from the BPCS operator workstation to the SIF logic solver, especially those used to prevent unintended logic changes to the SIF application software, should be validated. Attempts at changing logic in the SIF should be made from the BPCS operator workstation to verify that this action cannot take place. The security technique used to protect against changes to logic from the configuration station should also be tested. If this involves connecting the configuration station only when changes are to be made, verify that another PES station cannot perform this function. If password protection is the technique used, verify that the password cannot be easily discovered through normal “hacking” in computer software. This is especially important if the SIF display station is also used as the configuration station with key lock and or password protection.

Where a separate operator display station is provided for the SIF, tests should confirm that changes to logic in the SIF logic solver cannot be made from this station.

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[2] NFPA 8502, published by the National Fire Prevention Association.

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6.10 Final SIF test procedures

Verify that all inputs, outputs and logic are in correct state at end of test and ready for process startup to proceed. This includes removing all bypasses, jumpers, etc. and returning all final control elements to pre-startup positions. Verify that any temporary jumpers used for bypassing are accounted for by comparing to list provided for each SIF. See Annex J for example of a jumper control list.

Perform a final inspection on the logic solver and all SIF components. The intent of this inspection is to make sure all work on the SIF is complete and that the system can safely be returned to normal operation. The inspection should include, but not be limited to, the following items.

- Verify that all alarms are cleared. Exceptions might be low process variable alarms that cannot be satisfied until process has been advanced to some operation state other than out of service.
- Verify that all problems and failures identified have been addressed.
- Check any components and devices that were replaced to ensure proper working condition.
- Verify all switches and hand switches are in their proper positions.
- Visually inspect all SIF pressure and instrument gauges to insure proper working condition.
- Visually inspect tubing, wiring terminations, and wiring to insure that they are secure. This might include actually trying to pull wire from the connections.
- Verify that all final control elements are in the correct position for the process out of service state.
- Verify that all instrument air supply regulators are at their proper settings.
- Verify that field junction boxes and housings are secured and weather tight.
- Verify that all wiring conduit and conduit access plates are secure and weather tight.
- Verify that all process root valves to transmitters and switches are open and any bleed valves are closed.

7 On-line testing

Successful on-line testing requires planning, design provisions, and procedures. When possible, the SIF should be designed to minimize or eliminate the need for bypassing or jumpers for testing. Any installed equipment for on-line testing, such as bypasses or instrumentation, should be thoroughly tested, along with its associated logic during commissioning. Detailed test procedures are essential for on-line testing to ensure that the test is correctly implemented. It is important to emphasize that any on-line testing presents the risk of a process upset or unintentional shutdown as the result of an incorrectly performed test.

7.1 Preparation

Prior to any on-line testing a review of the tests to be conducted and the procedures for performing these tests should be carried out by persons from instrument/electrical maintenance, operations, and technical who are familiar with the process and the SIF. This group should review the following items at a minimum:

- Discuss the importance of operators on shift being given notification that a SIF system is about to be tested or worked on.

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- Review the SIF system description.
- Review the SIF system functional test procedure.
- Discuss whether the on-line test will affect other systems, such as the BPCS, alarms, or other SIFs.
- Discuss the work scope, exactly what will be checked, what flows, pressures, temperatures, levels, etc.
- Discuss why craftsman should notify the operator when activating each alarm.
- Discuss what devices will no longer function when bypassing the system.
- Review with Operations any special precautions required during the test.
- Discuss what operations and maintenance should do if an unplanned SIF trip occurs while the input being tested is in bypass.
- Discuss what operations and maintenance should do if the operator must initiate the SIF while the bypass is in place.
- Discuss what procedures will be used to ensure that the SIF is returned to service once the SIF testing is complete; e.g., automatic verification, independent review, etc.

7.2 When should on-line tests be performed

On-line testing should never be performed when it would compromise the safety of the process.

The SIF components should be calibrated based on the plant's Preventative Maintenance (PM) schedule for the process equipment. The calibrations should be performed according to the company calibration procedures.

On-line testing may be necessary where the normal operating cycle of the process between scheduled shutdowns is greater than the test interval used in evaluating the SIF for its integrity level. Maintaining the required integrity of the SIF requires that this test interval be maintained. Therefore, the testing of some SIF will require doing the testing on-line.

Testing SIF on-line introduces stress on both the process and those performing the testing. It is therefore imperative that on-line testing be performed under closely controlled conditions using procedures that have been proven. This section will outline guidelines for when such tests should be performed and how this can be accomplished without compromising the safety of the process.

7.2.1 Sensors

Process sensors that are going to require on-line testing should generally be installed with some level of redundancy to allow testing of one sensor while another is still making the necessary measurement. If on-stream reliability of the process is critical, a 2oo2 or 2oo3 voting of sensors may be used. The designer then determines whether bypasses will be used to facilitate testing. For either 2oo2 or 2oo3 voting, one sensor can be tested at a time without the need for bypasses. When on-line diagnostics are used to detect transmitter failure, the designer determines whether the voting logic will be changed. For example, the logic for the SIF may be reduced from 2oo3 voting to 1oo2 if a failed transmitter is voted toward the trip condition. In contrast, it would reduce from 2oo3 voting to 2oo2 if the failed transmitter is voted away from the trip condition. If a 1oo2 configuration is used for sensors, a bypass will be necessary to allow on-line testing of each sensor while maintaining measurement capability with the other sensor.

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Logic during such a test will reduce to 1oo1, which is a lower SIF integrity than the 1oo2, and appropriate precautions should be taken during the testing to ensure safety is not compromised.

The testing frequency for sensors can be more or less frequent than that of other SIF components depending on the MTTF of the components used and the voting configuration. Where analog sensors are installed in redundant configurations, the testing interval for individual sensors can often be extended due to diagnostic coverage provided by analog signal comparison and alarming on deviation of the signals. Testing and calibration of the sensors would then be performed when the deviation alarm is generated. Depending on the voting configuration, on-line testing may not be necessary to maintain SIF integrity. This assumes that common cause failures such as mis-calibration of all three sensors has been accounted for in the calibration procedures.

7.2.2 Logic solvers

Testing of logic solvers for SIF is not practical while the process is on-line. Therefore the full functionality of the logic solver should be tested and validated prior to placing the SIF in operation as a layer of protection for the process. Further testing of the logic solver should be performed at the scheduled down time for the process and any time the SIF is taken out of service for logic changes.

7.2.3 Final control elements

Final control elements often have limited on-line diagnostic capability. Consequently, final control elements generally contribute the greatest amount toward the probability to fail to function when a demand is placed on the SIF. These devices typically remain in one position for long periods of time without moving until they are called on to respond to a process demand. Final control elements may also be installed under process operating conditions that can be severe, e.g. corrosive, plugging, or polymerizing services. They also contain many moving parts which must function together to accomplish the desired action they are to perform. Since the test interval to achieve the required safety integrity is often shorter than the turnaround interval for the process, on-line testing of final control elements becomes a desirable alternative.

Whether simplex or redundant valves are utilized, on-line testing requires additional design provisions, e.g., full flow bypasses, partial stroke testing equipment, test instrumentation, etc., to allow testing to occur without process interruption.

Final control elements may have common components, which could render multiple devices unavailable when these common components fail. For example, if air were used to move valves, which are used for process isolation, the loss of air supply would be a potential common cause failure. If the air supply fails to provide the necessary pressure or volume to move either of the valves, the SIF will fail to accomplish its design function.

The testing interval required to achieve the SIF integrity is affected by the severity of the service the valve encounters. Temperature (high or low), erosion, corrosion, and polymerization are a few of the factors which may have an impact on the required testing interval. In many cases, on-line testing is required in order to achieve the SIF integrity. On-line testing may consist of a full functional test or a partial test of the valve failure modes. When on-line diagnostics or partial stroke testing is used to supplement full functional testing, an assessment of the failure modes detected by the diagnostics should be performed. The diagnostic coverage factor used in the SIL verification should be substantiated by failure modes and effects analysis (FMEA). Many users limit the coverage factor assumed in the SIL verification to a certain maximum, e.g. 60%.The SIL calculation is then performed by splitting the PFDavg calculation into two parts. A portion of the valve failure modes is tested at the partial stroke testing frequency. The remainder of the valve failure modes is tested at the full stroke testing frequency.

A visual inspection according to an approved procedure should be carried out regularly, e.g. every three months. See Annex O for a sample procedure or checklist for this visual inspection.

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7.2.4 HMI

Testing of the HMI during normal operation of the process should be done any time that there is an indication of a malfunction of the HMI display itself. This could result from a fault in an input to the display or a fault in the display component itself. When repairs are made or a HMI is replaced, all features of the original HMI specified for the SIF should be tested.

The HMI should also be tested on the same schedule as the logic solver.

7.2.5 Communications

Communications between the SIF and other systems should be tested on the same schedule as the logic solver and at any time that there is an indication of a malfunction of the communication link. If communication with another system has an impact on the safety integrity of the SIF, the test interval included in the integrity evaluation should be used. Any on-line testing of a communication link should not reduce the capability of the SIF to perform its function.

7.3 Performing on-line testing

7.3.1 Precautions

On-line testing should not be started unless it can be worked step by step to completion with no anticipated interruptions. Once the inputs or outputs are bypassed, a dedicated control system operator should monitor the process continuously using means independent of the SIF. The operator should be capable of initiating a manual trip of the SIF or other installed systems in the event of a process demand during the test. Once the manual block valves are opened or closed, a dedicated field operator should be available to open or close the block valves quickly if a process demand occurs. All personnel involved in on-line testing of the SIF components should be aware of the mitigation steps to take in case a process demand occurs while the testing is in progress. The following caution should be included at the beginning of all on-line test procedures:

CAUTION — THE OPERATOR (S) MUST FULLY UNDERSTAND AND BE PREPARED TO IMPLEMENT THE MITIGATION PLAN FOR THIS PROCESS IN THE EVENT THAT A TRUE TRIP DEMAND OCCURS DURING THE CONDUCT OF THIS PROCEDURE.

Similar to the off-line testing procedure, measure the power supply voltage, AC or DC, for the SIS components and verify if the power is within the acceptable range. Test values should be within ± 2% of normal values.

Check the line-to-ground voltage per line.

7.3.2 Sensors - Transmitters

Several examples of testing sensor (transmitter) logic on-line in SIS are shown in Annexes L, M, and V. In each of these procedures a slightly different approach is used but all of them accomplish the same result of verification of sensor operation and logic in the SIS.

7.3.3 Thermocouple test for 2oo3 configuration

See Annex Y for model procedure for performing a 2oo3 test of thermocouple operation and logic in SIF.

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7.3.4 Sensors – Switches

7.3.4.1 Level switch technique

See Annex K for example of a procedure for on-line testing of a level switch.

7.3.4.2 Pressure switches

Pressure switches can be tested on-line using the same procedure as off-line tests with provision for bypassing the input during the testing.

7.3.4.3 Temperature switches

Only the output portion of temperature switches can be generally tested on-line. Use the same procedure as off-line tests for the output portion of the switch with provision for bypassing the input during the testing.

7.3.5 Logic solvers

In general testing logic solvers while the process is in operation is not recommended. The logic solver is typically the most reliable portion of a SIF and once the application program is fully validated by the PSAT, there is no need to retest the logic solver unless changes have been made to the logic contained in the logic solver. When changes are made to the logic, the logic solver should be retested prior to placing the SIF back in operation.

Testing electromechanical based logic solvers on line would require extensive modifications to allow this testing. These modifications could result in a system with less integrity than one without the provisions for testing. It is therefore not considered a good practice to attempt testing electromechanical based logic solvers while the process is on-line.

Where the SIF is functioning during a startup of the process, a test of SIF logic typically occurs each time the process is started up. If more frequent test intervals than the normal process turnaround schedule is required to achieve the SIL required, credit might be taken for unplanned startups due to downtime forced by equipment or utility failures.

7.3.6 Final control elements

On-line testing of final control elements can be the most difficult testing associated with the SIF. Any test of the valve on-line may result in process disruption if the test is not properly conducted. Valve tests can consist of a full stroke using process bypasses or a partial stroke to a specific percentage of valve movement. Any valve test should be evaluated to determine what failure modes are detected during the test. Of particular significance with respect to partial stroking of valves is that the partial stroke does not determine whether the valve will function to its full open or closed position. This can only be determined by a full stroke test.

Some companies take credit for on-line valve tests when an unplanned trip of the system takes place. They verify that all valves went to their correct position as required by the trip condition and that all indications of valve position indicated this to be true. They then document what has occurred and count this as a test of the valves affected. When taking such credit, consideration should be given to the performance requirement of the operation of the valve (i.e. speed of response and shutoff performance). The documentation should include the rationale for acceptance of the performance based on additional in-line testing while the opportunity is available or noting that prior testing could lead one to believe the performance is adequate until the next scheduled test.

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Techniques have been devised to allow some measure of testing of final control elements, particularly valves. These include use of manual block valves around the SIF valve for use while the testing is being performed. A drawback of this approach is high capital cost and the chance of leaving them in the wrong position after a test has been performed. Using this technique requires special attention to operation of the manual valves before and during the test. Annex Z is an example of testing valves that have installed manual block valves for testing. A valve lineup procedure has been developed by one company to follow during testing involving manual block and bypass valves. The procedure follows:

VALVE LINE-UP ACTIVITIES

During the course of this test, the Technician Performing the Test will be instructed to have an Operator close the upstream manual valve associated with this system. Since the upstream manual block valve is Car Sealed, the Operator must first remove and dispose of the Car Seal before closing this valve. Closing the manual block valve shall be performed in accordance with all existing site procedures.

Upon completion of this test, the Technician Performing the Test shall inform the Operator the upstream manual block valve may be opened. Opening of the manual block valve shall be performed in accordance with all existing site procedures. The Operator must install and lock a new Car Seal on the manual block valve and record the Car Seal Number in the space provided at the end of this test.

Another technique involves testing only through the final solenoid valve on the final control element actuator. This is common practice by many companies today and allows validation of elements of the SIF except the movement of the final valve itself. In this type of testing, the air supply to the valve actuator from the final solenoid is shutoff to prevent venting the actuator and operating the valve when the solenoid is tripped. Since about half of the final control element failures probably involve the solenoid, this technique can account for about half of the potential failures of the final control element package.

Some companies use redundant solenoids on each SIF valve to improve the availability or reliability of the SIF. Dependent on the solenoid configuration, bypassing may be required to test each solenoid one at a time and to verify that the solenoid has vented. When the test is complete, the technician should verify that the solenoid has been returned to service. Simply testing that the solenoid coil has energized or de-energized is not a complete test, since the solenoid must move to a specified vent state for correct functioning. For example, a test of the solenoid coil will not detect that the vent port is plugged with debris, preventing the venting of the air from the process valve. The following provides an example of a test for dual solenoid which is implemented using a bypass valve on the air line and a defeat switch in the logic.

a) Turn the bypass valve slowly to “Bypass” while watching the pressure gauge to ensure air pressure remains unchanged.

b) The trip solenoids are now bypassed. Check ( )

c) With the system in trip condition, temporarily place the defeat switch to OFF. Both solenoid valves should trip.

Solenoid valves tripped. Check ( )

d) Return all bypass valves to normal operating position. Check ( )

Other techniques for testing solenoids but not the valve are shown in Annexes W and MM.

Another technique proposed and used by some companies involves doing a partial stroking of the final

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control element valve to verify movement at least begins when called for by the SIF. This movement does not ensure that the valve will go to its full open or closed position when a real demand is placed on the system but does give some indication that the valve will at least attempt to go to its tripped position. Several examples of procedures for performing a partial stroking test of a SIF valve are shown in Annexes DD, EE, HH, and LL.

The following guidelines have been suggested for on-line testing of valves:

- SIL 1 SIF systems typically do not require any on-line testing.

- At turnaround intervals of less than 3 years and a target SIL of 2, double block valves seldom need to be partial stroke tested unless a dirty process increases the valve failure rate beyond the value normally used in PFD calculations.

- For SIL 3 applications, the testing frequency must be less than three years and on-line testing of some type (i.e., partial stroke) must be performed. Fortunately, only about 10% or less of the installations in the process industries are SIL 3. This means that for a small percentage of shutdown systems or for turnaround periods greater than 3 years, some type of on-line testing of valves is typically required.

Some cautions should be noted with regard to partial stroke testing of SIF valves. These include:

- One user noted that a failure occurred in a process valve which had been partial stroke tested to a specific mechanical stop position for years. The valve only moved 1/4 of its full stroke when actually called upon to move to its full trip position.

- If positive isolation, i.e. tight shutoff, is required, a partial stroke test does not test this capability. Since a partial stroke test cannot detect all failure modes of the valve, full credit should not be given for partial stroke testing. The following application limitations should be considered when evaluating the use of partial stroke testing:

1) The service is clean. No dirt, polymerization products, deposition, crystallization, corrosive chemicals, etc.

2) No documented history of a test that revealed valve failure due to process-related seat failure.

3) It must not be a tight shutoff application. This specification indicates that the valve seating is extremely important, so the only valid test is a full seat test.

Partial stroke testing must consist of verification that the valve moved a set percentage of valve range. It is not considered a valid test to only confirm open or closed limit switch contacts. Percent movement of the valve should be confirmed using position indication, such as limit switches or positioners, or using visual observation. To prevent buildup of ridges on the valve stem at the percent range for the test, it is recommended that the percentage of travel periodically be changed.

Several companies now have a package, which allows assessment of the torque required to move the certain valve types during the stroke. This does not verify tight shutoff capability, but does provide some diagnostic coverage. A listing of some vendors providing these techniques is shown in Annex JJ**.**

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7.3.7 HMI

On-line testing of the HMI is not required unless changes have been made in the information presented to the operator. Any changes that modify information to the operator about the status of the SIF should be tested when they are made and verified as being appropriate.

7.3.8 Communications

Any changes made to communications from the SIF to any other system should be tested when the changes are made. It is not recommended that changes be made while the SIF is providing protection to the process as these change activities could result in nuisance trips of the SIF or result in program errors, which could render the SIF incapable of performing its function.

7.4 Inspection (observation techniques that enhance SIF availability)

Almost as important as testing of the SIF is having a program in place that monitors the apparent condition of components of the system and their capability to provide the performance required to meet the safety requirements. An example of a condition that could limit the performance capability of a SIF component would be corrosion buildup around the stem of a sliding stem valve used to isolate a process stream when called upon by the SIF. The buildup, if not noticed and tended to, could prevent the valve from stroking all the way or even at all when called upon to take action. Inspection activities, which monitor such a condition and others, which might occur, can enhance the safety integrity of the SIF. Considerations that should be a part of these inspection programs are discussed in clauses that follow.

7.4.1 General considerations

The physical condition of the components of a SIF should receive a thorough mechanical inspection on a regular scheduled basis. This is especially true for field components exposed to environmental conditions, changes, and things like corrosion, process spills, leaks, etc. This inspection should be documented and any action that is found to be necessary initiated immediately or scheduled for the first opportunity if that is satisfactory.

7.4.2 Responsible personnel

The process unit Operations Department should be responsible for scheduling the inspections. The inspections should be scheduled to coincide with the scheduled functional test at a minimum. A schedule of once each quarter or twice a year may be appropriate for processes where conditions tend toward potential problems. In very serious environmental conditions the inspection might be necessary more frequently.

Maintenance Craftspeople should be responsible for performing and documenting inspections. Documentation records should be maintained for reference. These records may provide information relative to MTTF values for components that are used for SIF evaluation calculations and might be useful in relating process changes to problems which occur.

The maintenance and operations departments should be responsible for following up on the repair of any deficiencies discovered during the inspection to ensure repairs are completed satisfactorily.

7.4.3 Evaluation criteria

Each component of a SIF should be in good condition with no visible physical defects, which could impact the performance or reliability of the system.

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The instrument craftsmen should complete a Safety Instrumented System Inspection Form during the course of the system inspection. See Annex O for an example inspection form.

Examine all parts of the SIF for damage, deterioration, missing parts, or other physical damage. The physical examination should include:

- All input devices to the SIS such as transmitters, switches, thermocouples
- All output devices such as solenoid valves, control valves, motor controllers
- System wiring with particular attention to terminations, junction boxes, conduit
- SIS logic system - electromechanical relays, PLC, TMR, etc.

If a defect is found during the inspection it should be corrected as soon as possible. If the defect cannot be corrected immediately, a work order should be generated to repair the defect as soon as practical. The nature of the defect should be described on the Safety Instrumented System Inspection Form.

The inspection should include, but not be limited to the following items.

- Verify that all components of the SIF are properly tagged and labeled.
- Visually inspect devices for excessive corrosion.
- Visually inspect all components, including alarm lights, to insure proper working condition.
- Visually inspect all SIF pressure and instrument gauges to insure proper working condition.
- Visually inspect tubing, wiring connections, and wiring to insure proper working condition.
- Inspect heat tracing if appropriate to ensure proper operation.
- Verify that all instrument air supply regulators are at their proper settings, bug screens in place and not plugged, etc.
- Verify that boxes and housings have proper seals and covers and are secure.
- Verify that all conduit and conduit access plates have proper seals and are secure.
- Verify that tubing and cables are properly routed and secure.

7.4.4 Sensors

The following inspection criteria, at a minimum, apply to field sensors:

- Are instruments tagged with a special tag identifying them as part of a SIF?
- Are process connections in good condition with respect to leaks, insulation, corrosion, etc?
- Are process root valves in correct position?
- Is instrument properly supported?
- Is required heat tracing and insulation in good condition?

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- Is conduit connection in good condition and covers in place?
- Are drains, seals, and covers in place, if required, and in good condition?
- Are process tubing lines properly supported?
- Is conduit properly supported?

7.4.5 Logic solvers

Logic solver cabinets should be inspected for proper ventilation or cooling, buildup of dust or other foreign material, proper closure hardware in good condition, absence of moisture, wiring and grounding connections secure, cabinet security devices in good working order, and proper operation of any lights that are meant to indicate a status condition of the logic solver itself. Some vendors of this equipment have recommended routine maintenance schedules that may offer other items that should be checked.

7.4.6 Final control elements

Control valves should be inspected for the following conditions as a minimum:

- Bug screens in place and not plugged up
- Tubing condition for air supply, connections to positioner or topworks; connections tight with no leaks
- Solenoids properly mounted with tubing and electrical connections in good condition
- Valve piping gaskets not leaking
- Valve stem not leaking
- Topworks in good condition; no cracks, leaks at gaskets, etc.
- No corrosion buildup around valve stem
- Instrument pressure gauges in good condition
- Any auxiliary equipment such as signal converters and positioners, in good condition
- Any other conditions which might hinder proper operation of the valve
- Appropriate tagging of valve is in place

7.4.7 Switches

Switches used as hardwired bypasses should be inspected for proper position, security measures in place, and wiring connections secure.

7.4.8 Wiring connections

Any critical wiring connections in junction boxes, scramble boxes, or other terminations should be checked for proper tightness, labeling and mechanical protection. The use of wire nuts for making connections in SIF is not recommended. Seals where required should be checked. Conduit covers should be in place. Conduit drains should be in place and working properly. Cabinet doors should be closed, water tight, and properly labeled.

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7.5 Testing documentation

7.5.1 SIF test procedures

A specific written test procedure should be available for each SIF included in the SIS. The procedures should be of sufficient detail to allow personnel who are not intimately familiar with the SIF to perform the appropriate testing. These should include:

- List of safety function(s) included in the SIF
- Equipment description and location for each safety function
- Functional logic for each safety function
- Inspection procedures to be followed
- Calibration and testing methods to be followed
- Frequency of calibration, testing, inspections, and maintenance activities
- Specify acceptable performance limits (± 2% of full range if no limits specified)
- Specify sequence of testing if required
- Specify who should perform test
- Specify state of process when test is performed
- If SIF logic is mirrored in the BPCS, test should show that SIF actuated final control device.
- Verification of operational state of SIF after test complete
- Test of internal and external diagnostics (WDT, etc.)
- Verify auxiliary service components are operational (fans, filters, batteries, UPS, etc.).
- Define a means of ensuring testing is performed and documented.

All test procedures should have system being tested, page numbers, and revision date on each page of procedure. The responsible person for maintaining each procedure should be identified in the procedure.

All drawings used to describe SIF should be referenced including P&IDs, loop drawings, logic sheets, etc.

7.5.2 Documentation of functional testing of SIF

Document the results of functional tests for all SIF components and systems.

Test documentation should include but not be limited to the following data:

- Date of inspection and testing
- Name (signature) of the person(s) performing the work
- Tested equipment serial number or other unique identifier, such as loop number, tag number, or, equipment number

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- Results of the inspection and test (as found and as left conditions)

Important: Confirm and document that alarm and/or shutdown trip devices and process actuators operate within specified tolerances. This can be accomplished individually as a component test or as part of the loop or system test.

Retain records of these functional tests and inspections in accordance with plant policy. It is recommended that at least the two most frequent records of functional testing of the SIF be kept at the plant site. If a regulating body such as OSHA requires records retention, the retention period in that regulation should be followed.

7.5.3 Documentation of SIF component calibration

Document each calibration of a SIF component. Calibration documentation should include the following data:

- Date of inspection and calibration
- Name of the person performing calibration
- Calibrated equipment serial number or other unique identifier, such as loop number, tag number, or equipment number
- Before and after results of the calibration; i.e., “As Found” and “As Left” condition
- Test equipment (by manufacturer and model/serial number) used for the calibration

Calibration records should be maintained to confirm that this work was completed and to build a historical database of SIF component performance.

NOTE These records become the basis for adjustment to the calibration interval specified for each safety system component. The frequency(s) of testing and calibration of the SIF or portions of the SIF is re-evaluated at a periodic interval set by the site. The re-evaluation frequency is based on historical data, plant experience, hardware degradation, software reliability, etc.

7.5.4 Off-line tests

A good example of a test documentation form for off-line testing documentation is shown in Annex AA.

7.5.5 On-line tests

The same forms used to document off-line testing can be used to document on-line testing with the proper notations provided. Special forms may be developed if the user desires.

7.5.6 How test results are analysed

The results of the calibration and testing should be reported to the site engineer responsible for the SIF for review and approval. If necessary, the site engineer will consult with the site safety and environmental personnel for his/her review and recommendation with regard to the impact on the safety and/or environmental issue(s).

8 Inspections

An example of a form for documenting results of an inspection program is shown in Annex O.

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9 Auditing

Audits should be performed to verify that the procedures related to SIF and, in particular, those outlined in the SIF testing document remain in force throughout the life of the SIF. Records of audits and their results should be documented and maintained in plant records. Two types of documents that might accomplish this audit may be found in Annex FF and GG.

10 References

This document was compiled from input provided by operating companies, manufacturing companies, consultants, and individual engineers who have experience in the application, design, installation, operation, and maintenance of SIF. The best practices and procedures of these companies and individuals were combined and edited to allow use without disclosing any proprietary information from any one company or individual.

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CAUTION — PRIOR TO APPLYING THE INFORMATION IN THIS ANNEX TO DEVELOP A PROCESS SPECIFIC PROCEDURE, THE USER SHOULD REVIEW AND UNDERSTAND THE GUIDANCE IN THE BODY OF ISA-TR84.00.03-2002.

Annex A — Model procedure for approval required for replacing individual components in SIF

Scenario: A SIF instrument or valve needs to be replaced.

The following guidance should be followed in replacing the SIS component:

1. An instrument or valve with the exact model number of the failed SIF component is available from plant stores or a commercial supplier.

   Instrument Craft Person can make this decision.

2. An instrument or valve with the exact model number of the failed SIF component is not available from stores or commercial supplier.

   CASE 1:

   A list of equivalent instruments or valves has been prepared and approved for look-up use at plant site.

   Instrument Craft Person selects component from the list.

   CASE 2:

   1. Functional and physical specifications for the SIF component to be replaced are available in the SIF documentation.

   2. A substitute component with specifications that are equal to or exceed those of the failed component is identified. Equivalent functional performance of the available substitute instrument or valve is certain.

   Maintenance Technical Staff approves substitute.

   CASE 3:

   1. Functional and/or physical specifications for the SIF component to be replaced are INCOMPLETE in the SIF documentation, or

   2. The substitute instrument or valve available requires a change of

      - piping or process equipment;

      - measurement technology; and/or

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- functional performance of the SIF.

Engineering personnel with responsibility for SIF integrity of this process approves substitute.

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Annex B — Model procedure for deferring scheduled testing of SIF

Decision to defer

The scheduled test of a SIF may be deferred provided certain guidelines are followed. The following guideline will insure all proposed deferrals are properly reviewed and approved prior to granting a deferral. Note that the personnel titles used may be different from location to location. The intent is to reflect approval positions and not exact titles.

Deferral request

Deferral request shall be transmitted from Operations to the Instrument Specialist prior to the scheduled time to test a SIF. The timing shall allow ample time for the Instrument Specialist to conduct a fact based deferral analysis.

Reason for the request

There are several potential reasons for deferring the test of a SIF.

A turnaround is scheduled shortly after the scheduled test and the risk of off-line testing is lower than on-line testing. Also, the off-line test may enable the final control element to be tested whereas an on-line test may not allow the final control element to be tested.

1. The process equipment that the system is safeguarding is out of service. The agreement in this case is that the SIF will be tested prior to the process equipment being activated.

Deferral length

Suggested maximum length of time for a deferral should not exceed one quarter. If additional time is needed for a deferral after one quarter, it is suggested the deferral analysis be revisited along with approvals.

Deferral analysis

A deferral analysis should be conducted prior to granting a deferral. This analysis should include prior test results. A record of successful tests of the SIF should be the minimum acceptable criteria for deferring a test. The Instrument Specialist should participate in this deferral analysis and his/her concurrence should be required prior to forwarding to the approving authorities noted below.

Approvals required for a deferral

SIL I and SIL II systems: Operating and Technical Area Superintendent.

Procedure No.
Revision Date
Page _ of _

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SIL III systems: Site Operations Manager and Control Systems Manager

Communication of deferral

The following should be made aware of any approved deferrals.

- Site Operations Manager
- Operating Area Superintendent
- Technical Manager Control Systems
- Technical Superintendent
- Engineering/Maintenance Manager
- Instrument Specialist
- Control Systems Engineer

Documentation of deferral

All deferrals should be documented with each of the items above captured.

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Annex C — Model procedure for testing turbine thrust position monitors

PROBE V-1234

1. Put **VT-1234** in the defeat position.

   Red defeat light on the face of **VT-1234A** should be on - verify.

2. Check calibration of **VT-1234**. Record findings below, make no adjustments until initial checks are made.

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Table C.1 — Turbine thrust position

Calibrate 0 – 30 mils. Active.

ANY FAILURES? _____

VT-1234	ORIGINAL CALIBRATION					FINAL CALIBRATION		
	GAP VOLTS	TEST PT VOLTS	FAILURE LIMITS TEST PT.	MONITOR INDICATION	SWITCH SETTING	TEST PT VOLTS	MONITOR INDICATOR	SWITCH SETTING
ACTIVE +40 MIL.								
ACTIVE +3O MIL.			8.4 TO 9.1 V					
DANGER VSHH-1234 ACTIVE +30			27 to 33 mils					
ALERT VSH-1234 ACTIVE +20								
0 MIL.			4.6 to 5.4 V					
ALERT VSH-1234 INACTIVE -25								
DANGER VSHH-1234 INACTIVE -30			-27to -33mils					
INACTIVE -30 MIL.			0.9 to 1.6 V					
INACTIVE -40 MIL.								

3. Using wobulator pass **VT-1234** through its alarm point in the **active** direction. Do not pass VT-1234 through its trip point at this time.

   a. Red danger light on **VT-1234A** should be off - verify.

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b. **PI-4321** - located on S/D box should read 20# - verify.

c. **PI-4331** - located on S/D box should read 20# - verify.

d. **VAHH-5001-3** located on local panel and **UJR-6001** should be clear - verify

e. Alert light on **VT-2345** should come on - verify.

f. **VAH/TAH 5001-1** located on local panel should come on - verify.

g. **XA-7000** - the common trouble alarm in the control room should come on - verify.

h. **VAH/TAH 5001-1** alarm on sequence of events recorder (**UJR-6001**) should print out.

i. Acknowledge **XA-7000**.

4. Using wobulator (TK-3) pass **VT-1234** through its trip point in the **active** direction.

a. Red danger light on **VT-1234A** should come on - verify.

b. **PI-4321** - located on S/D box should go to zero - verify.

c. **PI-4331** - located on S/D box should go to zero - verify.

d. **XA-7000** - the common trouble alarm in the control room should reflash - verify.

e. **VAHH-5001-3** located on local panel should come on - verify.

f. **VAHH-5001-3** on sequence of events recorder (**UJR-6001**) should print out as being in the trip condition - verify.

g. Alert light on **VT-1234A** should remain on - verify.

h. **VAH/TAH 5001-1** located on local panel should remain on - verify.

5. Using wobulator adjust **VT-1234** below its trip point and **not below its alarm point**, reset monitor.

a. Red danger light on **VT-1234A** should go off - verify.

b. **VAHH-5001-3** should clear - verify.

c. **VAHH-5001-3** on sequence of events recorder (**UJR-6001**) should print out as being normal - verify.

d. Alert light on **VT-1234A** should remain on - verify.

e. **VAH/TAH 5001-1** located on local panel should remain on - verify.

f. **XA-7000** - the common trouble alarm in the control room should remain on - verify.

6. Using **XV-5050A** reset system.

a. **PI-4321** - located on S/D box should read 20 psig.

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b. **PI-4331** - located on S/D box should read 20 psig.

7. Using wobulator (TK-3) adjust **VT-1234** below it's alarm point.

   a. Alert light on **VT-1234A** should go off – verify.

   b. **VAH/TAH 5001-1** located on local panel should clear - verify.

   c. **XA-7000** - the common trouble alarm in the control room should clear - verify.

   d. **VAH/TAH 5001-**1 alarm on sequence of events recorder (**UJR-6001**) should print out as being normal – verify.

   e. Red danger light on **VT-1234A** should remain off - verify.

   f. **PI-4321** - located on S/D box should read 20# - verify.

   g. **PI-4331** - located on S/D box should read 20# - verify.

   h. **VAHH-5001-3** located on local panel and **UJR-6001** should remain clear – verify.

8. Using wobulator (TK-3) pass **VT-1234** through its alarm point in the **inactive** direction. **Do not pass VT-1234 through its trip point at this time.**

   a. Red danger light on **VT-1234A** should be off - verify.

   b. **PI-4321** - located on S/D box should read 20# - verify.

   c. **PI-4331** - located on S/D box should read 20# - verify.

   d. **VAHH-5001-3** located on local panel and **UJR-6001** should be clear – verify.

   e. Alert light on **VT-1234A** should come on - verify.

   f. **VAH/TAH 5001-1** located on local panel should come on - verify.

   g. **XA-7000** - the common trouble alarm in the control room should come on - verify.

   h. **VAH/TAH 5001-1** alarm on sequence of events recorder (**UJR-6001**) should print out.

   i. Acknowledge **XA-7000**.

9. Using wobulator pass **VT-1234** through its trip point in the **inactive** direction.

   a. Red danger light on **VT-1234A** should come on - verify.

   b. **PI-4321** - located on S/D box should go to zero - verify.

   c. **PI-4331** - located on S/D box should go to zero - verify.

   d. **XA-7000** - the common trouble alarm in the control room should reflash - verify.

   e. **VAHH-5001-3** located on local panel should come on - verify.

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f. **VAHH-5001-3** on sequence of events recorder (**UJR-6001**) should print out as being in the trip condition - verify.

g. Alert light on **VT-1234A** should remain on - verify.

h. **VAH/TAH 5001-1** located on local panel should remain on - verify.

10. Using wobulator adjust **VT-1234** below its trip point and not below its alarm point, reset monitor.

a. Red danger light on **VT-1234A** should go off - verify.

b. **VAHH-5001-3** should clear - verify.

c. **VAHH-5001-3** on sequence of events recorder (**UJR-6001**) should print out as being normal - verify.

d. Alert light on **VT-1234A** should remain on - verify.

e. **MAH/TAH 5001-1** located on local panel should remain on - verify.

f. **XA-7000** - the common trouble alarm in the control room should remain on - verify.

11. Using **XV-5050A** reset system.

a. **PI-4321** - located on S/D box should read 20 psig.

b. **PI-4331** - located on S/D box should read 20 psig.

12. Using wobulator (TK-3) adjust **VT-1234** below its alarm point.

a. Alert light on **VT-1234A** should go off - verify.

b. **VAH/TAH 5001-1** located on local panel should clear - verify.

c. **XA-7000** - the common trouble alarm in the control room should clear - verify.

d. **VAH/TAH 5001-1** alarm on sequence of events recorder (**UJR-6001**) should print out as being normal – verify.

e. Red danger light on **VT-1234A** should remain off - verify.

f. **PI-4321** - located on S/D box should read 20# - verify.

g. **PI-4331** - located on S/D box should read 20# - verify.

h. **VAHH-5001-3** located on local panel and **UJR-6001** should remain clear – verify.

13. Put **HS-5001** (bypass switch for the PGC thrust & vibration S/D) in the bypass position.

14. Using wobulator pass **VT-1234** through its trip point in the inactive direction.

a. **VAHH-5001-3** located on local panel should come on - verify.

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b. **VAHH-5001-3** on sequence of events recorder (**UJR-6001**) should print out as being in the trip condition - verify.

c. Red danger light on **VT-1234A** should come on - verify.

d. **VY-5001** should not energize and the S/D box should not trip.

e. **PI-4321** - located on S/D box should read 20 psig.

f. **PI-4331** - located on S/D box should read 20 psig.

15. Using wobulator adjust **VT-1234** back to a normal operating range and reset monitor.

a. **VAHH-5001-3** should clear.

b. Red danger light on monitor should go off.

c. **VAHH-5001-3** on sequence of events recorder (**UJR-5001**) should print out as being normal - verify.

16. Put **HS-5001** (bypass switch for the PGC thrust & vibration S/D) back in the normal position.

17. Using wobulator (TK-3) pass **VT-1234** through its trip point in the inactive direction again.

a. **VAHH-5001-3** located on local panel should come on - verify.

b. **VAHH-5001-3** on sequence of events recorder (**UJR-6001**) should print out as being in the trip condition - verify.

c. Red danger light on **VT-1234A** should come on - verify.

d. **VY-5001** should energize and the S/D box should trip.

e. **PI-4321** - located on S/D box should read 0 psig.

f. **PI-4331** - located on S/D box should read 0 psig.

18. Put **VT-1234** back in service and reset it.

a. Alert light on **VT-1234A** should be off – verify.

b. **VAH/TAH 5001-1** located on local panel should clear - verify.

c. **VAH/TAH 5001-1** alarm on sequence of events recorder (**UJR-6001**) should print out as being normal – verify.

d. Red danger light on **VT-1234 A** should be off.

e. **VAHH-5001-3** should clear.

f. **VAHH-5001-3** on sequence of events recorder (**UJR-6001**) should print out as being normal - verify.

g. **XA-7000** the common trouble alarm in the control room should clear – verify.

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19. Put defeat switch for **VT-1234 A&B** back to its neutral position.
    a. Red defeat light for **VT-1234 A&B** should be off - verify.
20. Using **XV-5050A** reset system.
    a. **PI-4321** - located on S/D box should read 20 psig.
    b. **PI-4331** - located on S/D box should read 20 psig.

When test is complete, sign and date below.

SIGNATURE **DATE**

OPERATOR:_____ DATE: _____

CRAFTSMAN: _____ DATE: _____

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Annex D-1 — Model procedure for electronic over-speed trip testing

1. Isolate **PI-4501A** and **PI-4501B**.

CAUTION — DO NOT ATTEMPT TO LOOSEN OR REMOVE PI-4501A OR PI-4501B UNTIL THE FOLLOWING STEP HAS BEEN COMPLETED.

2. Have operator close block valves up-stream and down-stream of **SV-4501**.

CAUTION — BE SURE VALVES UP-STREAM AND DOWN-STREAM OF SV-4501 ARE COMPLETELY CLOSED BEFORE PROCEEDING!

3. Check the calibration of the following pressure gauges.

	PI-4501A		**BEFORE**	**AFTER**	
	GAUGE INPUT	**FAILURE LIMITS OUTPUT**	**GAUGE OUTPUT**	**GAUGE OUTPUT**	**Failed? (Mark with √)**
0%	**0 PSIG**	**0# TO 10#**			
50%	**100 PSIG**	**90# TO 110#**			
100%	**200 PSIG**	**180#TO 220#**			
	PI-4501B		**BEFORE**	**AFTER**	
	GAUGE INPUT	**FAILURE LIMITS OUTPUT**	**GAUGE OUTPUT**	**GAUGE OUTPUT**	**Failed? (Mark with √)**
0%	**0 PSIG**	**0# TO 10#**			
50%	**100 PSIG**	**90# TO 110#**			
100%	**200 PSIG**	**180#TO 220#**			

4. Put **PI-4501A** and **PI-4501B** back in service. **SV-4501 must remain isolated.**

5. Have operator slowly open block valve up stream of **SV-4501**.

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a. **PI-4501A** should read Governor oil pressure.

b. **PI-4501B** should read 0 PSIG.

6. Have Operator close block valve up-stream of **SV-4501** on compressor turbine.

CAUTION — BE SURE VALVES UP-STREAM AND DOWN-STREAM OF SV-4501 ARE COMPLETELY CLOSED BEFORE PROCEEDING!

7. Turn power to speed switch **OFF**.

a. **XA-4501**, power failure or low speed alarm should come on - verify.

b. **SV-4501** should not energize, **PI-4501A** should still be reading Governor oil pressure and **PI-4501B** should be reading about zero - verify.

c. **SAH-4501** on local annunciator panel should remain clear - verify.

d. **SAH-4501** on sequence of events recorder (**UJR-6001**) should remain clear.

e. **SAHH-4501** on local annunciator panel should remain clear - verify.

f. **SAHH-4501** on sequence of events recorder (**UJR-6001**) should remain clear.

8. Connect frequency generator to **SSH/SSHH-4501** and apply an input signal above the low speed setting for **XA-4501** and **NOT** above the setting of **SSH-4501**.

NOTE Use only, Dynalco Model F-15 frequency generator. Noisy signals present in other frequency generators may cause SAH-4501 and SAHH-4501 to come on at the same time.

9. Turn power to speed switch **ON**.

a. **XA-4501**, power failure or low speed alarm should clear - verify.

b. **SV-4501** should not energize, **PI-4501A** should still be reading Governor oil pressure and **PI-4501B** should be reading about zero - verify.

c. **SAH-4501** on local annunciator panel should remain clear - verify.

d. **SAH-4501** on sequence of events recorder (**UJR-6001**) should remain clear.

e. **SAHH-4501** on local annunciator panel should remain clear - verify.

f. **SAHH-4501** on sequence of events recorder (**UJR-6001**) should remain clear.

10. Lower frequency below the setting of **XA-4501**.

a. **XA-4501**, power failure or low speed alarm should come on - verify.

b. **SV-4501** should not energize, **PI-4501A** should still be reading Governor oil pressure and **PI-4501B** should be reading about zero - verify.

c. **SAH-4501** on local annunciator panel should remain clear - verify.

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d. **SAH-4501** on sequence of events recorder (UJR-6001) should remain clear.

e. **SAHH-4501** on local annunciator panel should remain clear - verify.

f. **SAHH-4501** on sequence of events recorder (UJR-6001) should remain clear.

RECORD FINDINGS BELOW

INST. NO.	**PROCESS SETTING**	DEVICE SETTING	FAILURE LIMITS **HERTZ**	BEFORE	FINAL	Failed? (Mark with √)
XA-4501	**3600 RPM DEC.**	**6000 HERTZ DEC.**	**5400 TO 6600 HERTZ**			

11. Raise input frequency above the low speed setting for **XA-4501** and **NOT** above the setting of **SSH-4501**.

    a. **XA-4501**, power failure or low speed alarm should clear - verify.

    b. **SV-4501** should not energize, **PI-4501A** should still be reading Governor oil pressure and **PI-4501B** should be reading about zero - verify.

    c. **SAH-4501** on local annunciator panel should remain clear - verify.

    d. **SAH-4501** on sequence of events recorder (**UJR-6001**) should remain clear.

    e. **SAHH-4501** on local annunciator panel should remain clear - verify.

    f. **SAHH-4501** on sequence of events recorder (**UJR-6001**) should remain clear.

12. Raise frequency above the setting of **SSH-4501** and not above the setting of **SSHH-4501**.

    a. **SAH-4501** on local annunciator panel should come on - verify.

    b. **SAH-**4501 on sequence of events recorder (**UJR-6001**) should print.

    c. **XA-4501** power failure or low speed alarm should remain clear - verify.

    d. **SAHH-4501** on local annunciator panel should remain clear - verify.

    e. **SAHH-4501** on sequence of events recorder (**UJR-6001**) should remain clear.

    f. **SV-4501** should not energize, **PI-4501A** should still be reading Governor oil pressure and **PI-4501B** should be reading about zero - verify.

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RECORD FINDINGS BELOW

INST. NO.	PROCESS SETTING	DEVICE SETTING	FAILURE LIMITS HERTZ	BEFORE	FINAL	Failed? (Mark with √)
SSH-4501	5474 RPM INC.	9123 HERTZ INC.	8667 TO 9579 HERTZ			

13. Raise the frequency above the setting of **SSHH-4501**.

    a. **SAH-4501** on local annunciator panel should remain on - verify.

    b. **SAH-4501** on sequence of events recorder (**UJR-6001**) should not change.

    c. **XA-4501** power failure or low speed alarm should remain clear - verify.

    d. **SAHH-4501** on local annunciator panel should come on - verify.

    e. **SAHH-4501** on sequence of events recorder (**UJR-6001**) should print.

    f. **SV-4501** should energize and the pressure should equalize across it. **PI-4501A** and **PI-4501B** should now be reading the same pressure somewhere below the Governor Oil Pressure

RECORD FINDINGS BELOW

INST. NO.	PROCESS SETTING	DEVICE SETTING	FAILURE LIMITS HERTZ	BEFORE	FINAL	Failed? (Mark with √)
SSHH-4501	5940 RPM INC.	9900 HERTZ INC.	9405 TO 10395 HERTZ			

14. Put **SSH-4501** and **SSHH-4501** back in service.

    a. **XA-4501** power failure or low speed alarm should remain clear - verify.

    b. **SAH-4501** should clear - verify.

    c. **SAH-4501** on sequence of events recorder (**UJR-6001**) should print out clear - verify.

    d. **SAHH-4501** should clear - verify.

    e. **SAHH-4501** on sequence of events recorder (**UJR-6001**) should print out clear - verify.

    f. **SV-4501** should de-energize - verify.

15. Have Operator line **SV-4501** back up using the following procedure.

    a. SLOWLY open block valve up-stream of SV-4501 first. PI-4501A should start coming up. If PI-4501B starts coming up STOP because SV-4501 is leaking through.

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NOTE 1 If SV-4501 leaks through have operator close block valve UP STREAM of SV-4501. Slowly open block valve DOWN STREAM of SV-4501 to bleed pressure and allow SV-4501 TO SEAT, PI-4501B SHOULD GO TO 0 PSIG.

NOTE 2 Have operator close block valve DOWN STREAM of SV-4501 and repeat step 10.

Once it is determined that **SV-4501** is not leaking through and the block valve is completely opened proceed to step b.

b) **SLOWLY** open block valve down-stream of **SV-4501**. **PI-4501B** should drop to near zero without affecting **PI-4501A**.

When section is complete, sign and date below.

SIGNATURE **DATE**

OPERATOR:_____ DATE: _____

CRAFTSMAN: _____ DATE: _____

Procedure No.
Revision Date
Page _ of _

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Annex D-2 — Model procedure for testing turbine overspeed trip

Event:	Turbine Overspeed
Equipment number:	100PT (TriSen) and Turbine Mechanical Overspeed Trip
Test objective:	When the main steam turbine speed reaches 4800 rpm, the TriSen turbine governor will interlock down the turbine by de-energizing the turbine trip solenoid. In addition, if the TriSen interlock fails to operate, the mechanical overspeed assembly in the turbine will engage and shutdown the turbine at 5200 rpm.
Test frequency:	12-24 months during process shutdown
Process trip setting:	4800 ± 100 rpm for the TriSen interlock 5200 ± 100 rpm for the turbine overspeed
Type test:	Test by overspeeding turbine
Equipment required for test:	Handheld tachometer
Pre-test conditions:	Process shutdown with turbine uncoupled from blower. Steam available to turbine from package boiler.

Procedure No.
Revision Date
Page _ of _

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Interlock test procedure

TriSen hi-hi speed

_____1. Notify the control room operator that a hi-hi turbine speed interlock test will be taking place.

_____2. Ensure that the turbine is uncoupled from the blower.

_____3. Valve in the package boiler steam to the turbine.

_____4. Bypass both Eye-Hi interlocks by rotating the bypass switch on each unit. This will allow the turbine solenoid to be energized without water in the steam drum.

_____5. Enable local control of the turbine by rotating the governor bypass switch to the manual position. This switch is located in the enclosure beside the turbine.

_____6. Adjust the manual speed control valve that measures the air being applied to the turbine steam actuator. 15 psig of air pressure corresponds to minimum turbine speed, and 3 psig of air pressure corresponds to maximum turbine speed.

_____7. Reset the turbine trip solenoid by pressing the “START” button on the TriSen.

_____8. Raise the trip flag on the turbine into the normal position.

_____9. Begin raising the speed of the turbine by slowly adjusting the air pressure with manual speed control valve.

____10. Monitor the speed indicator mounted by the turbine and the reading on the TriSen in the control room. In addition, monitor the turbine speed with the handheld tachometer.

____11. Slowly increase the turbine speed as it approaches 4800 rpm to better observe the speed indicators when the interlock trips the turbine solenoid.

____12. When the turbine solenoid trips, observe and document the resulting trip point (“as found” condition).

____13. Adjust the manual speed control valve to the minimum position.

____14. The initial interlock test passed / failed. (circle one)

____15. If the interlock test failed, what corrective action was required?

_____
_____
_____

Procedure No.
Revision Date
Page _ of _

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Turbine mechanical overspeed

_____1. Notify the control room operator that a turbine mechanical overspeed test will be taking place.

_____2. Ensure that the turbine is uncoupled from the blower.

_____3. Valve in the package boiler steam to the turbine.

_____4. Bypass both Eye-Hi interlocks by rotating the bypass switch on each unit. This will allow the turbine solenoid to be energized without water in the steam drum.

_____5. Enable local control of the turbine by rotating the governor bypass switch to the manual position. This switch is located in the enclosure beside the turbine.

_____6. Raise the TriSen hi-hi speed interlock setting to 5500 rpm (refer to the TriSen Users manual for instructions).

_____7. Adjust the manual speed control valve that measures the air being applied to the turbine steam actuator. 15 psig of air pressure corresponds to minimum turbine speed, and 3 psig of air pressure corresponds to maximum turbine speed.

_____8. Reset the turbine trip solenoid by pressing the “START” button on the TriSen.

_____9. Raise the trip flag on the turbine into the normal position.

____10. Begin raising the speed of the turbine by slowly adjusting the air pressure with manual speed control valve.

____11. Monitor the speed indicator mounted by the turbine and the reading on the TriSen in the control room. In addition, monitor the turbine speed with the handheld tachometer.

____12. Slowly increase the turbine speed as it approaches 5200 rpm to better observe the speed indicators when the mechanical overspeed trips down the turbine.

____13. When the turbine overspeed assembly engages, observe and document the resulting trip point (“as found” condition).

____14. Repeat the overspeed test two more times for a total of three tests. Observe and document the resulting trip points (“as found” condition).

____15. Adjust the manual speed control valve to the minimum speed position.

____16. Turn off the #1 and #2 Eye-Hi Interlock Bypass.

____17. Return the TriSen hi-hi speed interlock setting to 4800 rpm (refer to the TriSen Users manual for instructions).

____18. Enable TriSen control of the turbine by rotating the governor bypass switch to the TriSen Governor position.

____19. The initial interlock test passed / failed. (circle one)

____20. If the interlock test failed, what corrective action was required?

_____
_____
_____

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Post-test inspection and documentation

_____1. The interlock equipment has been returned to normal and is ready for service.

_____2. Record “as found” condition results here:

_____
_____
_____

Test and inspection completed by:

Name:_____ Date:_____

_____ _____

_____ _____

Procedure No.
Revision Date
Page _ of _

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Annex E — Model procedure for testing permissive start for turning gear motor

1. Have an electrician pull the **"T**" leads on the turning gear motor starter.

2. Check the setting of **PSH-1234**, log findings below.

INST. NO.	SWITCH SETTING PROCESS	**FAILURE LIMITS**	**AS FOUND**	**AS LEFT**	FAILED? (MARK WITH √)
PSH-1234	**xx PSIG** **DEC.**	**y TO** **yy PSIG DEC.**			

3. Put a signal on **PSH-1234** that is above its trip point.

   a. **PAH-1234** permissive start turning gear alarm, on local panel should be clear.

   b. **XA-2345** common trouble alarm in control room should be clear.

4. Turn the hand switch for the turning gear motor to the **RUN** position.

   a. The motor starter should pull in - verify.

5. Lower the signal on **PSH-1234** below its trip point

   a. The motor starter should drop out - verify.

   b. **PAH-1234** permissive start turning gear alarm, on local panel should go on

   c. **XA-2345** common trouble alarm in control room should go on.

6. Put **PSL-1234** back in service.

   a. **PAH-1234** permissive start turning gear alarm, on local panel should clear.

   b. **XA-2345** common trouble alarm in control room should clear.

7. Return the hand switch for the turning gear motor to the **OFF** position.

8. Have electrician replace "**T**" leads and put motor starter back in service.

Procedure No.
Revision Date
Page _ of _

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SIGNATURE **DATE**

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CRAFTSMAN: _____ DATE: _____

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Annex F — Model procedure for lube oil pumps autostart test

NOTE Operations and maintenance personnel involved should review and understand this procedure prior to start of checks. Coordination and communication between operations and maintenance is critical.

This procedure will require two operators and two instrument craft-persons. One operator will man the hand switch for **P-1234** and the other will man the local control panel on **K-2345** compressor deck. The instrument craft-persons should have the necessary test equipment and fittings for field testing on hand prior to start of tests.

Each time **P-1234** starts or stops it will cause a swing in **LIC-4321,** third stage seal oil pot level controller. The operator at the local control panel for **K-2345** must understand and implement the necessary action to prevent a low seal oil pot level trip.

This procedure will call for the hand switch for **P-1234** to be placed in the off position while connecting test equipment and checking switch settings, this will prevent unnecessary pump starts and level swings.

PSL-1234A LOW LUBE OIL PRESSURE AUX. PUMP START AND ALARM SWITCH.

1. Have operator place hand switch for **P-1234** in the off position.

2. Isolate **PSL-1234A** and connect calibrated pressure source to it.

3. Check the setting of **PSL-1234A**, log results below.

INST. NO.	**SWITCH SETTING** PROCESS	**FAILURE LIMITS**	**AS FOUND**	**AS LEFT**	FAILED? (MARK WITH √)
PSL-1234A	**xx PSIG DEC.**	**yy TO yyy PSIG**			

4. Raise the input to **PSL-1234A** above its setting.

5. Have operator return the hand switch for **P-1234** to the auto position.

6. Have operator place **LIC-4321,** third case seal oil pot level controller in manual.

Procedure No.
Revision Date
Page _ of _

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CAUTION — THE OPERATOR AT THE LOCAL CONTROL PANEL FOR K-2345 MUST CLOSELY MONITOR LIC-4321. IN THE NEXT STEP P-1234 WILL START, CAUSING L-4321, THIRD CASE SEAL OIL POT LEVEL TO RISE RAPIDLY. K-2345 WILL NOT TRIP ON A HIGH SEAL OIL POT LEVEL. A LOW SEAL OIL POT LEVEL WILL CAUSE K-2345 TO TRIP. DO NOT OVER CORRECT FOR A HIGH LEVEL, THIS COULD RESULT IN A LOW-LEVEL TRIP.

7. Slowly lower the input to **PSL-1234A** below its setting.

    a. **P-1234** should start.

CAUTION — DO NOT STOP P-1234 AT THIS TIME, P-1234 SHOULD NOT BE STOPPED UNTIL PSL-1234A IS BACK IN SERVICE AND THE OPERATOR IS NOTIFIED.

    b. **PAL-1234A** on local panel should come on.

    c. **XA-3456** common trouble alarm in control room should come on.

    d. **PAL-1234A** should print on alarm printer.

8. Put **PSL-1234A** back in service.

    a. **PAL-1234A** on local panel should clear.

    b. **XA-3456** common trouble alarm in control room should clear.

    c. **PAL-1234A** should print out as being normal

9. Notify operator that **PSL-1234A** is back in service.

CAUTION — THE OPERATOR AT THE LOCAL CONTROL PANEL FOR K-2345 MUST CLOSELY MONITOR LIC-4321. IN THE NEXT STEP P-3428 WILL STOP, CAUSING L-4321,THIRD CASE SEAL OIL POT LEVEL TO DROP RAPIDLY. K-2345 WILL NOT TRIP ON A HIGH SEAL OIL POT LEVEL. A LOW SEAL OIL POT LEVEL WILL CAUSE K-2345 TO TRIP. THE OPERATOR SHOULD TAKE STEPS TO PREVENT THE THIRD CASE SEAL OIL POT LEVEL FROM DROPPING BELOW ITS TRIP POINT.

10. Have operator place the hand switch for **P-3428** in the off position.

    a. **P-3428** should stop.

11. Have operator place the hand switch for **P-3428** in the auto position.

    a. **P-3428** should remain off.

12. Have operator place **LIC-4321**, third case seal oil pot level controller back in auto.

When test is complete, sign and date below.

SIGNATURE	**DATE**
OPERATOR:_____	DATE: _____
CRAFTSMAN: _____	DATE: _____

Procedure No.
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Page _ of _

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Annex G — Model procedure for testing first-out sequence alarms

NOTE The following steps are to verify the First-Out annunciator sequence for the SIS alarms.

Drive **LSH-1234** through its alarm point using calibrated current source.

LTH-1234 on local annunciator panel (if applicable) should flash normally.

LTH-1234 on operator console in the control room should be in alarm condition.

Pass **LSH-2345** through its alarm point using calibrated current source.

LTH-2345 on local annunciator panel (if applicable) should flash normally.

LTH-2345 on operator console in the control room should be in alarm condition.

LTH-1234 on local annunciator panel should flash rapidly

Press the acknowledge button for the annunciator panel.

LTH-2345 should remain on steady.

LTH-1234 should remain flashing,

Repeat procedure actuating LTH 2345 alarm first and verify proper first out indication.

Procedure No.
Revision Date
Page _ of _

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Annex H — Model procedure for functional testing of TMR-based SIS instrumentation

NOTE This procedure addresses a SIS with multiple SIF.

H.1 Purpose

The purpose of this annex is to provide a model for site development of administrative controls and procedures to ensure that the integrity of all TMR-based SIS instrumentation is maintained through functional testing following (1) changes and repairs and (2) on a routine basis through periodic SIS system testing.

H.2 Management of change restrictions

H.2.1 Approval - The Operations Department Manager pre-approves the SIS configuration station connection to the TMR logic solver whenever the associated process unit is not totally shutdown.

H.2.2 Qualifications - Only TMR qualified personnel perform SIS testing work.

H.2.3 Written test procedure

A written, step-by-step functional test procedure is required prior to approval of work on the TMR LOGIC SOLVER whenever;

1. The associated process unit is not totally shutdown, and

2. Forcing of inputs and outputs is used as part of the functional test work.

H.2.4 Re-enabling ESD points

All active SIS points must be re-enabled after completion of commissioning work. Enabled I/O must be checked against a master list at the completion of functional testing; and this check must be documented as evidence of responsible management of change. This documentation should be filed with plant SIS records.

H.3 Procedure

H.3.1 Functional testing of SIS system following field changes and repairs

H.3.1.1 Reference documents

Obtain the SIS reference documents and testing procedures that document the part of the SIS system that is affected by the repair or field change. This documentation typically includes:

--`,,,`,,`,,,,,,```,,,,,,

1. Loop Diagram

2. SIS Logic Diagram

3. TMR Ladder Listing and Dictionary with Cross Reference

4. SIS Schematics, if applicable

H.3.1.2 Procedures

The procedure used when making changes to the TMR Logic Solver software should follow company guidelines or practices.

H.3.1.3 Comparison with master

The installed, modified TMR Logic Solver SIS Logic program is compared to the MASTER Program, [<Filename>.UPL] using the Upload-and-Compare Utility function of the TMR configuration station if available. If no program changes are identified EXCEPT FOR THOSE PLANNED MODIFICATIONS, an input-output functional check of the existing and unchanged SIS Logic is not required at this time.

H.3.1.4 Program compare listing

Printout the Program Compare Listing and file it with the documentation of the sensor and process actuator functional checks.

H.3.1.5 Functional check

All modifications to SIS logic are FUNCTIONALLY CHECKED. A checkout procedure should be defined according to the following steps:

1. The state-of-digital and value-of-analog inputs that are read through the Communication Module from TMR Logic to the BPCS can be monitored adequately at the BPCS Operator Workstation. Signals originating within the TMR logic (analog outputs, digital outputs) and any input signals that are received by the TMR logic and not fed forward to the BPCS will require connecting the TMR configuration computer to the TMR logic. The TMR configuration computer is used to verify correct SIS program values when an analog input field transmitter range is altered.

2. To functionally check analog and digital inputs associated with the SIS change, confirm that the TMR logic is properly reading

    a. the state of the digital inputs, and

    b. the 0%, 50% and 100% of range signal of the analog input in both counts and engineering units to validate square root or linear signal.

3. No input points should be disabled unless it is necessary to disable an undesirable trip function. See H-2 for Management of Change restrictions.

4. To functionally check digital or analog outputs associated with the SIS change either:

    a. Simulate a TMR logic input signal that would cause the output value to change state or take a known analog value; or

    b. Disable the associated output register and enter a forcing value.

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NOTE It may become necessary to disable other associated points to allow this output to be transmitted to the field or to the BPCS. See H-2 for Management of Change restrictions.

c. Proper output device response must be field validated.

5. Operation of all SIS trip and pre-alarms and first out trip indications that are associated with the changed logic are validated.

6. All points that were disabled during this functional checkout are returned to the enabled state following commissioning.

H.3.1.6 Documentation - The following documentation steps are required:

1. TMR logic documentation is completed, backup copies made and, if any logic changes were implemented, an up-to-date copy of all modified TMR configuration station files are inserted in Master TMR Logic SIS manual.

2. As a minimum, a printout of the POINT DISABLED file taken just prior to disconnecting from the TMR Logic is reviewed to ensure that all points not documented as “permanently out-of-service” are re-enabled. Other manuals are to be updated in a timely manner.

3. A copy of the POINT DISABLED listing is sent to the Staff member responsible for the unit's TMR Logic system.

4. Only documented “permanently out-of-service” points are left disabled.

5. Printouts of Points Disabled file following each repair must be kept in the file containing the last completed unit SIS Documentation.

H.3.2 Periodic functional testing

H.3.2.1 Functional test plan

An SIS Functional Test Plan that includes a procedure and that defines documentation is prepared for each SIS system.

H.3.2.2 Functional test requirement

A functional test of the SIS system is completed on a periodic basis by TMR Logic-qualified personnel.

H.3.2.3 Test plan approval

Operations Department Manager approves the Functional Test Plan.

H.3.2.4 Functional test documentation

Documentation of the completed, SIS functional test results including

1. as found/as left sensor calibration data and

2. pass/fail system response data

is maintained in Process Unit files for at least three years for auditing purposes.

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H.3.2.5 Periodic functional tests

All SIS system inputs and outputs, both analog and digital (including those triggering BPCS alarms and first out indications), are functionally tested on a periodic basis not to exceed the test interval included in the SIS integrity evaluation. More frequent testing of most field devices is recommended. A procedure for establishment of the test frequency for each interlock is included in the plant's risk management program.

The functional test procedure includes the following:

1. TMR Logic outputs may be functionally tested by

    a. disabling the point,

    b. altering its value/state, then

    c. verifying proper action in the field/BPCS Displays/Alarm Displays/etc.

Associated TMR Logic points are disabled and altered as necessary to permit operation of each control valve that is tripped by TMR Logic. Each control valve is opened to 50% output then tripped (opened/closed). The proper SIS action of each field automated valve should be field verified. Each proven SIS action is documented. See H-2 for Management of Change restrictions where forcing of input and output points is done.

2. TMR Logic input signals (DI/AI) are emulated from the field sensor, valve, or device and are validated in the TMR Logic and BPCS. Where both field and control room mounted start-stop switches can trigger an input, correct operation of both must be proven and documented.

3. The installed TMR Logic is compared to the MASTER Program, [<Filename>.UPL] using the Upload-and-Compare Utility function if available. If no program changes are identified, an input-output functional check of the SIS Logic is not required at the scheduled SIS functional checkout.

Printout the Program Compare Listing and file this listing with the documentation of the sensor and process actuator functional checks.

H.3.2.6 Complete functional check

A complete, field input-to-SIS valve functional check of the TMR Logic is to be performed at least once every four years. This check is in addition to the periodic software-compare validation of Step H-3.2.5.

H.3.2.7 Correction of deficiencies

All deficiencies noted during the functional check are corrected unless they have no impact on SIS safety function integrity. Department Manager approval is obtained and documented in the Functional Checkout records if a deficiency is not corrected.

H.3.2.8 Deficiency report

A report is written by a Staff TMR Logic specialist (for the complete input-output check made on a nominal four year cycle and for other scheduled functional checks) documenting all deficiencies encountered during commissioning and defining actions planned to eliminate such deficiencies. This information is filed with the SIS documentation.

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Annex J — Example of a jumper control list

Jumper Identification Number	Installed On	Installed By	Date	Removed From	Removed By	Date

A copy of this list should be placed in SIF record file after each functional test is performed.

Procedure No.
Revision Date
Page _ of _

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Annex K — Model procedure for on-line test of a high level switch

Obtain the necessary work permit? Verify on test form.

Place the DEFEAT/BYPASS SWITCH for device being tested in the DEFEAT/BYPASS POSITION. Verify on test form.

Remove level switch cover and check for contamination.

Check if terminal connections are tight.

Close level switch block valves. Open drain valve(s) to depressure switch.

Level interlock check:

a. Set up drain and block valves to flood the float chamber. The alarm should now be on. Verify on.

b. Line up valves to empty the float chamber. The alarm should now be off. Verify off.

c. Open process valves to level switch.

Return the defeat/bypass switch to run/normal position.

Procedure No.
Revision Date
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Annex L — Model procedure for on-line testing of flow sensors in a 1oo2 configuration (high or low trip)

From instrument record system, confirm the following:

Transmitters span

Pre-alarm switch setting (if applicable)

Deviation alarm switch setting (if applicable)

Trip alarm switch setting

All confirm ok.

Defeat/bypass switch for one transmitter must be in the DEFEAT/BYPASS position before test begins. Controller(s) using the signals from either transmitter should be in manual position. Make sure that Operations is set up to monitor the controlled variables while the controllers are in MANUAL mode.

Obtain necessary work permit.
Remove d/p cell junction box cover and check for contamination.
Check that terminal connections are tight.
Check calibration for both transmitters:

a. Close block valves for one transmitter.

b. Connect test gage and pressure regulator to high side of d/p cell. Hook up test milliamp meter to output.

c. Check zero by opening equalizing valve, record as found setting.

d. Close equalizing valve and open up d/p cell high side to regulator and test gage.

e. Apply full transmitter span and record output.

f. Re-calibrate if necessary and record as left setting.

Pre-alarm, trip, and deviation alarm check.

a. Apply pressure that is above the setpoint pressure to the high side of one d/p cell.

b. Gradually reduce pressure until pre-alarm and deviation alarm (if applicable) come on, record as found setting and alarm status.

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c. Gradually reduce pressure until trip switch operates, record as found setting and alarm status.

d. Re-calibrate switch if necessary and record as left setting.

Repeat both tests for other d/p cell.

Testing of high flow transmitters can be done by raising pressure above high alarm and trip values and verifying alarm and trip status.

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Annex M — Model procedure for on-line testing of pressure sensors in a 2oo3 configuration (high or low trip)

Note that this variable must be bypassed or defeated in the SIF logic before testing.

Check deviation alarm (if applicable). The pre-alarm and the trip alarm should not come on during this check.

a. Lower the pressure of the # 1 transmitter by blocking process and venting transmitter. Deviation alarm on (__).

b. Restore pressure, clear the alarm.

c. Lower the pressure of the # 2 transmitter. Deviation alarm on (__).

d. Restore pressure, clear the alarm.

e. Lower the pressure of the # 3 transmitter. Deviation alarm on (__)

f. Restore pressure, clear the alarm.

The following steps involve a check of the logic voting system.

a. All alarms should be clear. If not correct the problem before starting this test.

b. Gradually lower the input pressure of one transmitter until it is below the trip setpoint. Record alarm conditions below.

c. Gradually lower the pressure of another transmitter until it is below the pre-alarm setpoint. Record alarm conditions below.

d. Continue to lower the input until it is below the trip setpoint. Record alarm conditions below.

e. Restore input to one transmitter and record the reset conditions below.

f. Restore input to the other transmitter and record the reset conditions below.

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Step	Deviation alarm		Pre-alarm		Trip	
b.	On ( )	Off ( )	On ( )	Off ( )	On ( )	Off ( )
c.	On ( )	Off ( )	On ( )	Off ( )	On ( )	Off ( )
d.	On ( )	Off ( )	On ( )	Off ( )	On ( )	Off ( )
e.	On ( )	Off ( )	On ( )	Off ( )	On ( )	Off ( )
f.	On ( )	Off ( )	On ( )	Off ( )	On ( )	Off ( )

Repeat the above procedure for the other two combinations of transmitters. Record data for as found and as left values for deviation, pre-alarm, and trip setpoints for each transmitter.

Transmitter Number	Deviation alarm – as found	Deviation alarm – as left	Pre-alarm as found	Pre-alarm as left	Trip setpoint – as found	Trip setpoint – as left

This procedure can be used for high deviation, pre-alarm, and trip setpoints also.

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Annex N — Model procedure for testing temperature switches

Perform the following steps for verification of switch input processing validation and trip check.

1. Set the calibrated temperature bath to allow simulation of the input temperature over the calibrated range of the temperature switch.

2. Place temperature switch in temperature bath.

3. Increase the simulated temperature until a High temperature pre-alarm and trip occurs as indicated by the loop documentation (if applicable). Verify and document that pre-alarm and trip occur at correct set point.

4. Decrease the simulated temperature until the High temperature trip and pre-alarm clears as indicated by loop documentation (if applicable). Verify and document that trip and pre-alarm clear at correct set point. Also verify that the SIF does not automatically reset.

5. Decrease the simulated temperature until a Low temperature pre-alarm and trip occurs as indicated by loop documentation (if applicable). Verify and document that pre-alarm and trip occurs at correct set point.

NOTE Increase the simulated temperature until the Low temperature trip and pre-alarm clears as indicated by loop documentation (if applicable). Verify and document that pre-alarm and trip clear at correct set point. Also verify that the SIF does not automatically reset.

Procedure No.
Revision Date
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Annex O — Example visual inspection form for SIF

The SIF system should be visually inspected on some predetermined schedule to see if there are any problems that should be addressed before or during the functional testing. Since the SIF will not be in bypass during this inspection, do not open enclosures or devices in order to perform this inspection. This inspection is intended to be a visual inspection to determine how well the SIF devices have held up during a period of operation. Examples of items to check are…

Gauges	Instrument Air Supplies
Tubing	Conduit
Instrument Mountings	Hand Switches
Isolation Valves	Enclosure Purges
Instrument Covers	Paper Supply for printers
Alarm Panel Test Lights	Bug Screens
Heat tracing	

Items that need to be addressed should be listed at the bottom of this form and reported to the operations and maintenance. These items then should be addressed and corrected at the first opportunity allowed by the process operation.

The inspection should include, but not be limited to the following items.

- Verify that all components of the SIF are properly tagged and labeled.
- Visually inspect devices for excessive corrosion.
- Visually inspect all components to insure proper working condition.
- Visually inspect all SIF pressure and instrument gauges to insure proper working condition.
- Visually inspect tubing and wiring to insure proper working condition.
- Verify that all instrument air supply regulators are at their proper settings.
- Verify that all shutdown components are painted red.
- Verify that boxes and housings have proper seals and are secure.

Procedure No.
Revision Date
Page _ of _

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- Verify that tubing and cables are properly routed and secure.

Visual checks:

Tagging:

a) Are all instruments in this system tagged with a special tag identifying them as “SIF Instrument”?

Yes ( ) No ( )

b) Tagging condition: Good ( ) Bad ( )

Process connections:

Valves	NA	( )	Insulation	NA	( )
Ok	[ ]		Ok	[ ]	
Leaks	[ ]		Repairs	[ ]	
Corroded	[ ]		Missing	[ ]	
Comments	[_____]		Comments	[_____]	
Piping			Heat Tracing	NA	[ ]
Ok [ ]	Bad	[ ]	Ok [ ]	Bad	[ ]
Comments	[_____]		Comments	[_____]	

Conduit system: OK ( ) Bad ( ) If bad check below.

Covers off	[ ]	Drains missing	[ ]	Supports gone	[ ]
Seal needed	[ ]	Flex bad	[ ]	Conduit broken	[ ]
Fitting bad	[ ]	Corrosion	[ ]	Other	[ ]

Details [_____]

Correction made? Yes ( ) No ( )

Control valve:

General

Bug screens	ok	[ ]	clean	[ ]	missing [ ]
Tubing condition	ok	[ ]	corroded	[ ]	

Comments [_____]

Trip solenoids None installed [ ]

Procedure No.
Revision Date
Page _ of _

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Bug screens ok [ ] clean [ ] missing [ ]

Tubing condition ok [ ] corroded [ ]

Comments [_____]

Piping gasket leak [ ] Valve gasket leak [ ]

Packing gland leak [ ] Sticky stem action [ ]

Topworks problem [ ]

Details [_____]

Positioner problem [ ]

Details [_____]

Signal system problem [ ]

Details [_____]

Auxiliary device problem [ ]

Details [_____]

Once inspection is complete, sign and date below.

?

SIGNATURE **DATE**

Operator/Craftsman: _____ Date: _____

Items needing attention:

_____
_____
_____
_____
_____
_____
_____

Procedure No.
Revision Date
Page _ of _

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Annex P — Model procedure for testing a permissive pressure logic point

PERFORM THE FOLLOWING STEPS TO TEST PASS #1 & #2 PILOT GAS LOW PRESSURE SHUTDOWN.

NOTE When the shutdown reset is activated, a 15 minute timer is activated allowing time for the pilot pressure to increase above its trip point. However, if the pressure is satisfied prior to that 15 minutes and stays acceptable for at least 15 seconds, another timer will arm the shutdown and make it active.

Steps:

1) **DECREASE** pressure at PT9110 to 1.98 Psig. Verify PXL9110 Activated. RECORD TRIP VALUE _____ PSIG.

Initials _____ Date _____

2) **VERIFY** Pilot Gas solenoid XY9111 status XL9111 indicates Tripped (de-energized) and valve XV9111 closed and HMI indication ZLC9111 indicates a closed valve.

Initials _____ Date _____

3) **ACTIVATE** HS9617 Reset. Start StopWatch.

Initials _____ Date _____

4) **VERIFY** Pilot Gas solenoid status XL9111 is Normal (energized), reset solenoid XY9111 Verify XV9111 Opens and HMI open indication ZLC9111 indicates an open valve.

Initials _____ Date _____

5) **WAIT** 15-minutes then verify XL9111 valve status alarmed and Valve XV9111 closed. Record Elapsed Time: _____ minutes.

Initials _____ Date _____

Procedure No.
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6) **VERIFY** Pilot Gas valve Position alarm ZLC9111 is alarmed and indicates a closed valve.

Initials _____ Date _____

7) **ACTIVATE** HS9617 Reset. **Start StopWatch.**

Initials _____ Date _____

8) **VERIFY** Pilot Gas solenoid status XL9111 is Normal (energized), reset solenoid XY9111, verify XV9111 Opens.

Initials _____ Date _____

9) **VERIFY** Pilot Gas valve Position alarm XA9111 is normal and ZLC9111 indicates an open valve.

Initials _____ Date _____

10) **INCREASE** the Pressure to Pilot Gas pressure transmitter PT9110 to above the trip point ~ 5Psig. Verify Reading on PI9110.

Initials _____ Date _____

11) **VERIFY** Shutdown alarm PXL9110 CLEARS.

Initials _____ Date _____

12) **AFTER** a 15 second delay Decrease the Pilot Gas pressure to 1.0 Psig. and VERIFY XL9111 indicated Tripped (de-energized). Record Elapsed time _____Min.

Initials _____ Date _____

13) **VERIFY** Pilot Gas valve Position alarm XA9111 is alarmed and ZLC9111 indicates a closed valve.

Initials _____ Date _____

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14) **INCREASE** the Pressure to PT9110 to above it max range (~18psig) and verify Transmitter failure alarm PA9110 Alarmed.

Initials _____ Date _____

15) **DECREASE** the Pressure to PT9110 to below zero (~-1psig) and verify Transmitter failure alarm PA9110 Alarmed.

Initials _____ Date _____

16) **INCREASE** the Pressure to PT9110 to above its trip point (~5.0psig) and verify shutdown alarm PXL9110 Cleared.

Initials _____ Date _____

Procedure No.
Revision Date
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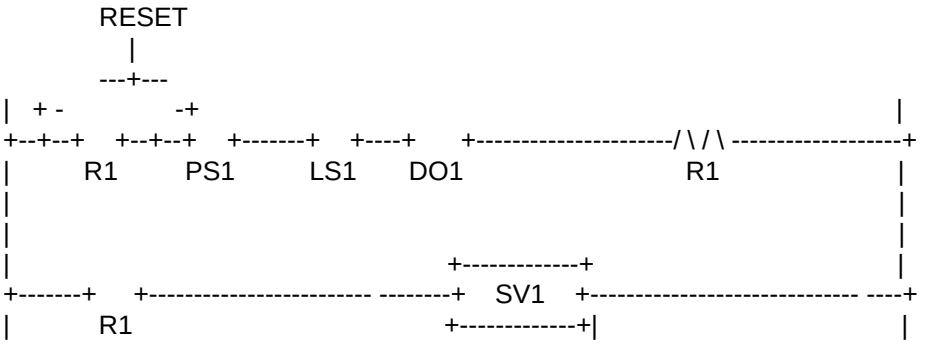
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NOTE This procedure was developed by an operating company for a specific application. It has been modified to remove any reference that might tie it to a specific company. This procedure should only be used as an example of how a user might develop a procedure specific to their SIS application. Any references to specific brands of instrumentation in the procedure are to clarify the intent of the procedure only and are in no respect meant to suggest these brands are acceptable or preferred for the user's specific application. The instrument identification numbers used in the procedures are for clarification purposes only and should in no way be taken as indicative of a particular company's instruments on a particular process.

CAUTION — PRIOR TO APPLYING THE INFORMATION IN THIS ANNEX TO DEVELOP A PROCESS SPECIFIC PROCEDURE, THE USER SHOULD REVIEW AND UNDERSTAND THE GUIDANCE IN THE BODY OF ISA-TR84.00.03-2002.

Annex Q — Model procedure for testing a simple SIF

This test procedure is for a process where high pressure could cause rupture of a vessel and release of a hazardous gas. The initiator is PT1. PS1 is the hardwired logic and the final control element is PV1. There is another PSM Critical interlock in this circuit for Low Level – LS1. The basic process control system also mirrors both interlocks by DO1. The simple circuit is shown in the following diagram.



Procedure No.
Revision Date
Page _ of _

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PSM critical interlock check method no. 1

Name of event:	Column High Pressure
Test objective:	When column pressure reaches 350 psig (increasing) interlock pressure automatic valve (PV1)
PSM critical device:	PT1 located on platform beside column at second level
Final control element:	Closes pressure automatic (PV1)
Test frequency:	12 months
Process trip setting:	350 psig + / - 20 psig
Type of test:	Simulate pressure on process side of transmitter to test loop
Test equipment required:	Hand pump with calibrated pressure gauge
Reference prints:	Instrument Dwg. Xxxxx Dwg. Yyyyy
	Electrical Dwg. Zzzzz Dwg. Qqqqq
Test to be conducted by:	Operations – qualified CCR and field operator
	E&I – qualified instrument technician
Pre-test conditions:	Process shutdown
	Column shutdown
	Steam off column

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Set-up requirements:

Operations: (Underlines next to each step are provided to assist you as check marks. They are not required to be used.)

CCR operator:

_____ Place the column pressure controller (PC1) on MANUAL and set valve position (PV1) to open.

Field operator:

_____ Verify the pressure valve (PV1) is open.

Instrument:

There is a PSM critical interlock (PS1) and a non-PSM critical interlock (DO1). We are testing the PSM critical interlock and therefore must bypass the non-PSM critical interlock. We must also bypass the Low Level PSM critical interlock.

_____ Bypass LS1

_____ Bypass DO1

Procedure No.
Revision Date
Page _ of _

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Procedure:

Instrument:

_____ 1. Connect a hand pump and calibrated gauge to the input of PT1. Apply 300 psig load to PT1.

_____ 2. Slowly increase the simulated pressure until the interlock occurs at 350 psig.

_____ 3. Document the observed trip point. Psig _____.

_____ 4. Inspect to assure the interlock system is in good condition. Inspect conduits, piping, identification tags, etc.

CCR operator:

_____ 1. Verify that the column high pressure interlock alarm and light activated (PA1).

_____ 2. Verify the pressure controller valve loading (PV1) is still indicating open.

Field operator:

_____ 1. Verify the pressure valve closed (PV1) when interlock activation occurred.

Post test inspection and documentation

CCR operator:

_____ 1. The initial interlock test passed/failed

Instrument:

_____ 1. The interlock equipment has been returned to normal and is ready for service.

_____ 2. If the initial interlock test failed, what corrective action was required?

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Annex R — Model procedure for testing a complex logic system

R.1 Preflash evaporator injection

R.1.1 Pre-test signature requirements

I have read and understand the scope and content of this test, and verify that it is safe to perform the test as described below.

_____

Operator (Signature) Date

I have reviewed this test document, met the prerequisites as detailed in plant policies, briefed all appropriate personnel, received a written work permit, and am ready to begin the test.

_____

Technician performing the test (Signature) Date

R.1.2 Test equipment requirements

- ? Two (2) Thermocouple Temperature Simulators (Type J)
- ? Or,
- ? Three (3) Thermocouple Temperature Simulators (Type J), if available.
- ? Bypass Enable Keyswitch Key for Pre-Flash Evaporator Injection (Located in Bypass
- ? Enable Keyswitch HS-2308).
- ? Two (2) Radios

  NOTE Do not operate radios in the computer room.

- ? NOTES:

⇒ All test equipment must be calibrated within one year of this test and have the proper certification from the on-site metrology laboratory.

⇒ Prior to its use, all test equipment must be compared to another identical instrument to ensure the test equipment is serviceable and ready for use.

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R.1.3 General

Reference: SIF Drawing(s) specific to this system

R.1.4 Valve line-up activities

Before beginning any portion of this test, the Technician Performing the Test shall have an Operator close the downstream manual injection system valve associated with this system. Since the downstream manual injection block valve is Car Sealed, the Operator must first remove and dispose of the Car Seal before closing this valve. Closing of the manual block valve shall be performed in accordance with all existing site procedures.

Upon completion of this test, the Technician Performing the Test shall inform the Operator the downstream manual block valve may be opened. Opening of the manual block valve shall be performed in accordance with all existing site procedures. The Operator must install and lock a new Car Seal on the manual block valve and record the Car Seal Number in the space provided at the end of this test.

NOTE See the Testing Tables for detailed instructions and sign-off for the valve line-up activities.

R.1.5 Inspection

Before beginning any portion of this test, the Technician Performing the Test shall ensure that the system is in a normal Off-line condition and NOT tripped. If the system is tripped, the Technician Performing The Test shall **STOP**, and perform the following:

- Contact Operations to confirm that the system is in a normal Off-line condition.
- Request that Operations Reset the system.
- Confirm that all conditions have returned to normal, the system is in a normal Off-line condition, and the system is NOT tripped.
- Confirm downstream manual block valves have been placed into the CLOSED position.

  Initial _____

R.1.6 Thermocouple input, trip, and bypass action

This section tests thermocouple input processing, thermocouple trip action, and thermocouple bypass action. This section requires that Thermocouple Temperature Simulators be connected to the thermocouple leads prior to beginning the test. At the conclusion of this section, all Thermocouple Simulators may be disconnected.

The Thermocouple Input Trip and Manual Reset system indicators are verified, and the Final Control Devices are tested. Since this system is de-energize to trip, the Final Control Devices will be checked to ensure they are de-energized and fail to the safe position during a trip, and are energized and return to the normal position after a Manual Reset.

A hardwired Bypass Enable keyswitch, located on the front door of the Triconex cabinet (the Triconex cabinet is located in the Computer Room), must be placed into the Bypass Enable position before inputs can be bypassed. Once enabled, the BPCS Bypass Set and Bypass Reset soft switches are used to bypass points for maintenance. The BPCS Bypass Set switch sets the triad, pair, or individual input into bypass (i.e. TE-2307X, TE-2307Y, and TE-2307Z are placed into bypass by BPCS switch HS-2307S). Individual thermocouples are not typically bypassed (i.e. the Operator is prevented from bypassing ONLY TE-2307Z).

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Table R-1-6A should be used to validate the Thermocouple Input, Trip, and Bypass Action. All BPCS points for this system can be found on BPCS schematic “PREFLASH."

Table R.1.6A — Thermocouple input, trip, and bypass action validation

Testing comment: The following section prepares the system for testing.

Step	Step Instructions	Expected Result(s)	Check (Initials)
1.0	Ensure system is NOT tripped.	Verify that “System Trip” lamp on switch HS-2306 is NOT lit. Verify that BPCS tag HXB-2306C is NOT in alarm.	
2.0	Remove the Car Seal from the DOWNSTREAM injection system manual block valve and dispose of the Car Seal. Close the DOWNSTREAM injection system manual block valve. Verify the UPSTREAM injection system manual block valve is Car Sealed. NOTE If the UPSTREAM injection system manual block valve is NOT Car Sealed, request the Operator install and lock a new Car Seal on this valve.	Request the Operator remove the Car Seal and close the DOWNSTREAM injection system manual block valve. Verify that Operations has performed this step. Record the Car Seal of the UPSTREAM injection system manual block valve below: UPSTREAM Car Seal Number: _____	
3.0	Verify that BPCS setpoint indicator is correct.	Verify that BPCS setpoint indicator TSP-2307 reads: 245.0 deg. F.	
4.0	Momentarily disconnect Thermocouple TE-2307X.	Verify that BPCS tag TXA-2307C, Thermocouple Burnout, is in alarm.	
5.0	Connect a Thermocouple Temperature Simulator to TE-2307X.	Verify that temperature readings are received on BPCS indicator TI-2307X.	

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Testing comment: The following section tests the X and Y thermocouples.

T/C X is driven high, then T/C Y is driven high.

6.0	Drive TE-2307X above the high trip setpoint: 245.0 deg. F.	N/A	
7.0	Momentarily disconnect Thermocouple TE-2307Y.	Verify that BPCS tag TXA-2307C, Thermocouple Burnout, is in alarm.	
8.0	Connect a Thermocouple Temperature Simulator to TE-2307Y.	Verify that temperature readings are received on BPCS indicator TI-2307Y.	
9.0	Drive TE-2307Y above the high trip setpoint: 245.0 deg. F.	Verify that “System Trip” lamp on switch HS-2306 is lit. Verify BPCS tag HXB-2306C is in alarm. Verify BPCS tag TAX-2307C, High Temperature Trip, is in alarm. Verify annunciator TAX-2307A is in alarm. Verify that solenoid valves are de-energized and valves are OPEN. XY-2307A, XV-2307A XY-2307B, XV-2307B XY-2307C, XV-2307C XY-2307D, XV-2307D Note actual temperature on simulator where trip occurred and document on the appropriate SIS Field Function Test Findings Form. Record all findings on the appropriate SIS Field Function Test Findings Form.	
10.0	Drive TE-2307X below the high trip setpoint: 245.0 deg. F.	Verify "OK to Reset" lamp on switch HS-2306 is lit and BPCS tag HXA-2306C is in alarm.	
11.0	Drive TE-2307Y below the high trip setpoint: 245.0 deg. F.	N/A	

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Testing comment: The following section tests the X and Y thermocouples.

T/C X is driven high, then T/C Y is driven high (Cont.).

12.0	Reset the system by positioning switch HS-2306 to the System Reset position. Return switch HS-2306 to the Normal position.	Verify that “System Trip” lamp on switch HS-2306 is NOT lit. Verify BPCS tag HXB-2306C is NOT in alarm. Verify BPCS tag TAX-2307C, High Temperature Trip, is NOT in alarm. Verify annunciator TAX-2307A is NOT in alarm. Verify that solenoid valves are energized and valves are CLOSED. XY-2307A, XV-2307A XY-2307B, XV-2307B XY-2307C, XV-2307C XY-2307D, XV-2307D Record all findings on the appropriate SIS Field Function Test Findings Form.	

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Testing comment: The following section tests the X and Y thermocouples.

The Bypass for T/C X and T/C Y is tested.

13.0	Confirm that Bypass Enable Keyswitch HS-2308 is NOT in the Bypass position (the Bypass Enable keyswitch is located on the front of the Triconex cabinet). Confirm that inputs can NOT be placed into bypass by selecting BPCS switch THS-2307S, Bypass Set.	Verify that BPCS tag TAB-2307C is NOT in alarm. Verify that annunciator HA-2308A is NOT in alarm.	
14.0	Place Bypass Enable key HS-2308 in the Bypass position (NOTE The Bypass Enable Keyswitch is located on the front of the Triconex cabinet).	Verify that “Bypass Enabled” lamp on switch HS-2306 is lit. Verify BPCS tag HXC-2308C is in alarm.	
15.0	Select BPCS switch THS-2307S, Bypass Set.	Verify that BPCS tag TAB-2307C is in alarm. Verify that annunciator HA-2308A is in alarm.	
16.0	Drive TE-2307X above the high trip setpoint: 245.0 deg. F.	N/A	
17.0	Drive TE-2307Y above the high trip setpoint: 245.0 deg. F.	Verify that “System Trip” lamp on switch HS-2306 is NOT lit. Verify BPCS tag HXB-2306C is NOT in alarm.	
18.0	Drive TE-2307Y below the high trip setpoint: 245.0 deg. F.	N/A	
19.0	Select BPCS switch THS-2307R, Bypass Reset.	Verify that BPCS tag TAB-2307C is NOT in alarm. Verify that annunciator HA-2308A is NOT in alarm.	
20.0	Disconnect Thermocouple Temperature Simulator from TE-2307Y. Restore Thermocouple TE-2307Y to its normal configuration.	N/A	

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Testing comment: The following section tests the X and Z thermocouples.

T/C X is high, then T/C Z is driven high.

21.0	Momentarily disconnect Thermocouple TE-2307Z.	Verify that BPCS tag TXA-2307C, Thermocouple Burnout, is in alarm.	
22.0	Connect a Thermocouple Temperature Simulator to TE-2307Z.	Verify that temperature readings are received on BPCS indicator TI-2307Z.	
23.0	Drive TE-2307Z above the high trip setpoint: 245.0 deg. F.	Verify that “System Trip” lamp on switch HS-2306 is lit. Verify BPCS tag HXB-2306C is in alarm. Verify BPCS tag TAX-2307C, High Temperature Trip, is in alarm. Verify annunciator TAX-2307A is in alarm. Note actual temperature on simulator where trip occurred and document on the appropriate SIS Field Function Test Findings Form.	
24.0	Drive TE-2307X below the high trip setpoint: 245.0 deg. F.	Verify "OK to Reset" lamp on switch HS-2306 is lit. Verify BPCS tag HXA-2306C is in alarm.	
25.0	Drive TE-2307Z below the high trip setpoint: 245.0 deg. F.	N/A	
26.0	Reset the system by positioning switch HS-2306 to the System Reset position. Return switch HS-2306 to the Normal position.	Verify that “System Trip” lamp on switch HS-2306 is NOT lit. Verify BPCS tag HXB-2306C is NOT in alarm. Verify BPCS tag TAX-2307C, High Temperature Trip, is NOT in alarm. Verify annunciator TAX-2307A is NOT in alarm.	

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Testing comment: The following section tests the X and Z thermocouples.

The Bypass for T/C X and T/C Z is tested.

27.0	Select BPCS switch THS-2307S, Bypass Set.	Verify that BPCS tag TAB-2307C is in alarm. Verify that annunciator HA-2308A is in alarm.	
28.0	Drive TE-2307X above the high trip setpoint: 245.0 deg. F.	N/A	
29.0	Drive TE-2307Z above the high trip setpoint: 245.0 deg. F.	Verify that “System Trip” lamp on switch HS-2306 is NOT lit. Verify BPCS tag HXB-2306C is NOT in alarm.	
30.0	Drive TE-2307X below the high trip setpoint: 245.0 deg. F.	N/A	
31.0	Select BPCS switch THS-2307R, Bypass Reset.	Verify that BPCS tag TAB-2307C is NOT in alarm. Verify that annunciator HA-2308A is NOT in alarm.	

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Testing comment: The following section tests the Y and Z thermocouples.

T/C Z is high, then T/C Y is driven high.

32.0	Disconnect Thermocouple Temperature Simulator from TE-2307X. Restore Thermocouple TE-2307X to its normal configuration.	N/A	
33.0	Momentarily disconnect Thermocouple TE-2307Y.	N/A	
34.0	Connect a Thermocouple Temperature Simulator to TE-2307Y.	Verify that temperature readings are received on BPCS indicator TI-2307Y.	
35.0	Drive TE-2307Y above the high trip setpoint: 245.0 deg. F.	Verify that “System Trip” lamp on switch HS-2306 is lit. Verify BPCS tag HXB-2306C is in alarm. Verify BPCS tag TAX-2307C, High Temperature Trip, is in alarm. Verify annunciator TAX-2307A is in alarm. Note actual temperature on simulator where trip occurred and document on the appropriate SIS Field Function Test Findings Form.	
36.0	Drive TE-2307Z below the high trip setpoint: 245.0 deg. F.	Verify "OK to Reset" lamp on switch HS-2306 is lit. Verify BPCS tag HXA-2306C is in alarm.	
37.0	Drive TE-2307Y below the high trip setpoint: 245.0 deg. F.	N/A	
38.0	Reset the system by positioning switch HS-2306 to the System Reset position. Return switch HS-2306 to the Normal position.	Verify that “System Trip” lamp on switch HS-2306 is NOT lit. Verify BPCS tag HXB-2306C is NOT in alarm. Verify BPCS tag TAX-2307C, High Temperature Trip, is NOT in alarm. Verify annunciator TAX-2307A is NOT in alarm.	

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Testing comment: The following section tests the Y and Z thermocouples.

The Bypass for T/C Y and T/C Z is tested.

39.0	Select BPCS switch THS-2307S, Bypass Set.	Verify that BPCS tag TAB-2307C is in alarm. Verify that annunciator HA-2308A is in alarm.	
40.0	Drive TE-2307Y above the high trip setpoint: 245.0 deg. F.	N/A	
41.0	Drive TE-2307Z above the high trip setpoint: 245.0 deg. F.	Verify that “System Trip” lamp on switch HS-2306 is NOT lit. Verify BPCS tag HXB-2306C is NOT in alarm.	
42.0	Drive TE-2307Y below the high trip setpoint: 245.0 deg. F.	N/A	
43.0	Drive TE-2307Z below the high trip setpoint: 245.0 deg. F.	N/A	
44.0	Select BPCS switch THS-2307R, Bypass Reset.	Verify that BPCS tag TAB-2307C is NOT in alarm. Verify that annunciator HA-2308A is NOT in alarm.	

Testing comment: The following section restores the system.

45.0	Place Bypass Enable key HS-2308 located in Bypass Enable Keyswitch HS-2308, in the Normal position (NOTE the Bypass Enable Keyswitch is located on the front of the Triconex cabinet).	Verify that “Bypass Enabled” lamp on switch HS-2306 is NOT lit. Verify BPCS tag HXC-2308C is NOT in alarm.	
46.0	Disconnect Thermocouple Temperature Simulators from TE-2307Y and TE-2307Z.	N/A	
47.0	Restore Thermocouples TE-2307Y and TE-2307Z to their normal configuration.	N/A	
48.0	Ensure the system has been returned to normal.	Verify all switch lamps for HS-2306 are NOT lit.	
49.0	Record all findings on the appropriate SIS Field Function Test Findings Form.	N/A	

R.1.7 Manual trip/Reset logic function validation

Manual Trip and Reset logic function validation is conducted by positioning the switch into the System Trip and Reset Positions. The Manual Trip and Reset system indicators are verified, and the Final Control Devices are tested. Since this system is de-energize to trip, the Final Control Devices will be

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checked to ensure they are de-energized and fail to the safe position during a trip, and are energized and return to the normal position after a Manual Reset.

Table R-1-7A should be used to validate the Manual Trip and Reset function. All BPCS points for this system can be found on BPCS schematic “PREFLASH."

Procedure No.
Revision Date
Page _ of _

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Table R.1.7A — Manual trip and reset logic functionality validation

Step	Step Instructions	Expected Result(s)	Check (Initial)
50.0	Initiate a Manual Trip by positioning switch HS-2306 to the System Trip position. Return switch HS-2306 to the Normal position. Request operations remove the bleeder cap between the four valves XV-2307A/B/C/D.	Verify that “System Trip” lamp on switch HS-2306 is lit. Verify BPCS tag HXB-2306C is in alarm. Verify the restriction orifice located by valves XV-2307A,B,C,&D, is leaking to ground. Verify that solenoid valves are de-energized and valves are OPEN. XY-2307A, XV-2307A XY-2307B, XV-2307B XY-2307C, XV-2307C XY-2307D, XV-2307D Record all findings on the appropriate SIS Field Function Test Findings Form.	
51.0	Initiate a Manual Reset by positioning switch HS-2306 to the System Reset position. Return switch HS-2306 to the Normal position.	Verify that “System Trip” lamp on switch HS-2306 is NOT lit. Verify BPCS tag HXB-2306C is NOT in alarm. Verify the restriction orifice located by valves XV-2307A,B,C,&D, is NOT leaking to ground. Verify that solenoid valves are energized and valves are CLOSED. XY-2307A, XV-2307A XY-2307B, XV-2307B XY-2307C, XV-2307C XY-2307D, XV-2307D Record all findings on the appropriate SIS Field Function Test Findings Form.	

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Testing comment: Restore the system to normal.

52.0	Ensure the system has been returned to normal. Request operations re-install the bleeder cap between the four valves XV-2307A/B/C/D.	Verify all switch lamps for HS-2306 are NOT lit.	
53.0	Record all findings on the appropriate SIS Field Function Test Findings Form.	N/A	
54.0	Open the DOWNSTREAM injection system manual block valve. Install and lock a new Car Seal on the DOWNSTREAM injection manual block valve.	Request the Operator open the DOWNSTREAM injection manual block valve and install and lock a new Car Seal onto the valve. Verify that Operations has performed this step. Record the new Car Seal on the DOWNSTREAM injection system manual block valve below: DOWNSTREAM Car Seal Number: _____	

R.1.8 Test completed: Time:_____ Date:_____

R.1.9 Signature identification log

Print Name	Signature

Procedure No.
Revision Date
Page _ of _

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R.1.10 Post test activities

R.1.10.1 Post test sign-offs

Test Equipment	Model No.	Equip. No.	Date

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R.1.10.2 Failure log

Step	Device	Failure Description*	Failure Corrected	Initials

* Attach additional sheets if necessary

R.1.11 Post-test signature requirements

I have verified that the system was returned to its normal operational condition and is ready for startup.

_____

Operator (Signature) Date

This completed test has been reviewed and all pertinent data has been captured for historical reference.

_____

Technician Performing the Test (Signature) Date

Procedure No.
Revision Date
Page _ of _

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Annex S — Model procedure for testing emergency stop switch

Procedure:

_____1. Verify that all interlocks are satisfied for operating condition. This may require forcing any startup permissive interlocks with either a current source or a HART communicator.

_____2. Notify the control room operator that a test of the emergency stop switch is going to take place.

_____3. When the control room operator is ready to begin the test, I/E technician will monitor the emergency stop relay in the interlock cabinet.

_____4. Have the control room operator change the emergency stop switch position to stop. Verify that the relay de-energizes when the switch changes position.

_____5. Verify that the alarms for process shutdown are actuated.

_____6. Verify that all valves go to the correct position (field operator).

_____7. Verify that HMI display indicates correct position for all valves.

_____8. Return the emergency stop switch to normal position.

_____9. Did the emergency stop switch shutdown the process correctly? Yes / No (circle one)

____10. If test of emergency stop switch was not successful, what was required to correct the situation?

_____
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Test performed by: _____ Date _____

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Procedure No.
Revision Date
Page _ of _

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Annex T — Model procedure for testing a relay implemented SIF

Vessel exit temperature interlock tests (Loop No. TS-1, TS-2, TS-3)

Responsibility: I = Instrument O = Operations E = Electrical

I/E:

____1. Bypass all necessary interlocks to reset Feed and Dump interlocks.

In relay cabinet A in building 100: Install jumpers between following terminals:

terminal P21	terminal 8 on relay AR11
terminal 3 on relay AR13	terminal 8 on relay AR9
jumper terminal 9 on relay AR9	terminal 6 on relay AR13
terminal 5 on relay AR5	terminal 4 on relay AR11
terminal 9 on relay AR2	terminal 8 on relay AR5
terminal 9 on relay AR5	terminal 6 on relay AR2
terminal 9 on relay AR7	terminal 2 on relay AR5
terminal P62	terminal 10 on relay AR5
terminal 9 on relay AR15	terminal 10 on relay AR11
terminal 11 on relay AR11	terminal 6 on relay AR15
terminal 5 on relay AR15	terminal 5 on AR10
terminal 9 on relay AR 16	terminal 8 on relay AR17
terminal 9 on relay 17	terminal 6 on relay AR24
terminal 11 on relay AR17	terminal 9 on relay AR29
terminal 5 on relay AR12	terminal 6 on relay AR29
terminal 8 on relay AR30	terminal 4 on relay AR31
terminal 5 on relay AR31	terminal 4 on relay AR17
terminal 9 on relay AR33	terminal 3 on relay AR1
terminal 4 on relay AR27	terminal 6 on relay AR34
terminal 9 on relay AR34	terminal 6 on relay AR35
terminal 8 on relay AR6	terminal 8 on Relay AR35
terminal 9 on relay AR35	terminal 6 on relay AR36

Procedure No.
Revision Date
Page _ of _

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terminal 9 on relay AR36	terminal 10 on relay AR10
terminal 11 on relay AR10	terminal 13 on relay AR10
terminal P41	terminal 6 on relay AR23
terminal P42	terminal 13 on relay AR25
terminal 14 on relay AR25	terminal 6 on relay AR25
terminal 9 on relay AR13	terminal 9 on relay AR8
terminal 3 on relay AR5	terminal 11 on relay AR1
terminal P33	terminal 5 on relay BR9

- Block AR20 Low Feed flow
- Block AR10 Dump System
- Block AR40

- Install jumper in section 4 of Bldg 100 480v switchgear from terminal UA-5 to terminal UE-11.
- Install a jumper in section 4 of Bldg 100 480v switchgear from terminal UA-5 to terminal UE-12.
- Rack Circulating Pump Breaker into the test position. (This will remove power from the motor.)
- Assure that sparge water HS-4544 is in the run position (no water flow).
- Install a jumper in relay cabinet A from terminal 5 on relay AR17 to terminal 6 on relay AR33.

E/I: 2. Take the necessary action to satisfy the following interlocks by establishing process conditions or driving the transmitters with test equipment.

LX-4711 Feed Off-Gas Separator Hi Hi Level

PX-4549 Low low Process Air Pressure

E/I: 3. Disconnect TE-4513 at the tag head and connect a thermocouple simulating device to the tag head and load to clear the interlock.

O: 4. Activate Dump System reset switch HS-4540

Place HS-2361 in normal position.

Activate HS-4593, HS-4594, HS-4541, HS-4571, and HS-4542 resets.

Push start button on circulating pump and observe run condition on BPCS.

The proper valves should now be reset.

O: 5. Verify the proper interlocks, audible alarms, or visual indications are not activated.

a. Verify the following valves are in proper run position.

HV-4508-1 Water valve #1 closed

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HV-4508-2 Water valve #2 closed

HV-4508-3 Water bleed valve open

HV-4503-1 Feed valve open

HV-4503-2 Feed valve open

HV-4503-3 Feed bleed valve closed

E/I: 6. Slowly lower the signal on TE-4513 until the low interlock occurs. Verify the interlock actuates at correct setting.

O: 7. Verify the Feed interlocks, audible alarms, and visual indications have occurred:

a. HV-4508-1 Water valve #1 open

HV-4508-2 Water valve #2 open

HV-4508-3 Water bleed valve closed

HV-4503-1 Feed valve closed

HV-4503-2 Feed valve closed

HV-4503-3 Feed bleed valve open

E: 8. Increase the signal on TE-4513 to clear interlock.

O: 9. Activate Feed reset switch HS-4542

The unit Feed valves should now be reset.

10. Verify that the Feed interlocks, audible alarms, or visual indications are not activated.

a. Verify the following valves are in proper run position.

HV-4508-1 Water valve #1 closed

HV-4508-2 Water valve #2 closed

HV-4508-3 Water bleed valve open

HV-4503-1 Feed valve open

HV-4503-2 Feed valve open

HV-4503-3 Feed bleed valve closed

E/I: 11. Slowly raise the TE-413 signal until the interlock occurs. Verify that the interlock occurs at the proper setpoint.

O: 12. Verify the Feed interlocks, audible alarms, and visual indications have occurred:

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a. HV-4508-1 Water valve #1 open

HV-4508-2 Water valve #2 open

HV-4508-3 Water bleed valve closed

HV-4503-1 Feed valve closed

HV-4503-2 Feed valve closed

HV-4503-3 Feed bleed valve open

E/I: 13. Move the jumper that goes from terminal 11 of AR10 to terminal 13 of AR10. Place it on terminal 11 of AR10 to terminal 6 of AR37. This will bypass TS2 interlock of TE-4513.

E: 14. Install a jumper from terminal P1 to terminal 10 of AR3.

- Block BR14
- Connect a voltmeter to terminal 6 on relay AR40. Verify the presence of voltage to this point.

O: 15. Activate the Feed reset switch HS-4542.

The unit valves should now be reset.

O: 16. Verify that the interlocks, audible alarms, or visual indications are not activated.

a. Verify the following valves are in proper run position.

HV-4508-1 Water valve #1 closed

HV-4508-2 Water valve #2 closed

HV-4508-3 Water bleed valve open

HV-4503-1 Feed valve open

HV-4503-2 Feed valve open

HV-4503-3 Feed bleed valve closed

b. Verify the presence of power on terminal 6 of AR10.

E/I: 17. Slowly raise the signal on TE-4513 until the interlock occurs. Verify that the interlock occurs at proper setpoint.

O: 18. Verify the interlocks, audible alarms, and visual indications have occurred.

a. HV-4508-1 Water valve #1 open

HV-4508-2 Water valve #2 open

HV-4508-3 Water bleed valve closed

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HV-4503-1 Feed valve closed

HV-4503-2 Feed valve closed

HV-4503-3 Feed bleed valve open

b. verify the loss of voltage on terminal 6 on relay AR40.

E/I: 19. To verify redundant relays on interlock, move the following jumpers:

- Move the jumper that goes from terminal 11 on relay AR17 to terminal 9 on relay AR24. Place it on terminal 10 on relay AR17 to terminal 8 on relay AR24.
- Move the jumper that goes from terminal 5 on relay AR31 to terminal 4 on relay AR17. Place it on terminal 5 on relay AR31 to terminal 6 on relay AR36.
- Remove the jumper that goes from terminal 9 on relay AR35 to terminal 6 on relay AR36.

E/I: 20. Repeat steps 2-4.

E/O: 21. Verify that the proper interlocks, audible alarms, and visual indications are not activated.

- Using terminal 6 on relay AR22 as a common point, verify the presence of voltage to neutral indicating Feed interlock is reset.
- Using terminal 6 on relay AR40 as a common point, verify the presence of voltage to neutral indicating LV-4586 and FV-2141 is reset.

E/I: 22. Slowly raise the TE-4513 signal until the interlock occurs. Verify the interlock occurs at the correct setpoint.

E/O: 23. Verify that the proper interlocks, audible alarms, and visual indications are activated.

- Using terminal 6 on relay AR22 as a common point, verify the presence of no voltage to neutral indicating Feed interlock.
- Using terminal 6 on relay AR$0 as a common point, verify the presence of no voltage indicating LV-4586 and FV-2141 interlock.

E/I: 24. To verify redundant feed interlock by the redundant dump relay block relay AR11 and unblock relay AR10.

E/I: 25. Repeat steps 20, 21, 22, and 23.

E/I: 26. To verify redundant preheater interlock by the redundant dump relay:

- Move the jumper from terminal 5 on relay AR15 to terminal 5 on relay AR10. Place it on terminal 5 on relay AR15 to terminal 4 on relay AR10.

27. Move jumper from terminal 5 on AR5 to terminal 4 on AR11. Place it from terminal 5 on AR5 to terminal 8 on AR4.

E/I: 28. Repeat steps 2, 3, 4, 15, 16, 17, and 18.

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E: 29. Remove all jumpers and return loops to their normal mode of operation.

Reference Drawings:

Schematics, ladder logic and wiring diagrams.

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Annex U — Model procedure for testing SIF watchdog timer

Description: Because the interlocks implemented in the SIF require a high level of integrity, a watch dog timer system has been implemented. This system will provide an external check of the operating condition of the SIF processor and its associated I/O cards. This is accomplished by utilizing a relay and an associated circuit, which must be periodically pulsed in order to stay energized. This pulsing signal is generated within the SIF configuration and is output to the WDT. If the external WDT detects a loss of pulsing signal, the WDT relay will de-energize. This will activate an alarm as well as certain interlocks. All hard-wired interlocks will be dropped out.

All three of the outputs are paralleled as inputs to the watchdog timer.

Output #2 is programmed with input #2. This input has only one field connection, which is the neutral side of the input. The intent of the input is to detect an input card failure. If this occurs, the input goes high which causes the output to go high. This prevents the external watchdog timer from pulsing and eventually causes it to trip.

Output #1 is unconnected in the BPCS logic. This point is to detect an output card failure, which will cause the point to go high and trip the timer.

Output #3 is programmed to pulse (square wave) the external watchdog timer. Timing between the pulse and the watchdog is critical to the watchdog relay staying energized. At least two pulses per timer interval are needed to keep the timer energized.

Procedure:

_____1. Put the interlock bypass switch in the SIF program to the bypass position.

_____2. Verify the interlock bypass alarm energizes on the BPCS.

_____3. Verify the process being protected by the SIF is running and the following safety interlock relays are energized: 5860-R, 1454-R, 5808-R, and 3105-R.

_____4. Hold in the SIF WDT test button in the SIF cabinet and using a stopwatch, measure the time required for the SIS WDT relay to de-energize.

_____5. Document the time required for the WDT circuit to the interlocks: _____ seconds (set point = 2 seconds, tolerance = ± 1.5 seconds).

_____6. Verify the WDT alarm sounds from the BPCS.

_____7. Verify the WDT safety relay, 5860-R, de-energized.

Procedure No.
Revision Date
Page _ of _

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Annex V-1 — Model procedure for on-line testing of sensor logic

Safety Instrumented System on-line testing procedure

SECTION 1 - GENERAL INFORMATION

Recommended Personnel required to accomplish this Trip System Test is **2 Technicians and 1 Operator.** Each step shall be completed and initialed by the Instrument Craftsman. An **Operations** representative shall track the actions of the procedure, participate in the procedure as described and manage the Bypass Switches, Keys and Bypass Log Book.

____ 1. Test Equipment List

(1) Fluke Multimeter

(2) Precision DC Milliamp/Voltage source

(1) Thermocouple Simulator

(1) Honeywell Smart Field Communicator

(1) Pneumatic hand pump with 0-15 psig test gauge

(1) Wallace & Tiernan Calibrator

(1) 24VDC Power Supply

____ 2. Obtain a Current version of the "SIS description" and "SIS Calibration Sheets" before continuing.

SECTION 2 - GENERAL SYSTEM CHECKOUT

____ 1. Lamp test all ICS matrix LED's on ICS Panel by pushing the Lamp test pushbutton in the lower right hand corner of the matrix. Replace all malfunctioning LED's.

SECTION 3 - TRIP SYSTEM CHECKOUT (TRIP ALARMS)

NOTE TDC controllers and alarms are located on TDC Hi-way's 1 and 2. Sequence of Events (SOE) Recorder points are located on the LCN Universal Station Console located in the Computer Room.

____ 1. At the ICS, panel matrix, place Output Bypass switch **HS-1253** in **"BYPASS."** Verify illumination of the amber LED's at the bypass key switches. Also verify **"I-1 System**

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Bypassed" lights at Shutdown Switches HS-1252 and HS-1291 are illuminated at the TDC console.

NOTE The Output bypass switch is used to allow testing of the trip alarms since the Input Bypass switch is before the Trip Alarm.

____ 2. Verify the Trip transmitter (TT-1244) matches the Pre-Alarm transmitter (TT-1245) at TDC point T1244DCC. **Operations Note:** Monitor TDC point T1245.CC. Manually Trip the East Riser Diversion at shutdown switch HS-1252 located at the TDC console **if:** the temperature (T1245.CC) drops below TSLL-1244 trip point **or** Control Room Annunciator Shutdown alarm "XA-1345A Riser #1 Catalyst Slide Valve" trips. Monitoring the alarm is necessary since the Output Bypass Switch is in Bypass which disables East Riser Diversion.

____ 3. Connect the necessary test equipment to simulate the process at the transmitter below. Calibrate transmitter, remove equipment, return to service, and fill out calibration sheet. Refer to the Calibration Sheets and using a Honeywell Smart communicator verify the transmitter **Fail Modes** are correct. Verify the Smart Communicator indicates the ID properly. Disconnect the Smart Communicator upon completion of the above verification.

____ A. **TT-1244**

____ 4. Follow this step to verify the alarms and TDC indication for **TT-1244.**

____ A. Connect voltage simulator to input jacks of TT-1244 trip card. Verify TDC indication for Transmitter **TT-1244 (Group 504).** Simulate the process to 0, 50, & 100% of calibrated range. Verify the TDC Displays within 2% and verify the units are correct. Fill out calibration sheet for TY-1244.

____ B. Test the Trip System/Process Control Transmitter high deviation alarm for TT-1244 & 1245.

____ 1. Set **TT-1244** equal to the process Temperature indicated **TT-1245.** Verify TDC alarm T1244DCC is not in alarm.

____ 2. Decrease **TT-1244** temperature and verify TDC alarm T1244DCC alarms as the temperature reaches 10% below **TT-1245.** Set **TT-1244** equal to the process temperature indicated by **TT-1245.** Verify TDC alarm T1244DCC clears.

____ 3. Increase **TT-1244** temperature and verify TDC alarm T1244DCC alarms as the temperature reaches 10% above **TT-1245.** Set **TT-1244** equal to the process Temp indicated by **TT-1245.** Verify TDC alarm T1244DCC clears.

____ 4. Verify alarms listed below in step "C" are clear.

____ C. Observing TT-1244 Trip Card LED, verify TSLL-1244 LED illuminates Red at the Calibration Sheet specified (V) setting. Verify the input LED on ICS panel extinguishes at **TSLL-1244** trip point. Verify the alarms listed below trip 2 minutes after **TT-1244** input LED extinguishes. Complete TSLL-1244 calibration sheet.

____ 1. Hi-way 1 TDC Trip Alarm **"T1244ZCC."**

____ 2. Control Room Annunciator Trip Alarm **"TALL-1244A"**

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____ 3. Sequence of Events Recorder Alarm "T1244ZCC"

____ D. Disconnect all test equipment from TY-1244.

____ E. Verify that **TSLL-1244** is in a non-trip condition (ICS panel matrix green input LED for **TSLL-1244** is illuminated). Verify the Trip transmitter (TT-1244) matches the Pre-Alarm transmitter (TT-1245) at TDC point T1244DCC.

____ F. Return Output Bypass switch **HS-1253** to **"Normal."**

____ 5. At the TDC console, place controller **TRC-1245** in **"Manual." Operations Note: Monitor the Trip Transmitter at TDC point "T1244DCC" and make adjustments to the process as needed at controller T1245.CC. Slide Valve differential pressure controller PDRC-1304 should remain in Automatic to maintain the DP if needed.**

____ 6. Connect the necessary test equipment to simulate the process at the transmitter below. Refer to the Calibration Sheets and using a Honeywell Smart communicator verify the transmitter **Fail Modes** are correct. Verify the Smart Communicator indicates the ID properly. Disconnect the Smart Communicator upon completion of the above verification. Calibrate transmitter, remove equipment, return to service, and fill out calibration sheet.

____ A. **TT-1245A**

____ 7. Connect the necessary test equipment to simulate the process at the transmitter below. Refer to the Calibration Sheets and using a Honeywell Smart communicator verify the transmitter **Fail Modes** are correct. Verify the Smart Communicator indicates the ID properly. Disconnect the Smart Communicator upon completion of the above verification. Calibrate transmitter, remove equipment, return to service, and fill out calibration sheet.

____ A. **TT-1245B**

____8. Follow this step to verify the Pre-alarms and TDC indication for **TT-1245.**

____ A. Connect simulator in marshalling cabinet (refer to loop sheet T1245.cc) Verify TDC indication for Transmitter **TT-1245A.** Apply 0, 50 and 100% to the TDC and verify the TDC displays accurately within 2% and the units are correct. Leave at 100% and verify alarms listed below in step "B" are clear. If transmitter “A” is selected check TDC on T1245.CC. If transmitter “B” is selected check TDC on T1245.BCC.

____ B. Observing TSL-1245 Moore Industries Alarm Card LED verify TSL-1245 Red LED extinguishes at the Calibration Sheet specified (V) setting. Complete the calibration sheet for TSL-1245 and adjust the trip card setting as needed. Verify alarms listed below are in alarm.

____ 1. Hi-way 1 TDC Pre-Alarm **"T1245LCC."**

____ 2. Control Room Annunciator Pre-Alarm **"TAL-1245A"**

____ C. Disconnect all test equipment. Verify the Pre-Alarm transmitter matches the Trip transmitter at TDC point T1244DCC.

____ D. Return controller **T1245.CC** to **"Automatic."**

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Comments _____

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Procedure No.
Revision Date
Page _ of _

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Annex V-2 — Model procedure for testing sensor logic

See Annex V-1 for preliminary information.

____ 1. At the ICS panel matrix, place **LSHH-1404/LSHH-1418** bypass switch **HS-1404** in **"Bypass."**

____ 2. Verify illumination of the amber LED's at the bypass keyswitches. Also verify **"I-1 System Bypassed"** lights at Shutdown Switches HS-1252 and HS-1291 are illuminated at the TDC console.

____ 3. Verify TDC Tag: L1404.CC & L1418.CC Level indications match. **Operations Note:** Monitor the Pre-alarm transmitter (L1417.CC) since the Trip transmitters will be out of service. Locate manual shutdown switch HS-1321, 1343 and 1436 on the TDC console. If the level indicated by L1417.CC increases above LSHH-1404/1418 trip setting, operations should Manually trip Riser #1 and 2 Regenerated Catalyst Slide valve by switching HS- 1321 and HS-1343 to SHUTDOWN.

____ 4. Follow this step to connect a Smart communicator and ID transmitters LT-1404 & 1418.

____ A. Disconnect the Power from the positive (+) terminal of transmitter LT-1418.

NOTE This must be done so that the Smart Communicator may communicate with LT- 1404.

____ B. Refer to the Calibration Sheets and using a Honeywell Smart communicator verify transmitter LT- 1404 **Fail Mode** is correct. Verify the Smart Communicator indicates the ID properly. Disconnect the Smart Communicator upon completion of the above verification.

____ C. Reconnect the Power to the positive (+) terminal of transmitter LT-1418.

____ D. Disconnect the Power from the positive (+) terminal of transmitter LT-1404.

NOTE This must be done so that the Smart Communicator may communicate with LT- 1418.

____ E. Refer to the Calibration Sheets and using a Honeywell Smart communicator verify transmitter LT- 1418 **Fail Mode** is correct. Verify the Smart Communicator indicates the ID properly. Disconnect the Smart Communicator upon completion of the above verification.

____ F. Reconnect the Power to the positive (+) terminal of transmitter LT-1404.

____ 5. Follow this step to verify the alarms for **LT-1404 & 1418.**

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____ A. Connect the necessary test equipment to simulate the process at the transmitters below. Calibrate transmitter, remove equipment, return to service, and fill out calibration sheet.

____ A. **LT-1404**

____ B. **LT-1418**

____ B. Test the Trip Transmitters high deviation alarm for LT-1404 & 1418.

____ 1. Connect simulators to wiring to control room.

____ 2. Set **LT-1404** to 50% of the calibrated range. Set **LT-1418** to 50% of the calibrated range. Verify TDC alarm L1402DCC is not in alarm (Group 210).

____ 3. Maintain **LT-1404** signal at **50%** of the calibrated range. Decrease **LT-1418** signal and verify TDC alarm L1402DCC alarms as the signal reaches 40% of the calibrated range of **LT-1418.** Set **LT-1418** to 50% of the calibrated range. Verify TDC alarm L1402DCC clears (Group 210).

____ 4. Maintain **LT-1418** signal at **50%** of the calibrated range. Decrease **LT-1404** signal and verify TDC alarm L1402DCC alarms as the signal reaches 40% of the calibrated range of **LT-1404.** Set **LT-1404** to 50% of the calibrated range. Verify TDC alarm L1402DCC clears (Group 210).

____ 5 Complete LSD-1402 Calibration Sheet.

____ 6 Remove simulators and reconnect.

____ C. Connect simulator to input jacks of LT-1404 & 1418 trip cards. Verify TDC indication for Transmitter **LT-1404 & 1418 (TDC tag: L1404.CC & L1418.CC Group 210).** Simulate the process to 0, 50, & 100% of calibrated range. Verify the TDC Displays within 2% and verify the units are correct. Leave at 50% and verify alarms listed below in step "E" are clear. Fill out calibration sheets for LY-1404 & 1418.

____ D. Observing LT-1404 Trip Card LED, decrease LT-1404 and verify the Ronan LED illuminates Red at the Calibration Sheet specified (V) setting. Verify alarms listed below are in alarm. Fill out LSLL-1404 calibration sheet. Return to 50% and verify alarms in step “E” clear.

____ E. Observing LT-1418 Trip Card LED, decrease LT-1418 and verify LSLL-1418 Ronan LED illuminates Red at the Calibration Sheet specified (V) setting. Verify alarms listed below are in alarm. Fill out LSLL-1418 calibration sheet. Return to 50% and verify alarms are clear.

____ A. Hi-way 1 TDC Trip Alarm **"L1403BCC."** Group 405

____ B. Control Room Annunciator Trip Alarm **"LALL-1403A"**

____ C. Sequence of Events Recorder Alarm **"L1403BCC"**

____ F. Observing LT-1404 Trip Card LED, increase LT-1404 and verify LSHH-1404 Ronan Trip Card LED illuminates Red at the Calibration Sheet specified (V) setting. Complete LSHH-1404 calibration sheet. Set **LT-1404** above **LSHH-1404** trip point. Verify alarms listed below in step "G" are clear.

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____ G. Observing LT-1418 Trip Card LED, increase LT-1418 and verify LSHH-1418 Ronan Trip Card LED illuminates Red at the Calibration Sheet specified (V) setting. Complete LSHH-1418 calibration sheet. Verify alarms listed below are in alarm.

____ A. Hi-way 1 TDC Trip Alarm **"L1403XCC."** Group 405

____ B. Control Room Annunciator Trip Alarm **"LAHH-1403A"**

____ C. Sequence of Events Recorder Alarm **"L1403XCC"**

____ H. Disconnect all test equipment.

____ I. Verify that **LSHH-1404 and LSHH-1418** are in a non-trip condition (ICS panel matrix green input LED's for these inputs are illuminated). Verify TDC indication for LT-1404 and 1418 match.

____ J. Return **LSHH-1404/LSHH-1418** bypass switch **HS-1404** to **"Normal."**

Comments _____

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Revision Date
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Annex V-3 — Model procedure for on-line testing sensor logic

____ 1. At the ICS panel matrix, place **PSLL-1328/1329** Input bypass switch **HS-1328** in **"Bypass."** Verify illumination of the amber LED's at the bypass key switches. Also verify **"I-1 System Bypassed"** lights at Shutdown Switches HS-1252 and HS-1291 are illuminated at the TDC console.

____ 2. Verify TDC Tag: P1328.CC & P1329.CC DP indications match. **Operations Note:** Monitor the Pre-alarm transmitter (P1326.CC) since the Trip transmitters will be out of service. Locate manual shutdown switch HS-1321 on the TDC console. If the (P1326.CC) DP across the Regenerated Catalyst Slide valve falls below PDSLL-1328/1329 Trip Setting, then a manual trip of the Regen Cat Slide valve may be necessary.

____ 3. Follow this step to connect a Smart communicator and ID transmitters PDT-1328 & 1329.

____ A. Disconnect the Power from the positive (+) terminal of transmitter PDT-1329.

NOTE This must be done so that the Smart Communicator may communicate with PDT-1328.

____ B. Refer to the Calibration Sheets and using a Honeywell Smart communicator verify the transmitter PDT-1328 **Fail Mode** is correct. Verify the Smart Communicator indicates the ID properly. Disconnect the Smart Communicator upon completion of the above verification.

____ C. Reconnect the Power to the positive (+) terminal of transmitter PDT-1329.

____ D. Disconnect the Power from the positive (+) terminal of transmitter PDT-1328.

NOTE This must be done so that the Smart Communicator may communicate with PDT -1329.

____ E. Refer to the Calibration Sheets and using a Honeywell Smart communicator verify the transmitter PDT-1329 **Fail Mode** is correct. Verify the Smart Communicator indicates the ID properly. Disconnect the Smart Communicator upon completion of the above verification.

____ F. Reconnect the Power to the positive (+) terminal of transmitter PDT-1328.

____ 4. Follow this step to verify the alarms for **PDT-1328 & 1329.**

____ A. Connect the necessary test equipment to simulate the process at the transmitter below. Calibrate transmitter, remove equipment, return to service, and fill out calibration sheet.

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____ A. **PDT-1328**

____ B. **PDT-1329**

____ B. Connect simulators to PT-1328 & PT-1329 wiring to control room. Test the Trip Transmitters high deviation alarm for PDT-1328 & 1329.

____ 1. Set **PDT-1328** to 50% of the calibrated range. Set **PDT-1329** to 50% of the calibrated range. Verify TDC alarm P1327DCC is not in alarm.

____ 2. Maintain **PDT-1329** signal at **50%** of the calibrated range. Decrease **PDT-1328** signal and verify TDC alarm P1327DCC (Group 185) alarms as the signal reaches 40% of the calibrated range of **PDT-1328.** Set **PDT-1328** to 50% of the calibrated range. Verify TDC alarm P1327DCC clears.

____ 3. Maintain **PDT-1328** signal at **50%** of the calibrated range. Decrease **PDT-1329** signal and verify TDC alarm P1327DCC (Group 185) alarms as the signal reaches 40% of the calibrated range of **PDT-1329.** Set **PDT-1329** to 50% of the calibrated range. Verify TDC alarm P1327DCC clears.

____ 4. Complete PDSD-1327 Calibration Sheet.

____ 5. Remove simulators and reconnect.

____ C. Verify TDC indication for Transmitter **PDT-1328 & 1329 (TDC tag: P1328.CC & P1329.CC).** Simulate 0, 50, & 100% of calibrated range. Verify the TDC Displays within 2% and verify the units are correct. Leave at 100% and verify alarms listed in step "F" are clear. Fill out calibration sheets for PY-1328 & 1329.

____ D. Observing PDT-1328 Trip Card LED, decrease PDT-1328 signal and verify PDSLL-1328 LED illuminates Red at the Calibration Sheet specified (V) setting. Complete PDSLL-1328 calibration sheet. Set **PDT-1328** DP above **PDSLL-1328** trip point.

____ E. Observing PDT-1329 Trip Card LED, decrease PDT-1329 signal and verify PDSLL-1329 LED illuminates Red at the Calibration Sheet specified (V) setting. Complete PDSLL-1329 calibration sheet. PDT-1329 should remain in the **trip** condition.

____ F. Verify PDT-1329 ICS EP-01, I-1 Green Input LED is extinguished. Decrease **PDT-1328** signal and verify **PDT-1328,** I-1 EP-01 Input LED extinguishes at **PDSLL- 1328** trip setting. Verify the alarms listed below trip 30 seconds after **PDT-1328** input LED extinguished.

____ A. Hi-way 1 TDC Trip Alarm **"P1342ZCC."** Group 404

____ B. Control Room Annunciator Trip Alarm **"PDALL-1342A"**

____ C. Sequence of Events Recorder Alarm **"P1342ZCC"**

____ G. Disconnect all test equipment from PDT-1328 & 1329, PDY-1328 & 1329 and PDSD-1327. Place transmitters **PDT-1328** and **PDT-1329** back in service.

____ H. Verify that **PDSLL-1328** and **PDSLL-1329** are in a non-trip condition (ICS panel matrix green input LED's are illuminated). Verify PDT-1328 & 1329 TDC Indications match (TDC point P1328.CC & P1329.CC).

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____ l. Return **PDSLL-1328/1329** bypass switch **HS-1328** to **"Normal."**

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Annex W — Model procedure for on-line final control element functional testing

Overview

This section has been developed to test I-1 SIF solenoids and/or valves on-line without initiating an actual trip.

SIF Trip valves which are normally open may not be actuated. The trip valves that are Normally Open, with latching solenoids are setup to allow solenoid valve testing. The solenoid valve wires will be lifted in the field at the GUA conduit fitting terminal strip. All defective or corroded terminal strips shall be replaced as required. A 24VDC power supply will be connected to the solenoid to trip the solenoid valve. The valve will not be tripped from the ICS Emergency Trip System. The ICS Output line monitor provides continuous testing of the Solenoid Circuit between the ICS cabinet and the solenoid valve. Therefore, it is not necessary that the final control element be tested from the ICS cabinet.

The trip valves that are Normally Open, having any type of trip solenoid valve other than a Manual reset solenoid are currently not setup to test the solenoid valves.

SIF Trip valves which may be blocked before and after the Trip Valve and are normally closed shall be actuated.

____ 1. Obtain Final Control Element Checkout Sheets for the following Solenoid valves.

____ HY-1224B

____ HY-1229B

____ FY-1247B

____ 2. An operations representative must be present through each step of this Section. Obtain the applicable permits as required to function each valve and/or solenoid.

____ 3. Follow this step to verify operation of trip valve HV-1224, "Emergency Steam to Riser #1 Feed Line."

____ A. Obtain a current copy of Loop Dwg H1224.CC and "Final Control Element Checkout Sheet" for HY-1224B.

____ B. Verify operations manually blocked the 3" manual valve after HV-1224.

____ C. Remove HY-1224B Solenoid valve GUA conduit fitting cover. Visually inspect the terminal connectors in the GUA fitting.

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____ D. Verify the wire colors match the Loop Drawing.

____ E. Replace terminal strip if defective or corroded. Reconnect the Reset solenoid and field wires to the terminal strip if terminal strip replacement was done. Initial this step if terminal strip replacement was required. If replacement is required but material is not available then write comments in the "Final Control Element Checkout Sheet."

____ F. Disconnect the TRIP Solenoid Valve Wires from the GUA terminal block.

____ G. Verify the Output Line Monitor Fault RED LED is illuminated on the "Alarms Matrix" located on the front of the ICS, "Common Services Panel."

____ H. Verify Annunciator alarm "XA-5842A" and Hiway 1 TDC alarm "X5842BCC" are in the alarm condition.

____ I. To apply 24VDC to the Trip coil, connect the 24VDC power supply to the lifted wires.

____ J. Verify HV-1224 trips to the open position.

____ K. Disconnect the power supply from the Trip Solenoid valve, re-terminate the trip solenoid valve wires to the terminal strip and verify the valve remains in the Open position.

____ L. Verify the Output Line Monitor Fault RED LED is extinguished on the " Alarms Matrix" located on the front of the ICS, "Common Services Panel."

____ M. Verify Annunciator alarm "XA-5842A" and Hiway 1 TDC alarm "X5842BCC" are clear.

____ N. Disconnect the Reset Solenoid Valve Wires from the GUA terminal block.

____ O. To apply 24VDC to the Reset Coil, connect the 24VDC power supply to the lifted wires.

____ P. Verify HV-1224 Resets to the Closed position.

____ Q. Disconnect the power supply from the Trip Solenoid valve and re-terminate the reset solenoid valve wires to the terminal strip. Verify the valve remains in the closed position. Replace the GUA fitting cover.

____ R. Verify operations opened the 3" manual valve after trip valve HV-1224.

____ S. Complete "Final Control Element Checkout Sheet" for solenoid HY-1224B.

____ 4. Follow this step to verify operation of trip valve HV-1229, "Emergency Lift Steam to Riser #1."

____ A. Obtain a current copy of Loop Dwg H1229.CC and "Final Control Element Checkout Sheet" for HY-1229B.

____ B. Verify operations manually blocked the 3" manual valve after HV-1229.

____ C. Remove HY-1229B Solenoid valve GUA conduit fitting cover. Visually inspect the terminal connectors in the GUA fitting.

____ D. Verify the wire colors match the Loop Drawing.

____ E. Replace terminal strip if defective or corroded. Reconnect the Reset solenoid and field wires to the terminal strip if terminal strip replacement was done. Initial this step if

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terminal strip replacement was required. If replacement is required but material is not available then write comments in the "Final Control Element Checkout Sheet."

____ F. Disconnect the TRIP Solenoid Valve Wires from the GUA terminal block.

____ G. Verify the Output Line Monitor Fault RED LED is illuminated on the " Alarms Matrix" located on the front of the ICS, "Common Services Panel."

____ H. Verify Annunciator alarm "XA-5842A" and Hiway 1 TDC alarm "X5842BCC" are in the alarm condition.

____ I. To apply 24VDC to the Trip coil, connect the 24VDC power supply to the lifted wires.

____ J. Verify HV-1229 trips to the open position.

____ K. Disconnect the power supply from the Trip Solenoid valve, re-terminate the trip solenoid valve wires to the terminal strip and verify the valve remains in the Open position.

____ L. Verify the Output Line Monitor Fault RED LED is illuminated on the "Alarms Matrix" located on the front of the ICS, "Common Services Panel."

____ M. Verify Annunciator alarm "XA-5842A" and Hiway 1 TDC alarm "X5842BCC" are clear.

____ N. Disconnect the Reset Solenoid Valve Wires from the GUA terminal block.

____ O. To apply 24VDC to the Reset Coil, connect the 24VDC power supply to the lifted wires.

____ P. Verify HV-1229 Resets to the Closed position.

____ Q. Disconnect the power supply from the Trip Solenoid valve and re-terminate the reset solenoid valve wires to the terminal strip. Verify the valve remains in the closed position. Replace the GUA fitting cover.

____ R. Verify operations opened the 3" manual valve after trip valve HV-1229.

____ S. Complete "Final Control Element Checkout Sheet" for solenoid HY-1229B.

____ 5. Follow this step to verify the operation of trip valve FY-1247B, "Recycle Sourwater."

____ A. Verify operations removed the Car Seal from the "3- way Manual Bypass Valve" at FV-1247.

NOTE Observe FV-1247 for valve movement while completing the next step. FV-1247 should remain in the same position while turning the "3-way Manual Bypass Valve" to the Bypass Position.

____ B. Switch the "3-way Manual Bypass Valve" at FV-1247 to the "BYPASS" position.

____ C. Remove FY-1247B Solenoid valve GUA conduit fitting cover. Visually inspect the terminal connectors in the GUA fitting.

____ D. Replace terminal strip if defective or corroded. Initial this step if terminal strip replacement was required. If replacement is required but material is not available then write comments in the "Final Control Element Checkout Sheet."

____ E. Disconnect the Solenoid Valve Wires from the GUA terminal block.

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____ F. Verify the Output Line Monitor Fault RED LED is illuminated on the "Alarms Matrix" located on the front of the ICS, "Common Services Panel."

____ G Verify Annunciator alarm "XA-5842A" and Hiway 1 TDC alarm "X5842BCC" are in the alarm condition.

____ H. To apply 24VDC to the Trip coil, connect the 24VDC power supply to the lifted wires.

____ I. Verify solenoid valve FV-1247 vents and the pressure gauge located on the "3-way Manual Bypass Valve" local panel decreases to 0 PSIG.

____ J. Disconnect the power supply from the Trip Solenoid valve and re-terminate the solenoid valve wires to the terminal strip.

____ K. Verify the Output Line Monitor Fault RED LED is extinguished on the "Alarms Matrix" located on the front of the ICS, "Common Services Panel."

____ L. Verify (AN-01) Annunciator alarm "XA-5842A" and Hiway 1 TDC alarm "X5842BCC" are clear.

____ M. Manually reset the solenoid valve and verify the pressure gauge located on the "3-way Manual Bypass Valve" local panel returns to the signal output from E/P (FY-1247A).

____ N. Return the "3-way Manual Bypass Valve" at FV-1247 to the "NORMAL" position.

____ O. Verify operations replaced the Car Seal on the "3-way Manual Bypass Valve" control panel at FV-1247.

____ P. Complete "Final Control Element Checkout Sheet" for solenoid FY-1247B.

Comments _____

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Annex X — Model procedure for on-line testing of compressor SIF

GENERIC GUIDELINES

This is the on-line test procedure for the Wet Gas Compressor shutdown system. It is expected that this system will be tested yearly according to the accompanying procedure. All testing must be done in strict adherence to all the instructions and requirements of this test procedure. All test equipment must be verified before using for the function test. All test results must be recorded on the Control Systems function test worksheet. This form must be dated and signed and must be forwarded to the Control Systems CSE at the completion of the test.

In addition to this Testing procedure, there is a written Mitigation Plan and a Specific Maintenance Procedure for this SIF. Craftsmen must be familiar with the mitigation plan and the testing and maintenance procedures before commencing testing.

Testing of this system and any repair/maintenance items require the implementation of the Mitigation Plan or the unit must be shut down.

If maintenance is required based on what is found during the test, the craft must perform maintenance in strict adherence to the maintenance procedures for this system. For example, if any device is recalibrated or replaced, fill out calibration sheets. Document all other maintenance in field notes attached to the function test worksheet.

NOTES FOR ON-LINE TEST PREPARATION

The Wet Gas Compressor System cannot be fully tested on-line because the two shutdown outputs, Motor Stop Contacts and the Discharge Trip Valve, cannot be allowed to operate while the unit is running. The following procedures are designed to give the tester the best possible assessment of the functionality of each shutdown loop without actually initiating a shutdown of the compressor. These procedures should only be used for a standard yearly function test of the system. A full inspection should occur at the three year interval during turnaround.

1) Override ICS trip outputs

Since there is not a bypass switch for the compressor motor contacts, X-11871, or a bypass valve around the compressor discharge trip valve, XV-11855, these outputs must be defeated using the keyswitch output override key. This key is located at the lower right hand corner of the system test tray on the ICS panel. Turn this keyswitch to the OVERRIDE position - indicated by override LEDs on output modules and bypass light on Control Board Handswitch. The ICS shutdown system can no longer perform the trip of the compressor and trip of the discharge valve. However, the manual shutdown switches will still shutdown the machine, but not trip the discharge valve.

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2) Defeat the ICS auto-test system

The auto-test system routinely tests the operation of the ICS cabinet by testing the input modules, logic modules, and output modules. These tests will activate the LEDs on the face of the I/O cards, making it difficult to analyze the results of the function test being performed. Therefore, the auto-test should be defeated. To defeat the auto-test sequence, turn the auto-test keyswitch from the AUTO to MANUAL position.

Audit performed by: _____ Date: _____

Control systems representative: _____ Date: _____

Operations representative: _____ Date: _____

For the on-line function test, the actual Trip Outputs and the Shutdown Handswitches cannot be tested. Further, the ICS Auto-Test System is continually checking the logic. Therefore, only the Shutdown Inputs and Input Bypasses need be verified by this function test.

1) L-11609 East First Stage Dry Drum High Level Trip

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in “Output Override” _____

   Override LEDs on Output Modules

   are illuminated _____

   Bypass Light on HS-11871-A

   is illuminated _____

   Bypass Light on HS-11855 is

   illuminated _____

   NA-11555A in alarm _____
2. Check calibration for LT-11609. _____
3. Check that all S/D components are painted

   red and all have a red tag. _____

B. Function Test (Craftsman/Inspector)

1. Verify LY-11609 Analog Input Trip Setting by

   selecting the toggle switch to “A” and pressing

   the meter pushes button. Read the trip setting off

   of the Analog Display Module and record this

   value as the “As Found” value under the “ICS

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Trip Card" column. _____

2. Simulate signal to check trip setting. _____

3. Verify trip indicators. _____

   LAHH-11609 in alarm _____

   ICS Output Cards LED

   changed state _____

4. Set bypass key switch to "ENABLE" position

   and move toggle switch on LY-11609 input card

   to the "BYPASS" position. _____

5. Verify Input Bypass indicator. _____

   Bypass LED on Input Card

   is illuminated _____

6. Simulate signal to check trip. _____

7. Verify trip indicator. _____

   LAHH-11609 in alarm _____

8. Return system to ready to operate mode. _____

   Disconnect field test equipment _____

   Verify NOT in S/D condition _____

   Return LY-11609 bypass toggle

   switch to the center position. _____

9. Complete required forms. _____

   Malfunction Sheet _____

   DPMC-3319 _____

2) L-11608 West First Stage Dry Drum High Level Trip

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in "Output Override." _____

   Override LEDs on Output Modules are illuminated _____

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Bypass Light on HS-11871-A is illuminated _____

Bypass Light on HS-11855 is illuminated _____

NA-11555A in alarm _____

2. Check calibration for LT-11608. _____

3. Check that all S/D components are painted red and all have a red tag. _____

B. Function Test (Craftsman/Inspector)

1. Verify LY-11608 Analog Input Trip Setting by selecting the toggle switch to “A” and pressing the meter push button. Read the trip setting off of the Analog Display Module and record this value as the “As Found” value under the “ICS Trip Card” column. _____

2. Simulate signal to check trip setting. _____

3. Verify trip indicators. _____

LAHH-11608 in alarm _____

ICS Output Cards LED changed state _____

4. Set bypass key switch to “ENABLE” position and move toggle switch on LY-11608 input card to the “BYPASS” position. _____

5. Verify Input Bypass indicator. _____

Bypass LED on Input Card is illuminated _____

6. Simulate signal to check trip. _____

7. Verify trip indicator. _____

LAHH-11608 in alarm _____

8. Return system to ready to operate mode. _____

Disconnect field test equipment _____

Verify NOT in S/D condition _____

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Return LY-11608 bypass toggle
switch to the center position. _____

9. Complete required forms. _____

Malfunction Sheet _____

DPMC-3319 _____

3) L-11621 Second Stage Dry Drum High Level trip

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in “Output Override." _____

Override LEDs on Output Modules are illuminated _____

Bypass Light on HS-11871-A is illuminated _____

Bypass Light on HS-11855 is illuminated _____

NA-11555A in alarm _____

2. Check calibration for LT-11621. _____

3. Check that all S/D components are painted
red and all have a red tag. _____

B. Function Test (Craftsman/Inspector)

1. Verify LY-11621 Analog Input Trip Setting by selecting the toggle switch to “A” and pressing the meter push button. Read the trip setting off of the Analog Display Module and record this value as the “As Found” value under the “ICS Trip Card” column. _____

2. Simulate signal to check trip setting. _____

3. Verify trip indicators. _____

LAHH-11621 in alarm _____

ICS Output Cards LED
changed state _____

Procedure No.
Revision Date
Page _ of _

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4. Set bypass key switch to “ENABLE” position
   and move toggle switch on LY-11621 input card
   to the “BYPASS” position. _____
5. Verify Input Bypass indicator. _____
   Bypass LED on Input Card
   is illuminated _____
6. Simulate signal to check trip. _____
7. Verify trip indicator. _____
   LAHH-11621 in alarm _____
8. Return system to ready to operate mode. _____
   Disconnect field test equipment _____
   Verify NOT in S/D condition _____
   Return LY-11621 bypass toggle
   switch to the center position. _____
9. Complete required forms. _____
   Malfunction Sheet _____
   DPMC-3319 _____

4) L-11843 First Stage Suction Boot High Level Trip

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in “Output Override." _____
   Override LEDs on Output Modules are illuminated _____
   Bypass Light on HS-11871-A is illuminated _____
   Bypass Light on HS-11855 is illuminated _____
   NA-11555A in alarm _____
2. Check calibration for LT-11843. _____
3. Check that all S/D components are painted
   red and all have a red tag. _____

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B. Function Test (Craftsman/Inspector)

1. Verify LY-11843 Analog Input Trip Setting by selecting the toggle switch to “A” and pressing the meter push button. Read the trip setting off of the Analog Display Module and record this value as the “As Found” value under the “ICS Trip Card” column. _____
2. Simulate signal to check trip setting. _____
3. Verify trip indicators. _____
   LAHH-11843 in alarm _____
   ICS Output Cards LED changed state _____
4. Set bypass key switch to “ENABLE” position and move toggle switch on LY-11843 input card to the “BYPASS” position. _____
5. Verify Input Bypass indicator. _____
   Bypass LED on Input Card is illuminated _____
6. Simulate signal to check trip. _____
7. Verify trip indicator. _____
   LAHH-11843 in alarm _____
8. Return system to ready to operate mode. _____
   Disconnect field test equipment _____
   Verify NOT in S/D condition _____
   Return LY-11843 bypass toggle switch to the center position. _____
9. Complete required forms. _____
   Malfunction Sheet _____
   DPMC-3319 _____

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5) L-11857 Second Stage Suction Boot High Level Trip

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in “Output Override." _____

   Override LEDs on Output Modules are illuminated _____

   Bypass Light on HS-11871-A is illuminated _____

   Bypass Light on HS-11855 is illuminated _____

   NA-11555A in alarm _____

2. Check calibration for LT-11857. _____

3. Check that all S/D components are painted

   red and all have a red tag. _____

B. Function Test (Craftsman/Inspector)

1. Verify LY-11857 Analog Input Trip Setting by selecting

   the toggle switch to “A” and pressing the meter push button.

   Read the trip setting off of the Analog Display Module and record

   this value as the “As Found” value under the “ICS

   Trip Card” column. _____

2. Simulate signal to check trip setting. _____

3. Verify trip indicators. _____

   LAHH-11857 in alarm _____

   ICS Output Cards LED changed state _____

4. Set bypass key switch to “ENABLE” position

   and move toggle switch on LY-11857 input card

   to the “BYPASS” position. _____

5. Verify Input Bypass indicator. _____

   Bypass LED on Input Card is illuminated _____

6. Simulate signal to check trip. _____

7. Verify trip indicator. _____

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LAHH-11857 in alarm _____

8. Return system to ready to operate mode. _____

   Disconnect field test equipment _____

   Verify NOT in S/D condition _____

   Return LY-11857 bypass toggle

   switch to the center position. _____

9. Complete required forms. _____

   Malfunction Sheet _____

   DPMC-3319 _____

6) L-11895 Overhead Seal Oil Tank Low Level Trip

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in “Output Override." _____

   Override LEDs on Output Modules are illuminated _____

   Bypass Light on HS-11871-A is illuminated _____

   Bypass Light on HS-11855 is illuminated _____

   NA-11555A in alarm _____

2. Check calibration for LSLL-11895. _____

3. Check that all S/D components are painted

   red and all have a red tag. _____

B. Function Test (Craftsman/Inspector)

1. Simulate signal to check trip setting. _____

2. Verify trip indicators _____

   LALL-11895 in alarm _____

   ICS Output Cards LED changed state _____

3. Set bypass key switch to “ENABLE” position

   and move toggle switch on LSLL-11895 input

   card to the “BYPASS” position. _____

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4. Verify Input Bypass indicator. _____

   Bypass LED on Input Card is illuminated _____

5. Simulate signal to check trip. _____

6. Verify trip indicator. _____

   LALL-11895 in alarm _____

7. Return system to ready to operate mode. _____

   Disconnect field test equipment _____

   Verify NOT in S/D condition _____

   Return LSLL-11895 bypass toggle

   switch to the center position. _____

8. Complete required forms. _____

   Malfunction Sheet _____

   DPMC-3319 _____

7) P-11876 C-6800 Low Lube Oil Pressure Trip

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in “Output Override." _____

   Override LEDs on Output Modules are illuminated _____

   Bypass Light on HS-11871-A is illuminated _____

   Bypass Light on HS-11855 is illuminated _____

   NA-11555A in alarm _____

2. Check calibration for PT-11876. _____

3. Check that all S/D components are painted

   red and all have a red tag. _____

B. Function Test (Craftsman/Inspector)

1. Verify PT-11876 Analog Input Trip Setting by selecting

   the toggle switch to “A” and pressing the meter push button.

   Read the trip setting off of the Analog Display Module and record

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this value as the “As Found” value under the “ICS Trip Card” column. _____

2. Simulate signal to check trip setting. _____
3. Verify trip indicators _____

   PALL-11876 in alarm _____

   ICS Output Cards LED changed state _____
4. Set bypass key switch to “ENABLE” position and move toggle switch on PT-11876 input card to the “BYPASS” position. _____
5. Verify Input Bypass indicator. _____

   Bypass LED on Input Card is illuminated _____
6. Simulate signal to check trip. _____
7. Verify trip indicator. _____

   PALL-11876 in alarm _____
8. Return system to ready to operate mode. _____

   Disconnect field test equipment _____

   Verify NOT in S/D condition _____

   Return PT-11876 bypass toggle switch to the center position. _____
9. Complete required forms. _____

   Malfunction Sheet _____

   DPMC-3319 _____

8) N-11555-AA/AB High Axial Vibration Trip

NOTE These loops must be audited by maintenance.

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in “Output Override." _____

   Override LEDs on Output Modules are illuminated _____

   Bypass Light on HS-11871-A is illuminated _____

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Bypass Light on HS-11855 is illuminated _____

NA-11555A in alarm _____

2. Check condition of vibration monitors and wiring harness. _____

3. Check that all S/D components are painted

red and all have a red tag. _____

B. Function Test (Craftsman/Inspector)

1. Simulate signals to check trip settings. _____

2. Verify trip indicators. _____

NAHH-11555-D in alarm _____

ICS Output Cards LED changed state _____

3. Set bypass key switch to “ENABLE” position

and move toggle switch on NIS-11555-AA/AB

input card to the “BYPASS” position. _____

4. Verify Input Bypass indicator. _____

Bypass LED on Input Card is illuminated _____

5. Simulate signal to check trip. _____

6. Verify trip indicator. _____

NAHH-11555-D in alarm _____

7. Return system to ready to operate mode. _____

Disconnect field test equipment _____

Verify NOT in S/D condition _____

Return NIS-11555-AA/AB bypass

toggle switch to the center position. _____

8. Complete required forms. _____

Malfunction Sheet _____

DPMC-3319 _____

9) N-11555-Z1/6 C-6800 High Radial Vibration Trip

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These loops must be audited by maintenance.

A. Preparation (Craftsman)

1. Ensure ICS Cabinet is in “Output Override." _____

   Override LEDs on Output Modules are illuminated _____

   Bypass Light on HS-11871-A is illuminated _____

   Bypass Light on HS-11855 is illuminated _____

   NA-11555A in alarm _____

2. Check condition of vibration monitors and wiring harness. _____

3. Check that all S/D components are painted

   red and all have a red tag. _____

B. Function Test (Craftsman/Inspector)

1. Simulate signals to check trip settings. _____

2. Verify trip indicators. _____

   NAHH-11555-C in alarm _____

   ICS Output Cards LED changed state _____

3. Set bypass key switch to “ENABLE” position and move

   toggle switch on NIS-11555-Z1-6 input card to the “BYPASS” position. _____

4. Verify Input Bypass indicator. _____

   Bypass LED on Input Card is illuminated _____

5. Simulate signal to check trip. _____

6. Verify trip indicator. _____

   NAHH-11555-C in alarm _____

7. Return system to ready to operate mode. _____

   Disconnect field test equipment _____

   Verify NOT in S/D condition _____

   Return NIS-11555-Z1-6 bypass

   toggle switch to the center position. _____

8. Complete required forms. _____

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Malfunction Sheet _____

DPMC-3319 _____

Restoring the System to Normal Operation

This completes this SIS Inspection. Ensure that all shutdown inputs are in the normal “run” condition. Return the bypass toggle switches on each input module to the center position and turn the bypass keyswitch to the “OFF” position. Return the Output Override Keyswitch to the “NORMAL” position. Return the ICS Auto-Test keyswitches to the “NORMAL” and “AUTO” positions.

Comments _____

_____

_____

_____

_____

_____

_____

_____

_____

CRAFTSMAN SIGNATURE: _____

DATE*:* _____

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NOTE This procedure was developed by an operating company for a specific application. It has been modified to remove any reference that might tie it to a specific company. This procedure should only be used as an example of how a user might develop a procedure specific to their SIS application. Any references to specific brands of instrumentation in the procedure are to clarify the intent of the procedure only and are in no respect meant to suggest these brands are acceptable or preferred for the user's specific application. The instrument identification numbers used in the procedures are for clarification purposes only and should in no way be taken as indicative of a particular company's instruments on a particular process.

CAUTION — PRIOR TO APPLYING THE INFORMATION IN THIS ANNEX TO DEVELOP A PROCESS SPECIFIC PROCEDURE, THE USER SHOULD REVIEW AND UNDERSTAND THE GUIDANCE IN THE BODY OF ISA-TR84.00.03-2002.

Annex Y — Model procedure for on-line testing of 2oo3 temperature elements

SAFETY CRITICAL

****

TASK NO:

TAG NO.: MT284-HCO

PID NO: 901-198-25A, 28A, 30B, 31A

LOGIC DIA.: 901-191-856, 857, 859

SERVICE:

ACETYLENE CONVERTERS M-R-03D, HIGH OPERATING BED TEMPERATURE CUTOUT

****

System description:

This is a 2 out of 3 trip logic system. High operating bed temperature trip will operate all valves listed below.

Final control elements:

MR011-BV (closes), MR014-BV (opens), MR015-BV (closes),

MR065-BV (closes).

NOTE:

---

1. The thermocouples used in this trip circuit are upscale burnout.

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2. MT284-HCOA is the common alarm for this trip system.

3. Defeat alarm: MT282-DSA

   Discrepancy alarm: MT287-DIA

   High temp alarm: MT283-HA

4. TDC point alarms are on Console 3A, group C-8.

5. Before proceeding, verify that no other potential trip alarm conditions exist for M-R-03D by observing alarm panel status. If an abnormal condition exists, turn to appropriate inspection procedure and correct problem. Defeat switch common alarm must be OFF.

   CHECK On ( ) Off ( )

Access the INSTRUMENT RECORD SYSTEM and confirm the following:

Transmitter range = [0 to 1100 deg F]

High alarm setpoint = [400 deg F]

High confirmed CHECK = Yes ( ) No ( )

NOTIFY OPERATIONS

****

INSPECTION APPROVAL_____ _____

Time and Date Initials Operations Supervisor

CAUTION:

Individual defeat switches MT242, MT243, MT244, MT245, MT246, MT247, MT248, MT249, MT250, MT251-DS or the Master defeat switch, MR03D-DS must be in defeat position before inspection begins. Verify defeat position by observing red light and defeat alarm. Shutdown of all acetylene converters will occur if switches are not in Defeat position.

NOTICE:

Remind Console Operator to follow precaution plan for “Defeat of any Safety Critical System”, and also to log this defeat in the “Safety Critical System Defeat Log."

Check ( )

1. Did you obtain necessary work permit ? Yes ( ) No ( )

   Which type ? Hot work ( ) Instrument ( )

2. This check cannot be done if M-R-03D is in “REGEN” mode.

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3. If M-R-03D is in “Stand-by” mode, have Operations put it in “On-line” mode.

4. Control room check:

   a. Go to the TDC Console, record the current readings listed below.

Point temperatures:

1st set	2nd set	3rd set	TDC point
degF	degF	degF	degF
MT242 [ ]	MT310 [ ]	MT319 [ ]	MT328 [ ]
MT243 [ ]	MT311 [ ]	MT320 [ ]	MT329 [ ]
MT244 [ ]	MT312 [ ]	MT321 [ ]	MT330 [ ]
MT245 [ ]	MT313 [ ]	MT322 [ ]	MT331 [ ]
MT246 [ ]	MT314 [ ]	MT323 [ ]	MT332 [ ]
MT247 [ ]	MT315 [ ]	MT324 [ ]	MT333 [ ]
MT248 [ ]	MT316 [ ]	MT325 [ ]	MT334 [ ]
MT249 [ ]	MT317 [ ]	MT326 [ ]	MT335 [ ]
MT250 [ ]	MT318 [ ]	MT327 [ ]	MT336 [ ]
MT251 [ ]	MT288 [ ]	MT289 [ ]	MT337 [ ]

   b. Compare the readings. If there is any transmitter which needs to be repaired or replaced, do it first before continuation of this inspection.

   c. Verify the high alarm set point at the TDC console.

Check OK ( )

   d. Verify the high cutout set point at the TDC console.

Check OK ( )

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5. Remove thermocouple head cover and check condition for contamination.

MT242	Ok ( )	Bad ( )	MT310	Ok ( )	Bad ( )
MT243	Ok ( )	Bad ( )	MT311	Ok ( )	Bad ( )
MT244	Ok ( )	Bad ( )	MT312	Ok ( )	Bad ( )
MT245	Ok ( )	Bad ( )	MT313	Ok ( )	Bad ( )
MT246	Ok ( )	Bad ( )	MT314	Ok ( )	Bad ( )
MT247	Ok ( )	Bad ( )	MT315	Ok ( )	Bad ( )
MT248	Ok ( )	Bad ( )	MT316	Ok ( )	Bad ( )
MT249	Ok ( )	Bad ( )	MT317	Ok ( )	Bad ( )
MT250	Ok ( )	Bad ( )	MT318	Ok ( )	Bad ( )
MT251	Ok ( )	Bad ( )	MT288	Ok ( )	Bad ( )
MT319	Ok ( )	Bad ( )	MT328	Ok ( )	Bad ( )
MT320	Ok ( )	Bad ( )	MT329	Ok ( )	Bad ( )
MT321	Ok ( )	Bad ( )	MT330	Ok ( )	Bad ( )
MT322	Ok ( )	Bad ( )	MT331	Ok ( )	Bad ( )
MT323	Ok ( )	Bad ( )	MT332	Ok ( )	Bad ( )
MT324	Ok ( )	Bad ( )	MT333	Ok ( )	Bad ( )
MT325	Ok ( )	Bad ( )	MT334	Ok ( )	Bad ( )
MT326	Ok ( )	Bad ( )	MT335	Ok ( )	Bad ( )
MT327	Ok ( )	Bad ( )	MT336	Ok ( )	Bad ( )
MT289	Ok ( )	Bad ( )	MT337	Ok ( )	Bad ( )

6. Thermocouple burnout check:

   a. Disconnect thermocouple input one at a time at head for below listed thermocouples.

   b. When any sensor failure occurs, the point temperature will read upscale for thermocouple open circuit failures. The discrepancy alarm will also come on. Disconnect each thermocouple sensor one at a time as listed in the following table and verify this action.

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Discrepancy alarm

MT242	On ( )	Off ( )
MT243	On ( )	Off ( )
MT244	On ( )	Off ( )
MT245	On ( )	Off ( )
MT246	On ( )	Off ( )
MT247	On ( )	Off ( )
MT248	On ( )	Off ( )
MT249	On ( )	Off ( )
MT250	On ( )	Off ( )
MT251	On ( )	Off ( )
MT310	On ( )	Off ( )
MT311	On ( )	Off ( )
MT312	On ( )	Off ( )
MT313	On ( )	Off ( )
MT314	On ( )	Off ( )
MT315	On ( )	Off ( )
MT316	On ( )	Off ( )
MT317	On ( )	Off ( )
MT318	On ( )	Off ( )
MT288	On ( )	Off ( )
MT319	On ( )	Off ( )
MT320	On ( )	Off ( )
MT321	On ( )	Off ( )
MT322	On ( )	Off ( )
MT323	On ( )	Off ( )
MT324	On ( )	Off ( )

Procedure No.
Revision Date
Page _ of _

--`,,,`,-`-`,,,,`-`-`,,`-`-`-,,`,,`,`,,`---

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MT325	On ( )	Off ( )
MT326	On ( )	Off ( )
MT327	On ( )	Off ( )
MT289	On ( )	Off ( )

7. Perform 2 out of 3 voting logic check:

   a. Disconnect 1st input. Only the discrepancy alarm should come on. The high alarm and the cutout alarm should not come on.

   b. Disconnect 2nd input. The high alarm and the cutout alarm should come on.

   c. Record condition of cutout alarm below.

   d. Reconnect both inputs. Record condition of the cutout alarm below.

   e. Repeat procedures above for all combinations in the table below.

MT242	MT310	MT319	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X			X	On ( ) Off ( )

MT243	MT311	MT320	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

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MT244	MT312	MT321	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

MT245	MT313	MT322	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

MT246	MT314	MT323	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

MT247	MT315	MT324	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

Procedure No.
Revision Date
Page _ of _

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MT248	MT316	MT325	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

MT249	MT317	MT326	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

MT250	MT318	MT327	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

MT251	MT288	MT289	Reconnect	Cutout alarm
X	X			On ( ) Off ( )
	X	X		On ( ) Off ( )
X		X		On ( ) Off ( )
			X	On ( ) Off ( )

8. Final control elements check:

   a. Notify Operations that you are ready for the final control elements trip actuation. Have Operations prepare the final control elements for trip actuation check.

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b. As per Operations procedure for final control elements check, simulate a trip condition. Change the status of the defeat switch and observe the actuation of the valve. Record status below.

Defeat	MR011-BV actuation	MR014-BV actuation
ON	Yes ( ) No ( )	Yes ( ) No ( )
OFF	Yes ( ) No ( )	Yes ( ) No ( )

Defeat	MR015-BV actuation	MR065-BV actuation
ON	Yes ( ) No ( )	Yes ( ) No ( )
OFF	Yes ( ) No ( )	Yes ( ) No ( )

9. Transmitter calibration: Type K Thermocouple

a. Disconnect thermocouple leads from the terminals.

b. Connect a millivolt source (Transmation or equivalent) to the input of the transmitter.

c. Connect a milliamp meter to the output of the transmitter.

d. Check transmitter zero and span. Record as found values below.

e. Re-calibrate, if necessary and record as left values.

f. Proceed to next transmitter

until all transmitter listed have been checked.

	MT242-T	MT310-T	MT319-T	MT328-T
As found LRL, ma dc	[ ]	[ ]	[ ]	[ ]
As left LRL, ma dc	[ ]	[ ]	[ ]	[ ]
As found URL, ma dc	[ ]	[ ]	[ ]	[ ]
As left URL, ma dc	[ ]	[ ]	[ ]	[ ]

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	MT243-T	MT311-T	MT320-T	MT329-T
As found LRL, ma dc	[ ]	[ ]	[ ]	[ ]
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	MT244-T	MT312-T	MT321-T	MT330-T
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	MT245-T	MT313-T	MT322-T	MT331-T
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	MT246-T	MT314-T	MT323-T	MT332-T
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Procedure No.
Revision Date
Page _ of _

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	MT247-T	MT315-T	MT324-T	MT33-T
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	MT248-T	MT316-T	MT325-T	MT334-T
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	MT249-T	MT317-T	MT326-T	MT335-T
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	MT250-T	MT318-T	MT327-T	MT336-T
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Procedure No.
Revision Date
Page _ of _

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	MT251-T	MT288-T	MT289-T	MT337-T
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10. Replace all covers.

11. Visual checks:

    Tagging:

    a. Are all instrument in this task tagged with a special tag identifying them as “Critical Instrument”?

    Yes ( ) No ( )

    As “Critical Instrument” ( )

    As “Safety Critical Instrument” ( )

    b. Tagging condition: Good ( ) Bad ( )

    Conduit system: OK ( ) Bad ( ) If bad check below.

    Covers off [ ] Drains missing [ ] Supports gone [ ]

    Seal needed [ ] Flex bad [ ] Conduit broken [ ]

    Fitting bad [ ] Corrosion [ ] Other [ ]

    Details [_____]

    Correction made? Yes ( ) No ( )

    Block valve: MOV MR011-BV

    Piping gasket leak [ ] Valve gasket leak [ ]

    Packing gland leak [ ] Sticky stem action [ ]

    Topworks problem [ ]

    Details [_____]

    Block valve: MOV MR014-BV

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Piping gasket leak [ ] Valve gasket leak [ ]

Packing gland leak [ ] Sticky stem action [ ]

Topworks problem [ ]

Details [_____]

Block valve: MOV MR015-BV

Piping gasket leak [ ] Valve gasket leak [ ]

Packing gland leak [ ] Sticky stem action [ ]

Topworks problem [ ]

Details [_____]

Block valve: MOV MR065-BV

Piping gasket leak [ ] Valve gasket leak [ ]

Packing gland leak [ ] Sticky stem action [ ]

Topworks problem [ ]

Details [_____]

12. Verify that ALL cutout alarms are now OFF.

Check On ( ) Off ( )

13. Return ALL individual defeat switches and Master Defeat switch to in SERVICE position.

Check ( )

14. Notify Operations - Inspection complete.

--- --- ---

Time and Date Initials Tech. Initials Maint. Supvr.

****

RECOMMENDED CORRECTIVE ACTION (comment below)

_____

_____

_____

_____

Procedure No.
Revision Date
Page _ of _

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Annex Z — Model procedure for testing final control elements when manual bypass valves are provided

Converter Output Trip Verification

This procedure will test the trip outputs by opening the T/C (Upscale Burnout). Two thermocouple inputs will be disconnected to simulate a trip condition and the solenoids and trip indications will be verified. This test will cause a total system trip.

End Device Isolation

In order to validate that the interlock will perform its associated trip action when required, it is necessary to periodically test the end control devices such as control valves, block valves, and motor operated valves. However, in an on-line testing situation the unit operations cannot be altered or upset. Therefore, appropriate provisions should be made to isolate these end devices. This following section is intended to cover the methods necessary to perform this isolation in a safe manner.

Valve Isolation

Valves should be isolated in accordance with plant operating guidelines and safety guidelines.

WARNING!

Once the following valves are bypassed, the Converters cannot be tripped automatically by the SIF. Therefore, the Control Room Operator should monitor closely all critical process variables and notify the Field Operator immediately if an upset condition occurs so that he can remove all bypasses and allow the SIF to trip the converters.

The following steps should be taken:

1. Before attempting to perform this critical portion of the on-line test, verify with the Operations Representative that it is safe to isolate and test the affected equipment.

   Initials _____ Date: _____

2. Isolate the Shutdown Solenoid Valve (XV-5318) to the Hydrogen Feed Control Valve (FV-5318). This is accomplished as follows:

- Remove the car-seal from hand operated valve HS-5318 located on the bypass panel by the control valve.

- Turn hand valve HS-5318 until the solenoid valve is isolated.

Procedure No.
Revision Date
Page _ of _

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- Connect instrument air supply to test port on bypass panel and apply air pressure.

Initials _____ Date: _____

3. Isolate the Shutdown Solenoid Valve (XV-5324) Hydrogen Feed Block Valve (FV-5324). This is accomplished as follows:

- Remove the car-seal from hand operated valve HS-5324 located on the bypass panel by the block valve.
- Turn hand valve HS-5324 until the solenoid valve is isolated.
- Connect instrument air supply to test port on bypass panel and apply air pressure.

Initials _____ Date: _____

4. Isolate the Shutdown Solenoid Valve (XV-5325) to the Hydrogen Feed Control Valve (FV-5325). This is accomplished as follows:

- Remove the car-seal from hand operated valve HS-5325 located on the bypass panel by the control valve.
- Turn hand valve HS-5325 until the solenoid valve is isolated.
- Connect instrument air supply to test port on bypass panel and apply air pressure.

Initials _____ Date: _____

5. Isolate the Shutdown Solenoid Valve (XV-5323) Hydrogen Feed Block Valve (FV-5323). This is accomplished as follows:

- Remove the car-seal from hand operated valve HS-5323 located on the bypass panel by the block valve.
- Turn hand valve HS-5323 until the solenoid valve is isolated.
- Connect instrument air supply to test port on bypass panel and apply air pressure.

Initials _____ Date: _____

6. Place Converter Inlet Motor Operated Valve MOV-5379 in Test Bypass. This is accomplished by placing the MOV-5379C S/D Bypass Test switch located on the local bypass panel in the “Bypass” position. The amber shutdown bypass light located at the bypass panel box will illuminate to indicate that the Shutdown/Bypass switch is in the bypass position. V5379S in TDC will also indicate MOV-5379 bypassed.

Initials _____ Date: _____

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7. Place Converter Outlet Motor Operated Valve MOV-5390 in Test Bypass. This is accomplished by placing the MOV-5390C S/D Bypass switch located on the local bypass panel in the “Bypass” position. The amber shutdown bypass light located at the bypass panel box will illuminate to indicate that the Shutdown/Bypass switch is in the bypass position. V5390S in TDC will also indicate MOV-5390 Bypassed.

Initials _____ Date: _____

8. Isolate the Shutdown Solenoid Valve (XV-5386) Temperature Control Valves (TV-5386A & TV-5386B). This is accomplished as follows:

- Remove the car-seal from hand operated valve HS-5386 located by the control valve under the Converter fin fans.
- Turn hand valve HS-5386 until the solenoid valve is isolated.
- Connect instrument air supply to test port on bypass panel and apply air pressure.

Initials _____ Date: _____

9. Isolate the Converters Flare Vent Valves (V-5379 and V-5376). This is accomplished as follows:

- Remove the car-seal and close the manual block valve located directly upstream of the automatic block valves (V-5379 and V-5376).

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Annex AA — Example of a testing documentation form for off-line tests

(Example on following page.)

Procedure No.
Revision Date
Page _ of _

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INST. NO.	SERVICE	PROCESS SETTING	DEVICE SETTINGS	FAILURE LIMITS	AS FOUND	AS LEFT	Failed? (Mark with •)
XV-5083	**LEVEL, 1ST. STG. SUCTION DRUM.**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-7092	**LEVEL, . STG. SUCTION DRUM.**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-7104	**LEVEL, 3RD. STG. SUCTION DRUM**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-7128	**LEVEL, 4 TH. STG. SUCTION DRUM.**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-7132	**LEVEL, 4 TH. DISC. SUCTION DRUM**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-8505	**LUBE OIL PRESSURE**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-8506	**TRIP RELAY FOR MANUAL S/D**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-8511	**MAIN HEADER TRIP RELAY**		**15# Dec.**	**13.5# DEC. TO 16.5# DEC.**			
XV-8701	**LEVEL, 1ST. CASE SEAL OIL POT.**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-8702	**LEVEL, 2ND. CASE SEAL OIL POT.**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-8703	**LEVEL, 3RD. CASE SEAL OIL POT.**		**TRIP 3# DEC. RESET 10 INC.**	**TRIP 3# DEC. RESET 10 INC.**			
XV-8909	**LOW GOV. OIL PRESS. S/D RELAY**		**15# Dec.**	**13.5# DEC. TO 16.5# DEC.**			
XV-8910	**LOW SUCT. DRUM PRESS. S/D RELAY**		**15# Dec.**	**13.5# DEC. TO 16.5# DEC.**			
PI-5083	OUTPUT OF LS-5083 ON S/D BOX		**0# 20#**	**0# TO 2# 18# TO 22#**			
PI-7092	OUTPUT OF LS-7092 ON S/D BOX		**0# 20#**	**0# TO 2# 18# TO 22#**			
PI-7104	OUTPUT OF LS-7104 ON S/D BOX		**0# 20#**	**0# TO 2# 18# TO 22#**			

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Annex BB — Model SIF testing policy statement

The policy related to SIF testing shall apply to the SIF installed at this facility unless approved in writing by the facility safety review committee.

Policy Statement:

1. There is a requirement that our Safety Instrumented Functions be tested from the sensor all the way through the final control element. Some systems may require on-line test capability since they are normally operated longer than the one-year nominal test interval.

   It is understood that in some applications, exercising the final control element (control valve, motor, etc.) is not practical while the unit is running. In these applications, provisions shall be made to test the system all the way through the solenoid valve or motor starter interface relay. These final control elements shall then be exercised at the first opportunity (i.e., during unit turnaround).

   Any by-pass system installed to enable on-line testing will have safeguards installed to ensure the system is not accidentally defeated or left in the by-pass position. This shall include alarming when in the bypass position, use of key lock switches, written procedures regarding bypasses, etc.

2. If a SIF has failed its proof test in two consecutive tests due to the same problem, a recommendation shall be made to location management for a specific corrective action plan. One part of this plan is a root cause analysis of the problem. Note that just replacing a failed component is not sufficient. If further data is needed to identify the problem or to assure that the problem has been eliminated by the corrective action, an adjustment in the proof-testing interval may be recommended.

3. The following will be used in the future as a definition of a "Failed Proof Test." (Note that Proof Test and Functional Test are the same test.) A Failed Proof Test is defined as a test result indicating that the system is not functioning within the defined process variable tolerance and may not be performing to its designed specifications. A default value of +/- 10 percent of the process variable setpoint shall be used unless the test procedure specifies a more specific tolerance value.

   E.g., a pressure transmitter was calibrated from 0-100 psi with an 80-psi high pressure trip setting. If this system tripped within 10% of 80 psi (e.g., between 72 psi and 88 psi), this system has successfully passed its proof test. The intent is that the proof test be conducted before any repairs or modifications are made to the system.

   The following definitions apply to redundant inputs. On systems with a 1oo2 input architecture, if one of the transmitters passes the above proof test, then the system is defined as passing. In this case, one of the transmitters may have failed but the system would still have functioned as designed. On systems with a 2oo3 input architecture, if two of the transmitters pass the proof test requirements, the

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system is defined as passing.

4. Reports outlining the results of proof tests shall be sent to the facility safety review committee within 30 days of a test. The report shall state the systems performance as well as any deficiency. These reports shall be filed with the SIF documentation for a period of three years.

5. All SIF are required to be functionally tested in accordance with a test schedule based on the SIL determination criteria for the SIF. The test schedule should indicate the month (schedule month) and year in which the next function test is to be performed. The test due date is the last day of the scheduled month. A test performed any time within the scheduled month is considered "in compliance."

   If a test is performed prior to its scheduled month, the test is considered as being "in-compliance." But the system must be either retested in its originally scheduled month or the scheduled month must be changed to the month in which the test was actually performed. If changed, the new scheduled month will then be used as the basis for scheduling subsequent tests.

   If a test is performed after its scheduled month, the test is considered "out of compliance with proof testing interval" until the test is performed unless the test is formally deferred (see Annex B). The scheduled month, though, would not need to be changed for subsequent tests because it would still fall within the required test interval in the next test cycle. The scheduled month may be changed to the month in which the test was actually performed to take advantage of the entire allowed test interval, if so desired.

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Annex CC — Possible SIF performance metrics

The following metrics may be good indicators of SIF performance. These metrics could be tracked and reported on a quarterly or annual basis using a spreadsheet format.

- SIF Availability calculated using one of the approved methods in ISA-TR84.00.03-2002 and SIF test results. Only the number of SIF functional tests performed and number of SIF tests failed are required. These numbers could be accumulative totals for the past three year period.
- Number of SIF identified and classified by SIL by PHA.
- Number of SIF evaluated against SIL requirements.
- Number of SIF that meet SIL requirements.
- Number of SIF successful trips and, where feasible, estimated $ savings.
- Number of unsuccessful trips and actual $ cost.
- Number of covert failures discovered during testing that could have resulted in high consequence event if a SIF demand had occurred and, where feasible, estimated potential $ impact.

SIF Availability Calculations

The SIF performance capability should be defined by one of the three calculational techniques outlined in ISA-TR84.00.02-2002. A technique should be selected and all SIF evaluated using the same technique.

Failure Mode Concepts

Failures in SIF can occur both overtly and covertly. Overt failures typically reveal themselves by tripping all or part of the SIF. An example would be a normally open fail closed trip valve closing when its solenoid valve fails resulting in a process upset. The operator would be quickly aware of the failure. If the process is still running, the operator is aware of the failure and can perform mitigating actions to simulate the SIF function and respond to demands while the SIF is inoperable. So, overall availability of the safety function is not greatly affected by overt failures unless the failures are very frequent (MTBF < 1 year).

Covert failures do not reveal themselves and do not affect the operation of the process. They are potentially hazardous because they may not allow the SIF to perform a safety function should a hazardous demand occur. The operator is unaware that the SIF is inoperable and is not in state of readiness to respond to a demand should one occur. Some covert failure modes can be turned into overt failure modes by using system diagnostics to reveal the failure. However, system function testing is generally required to reveal and correct covert failures. By their nature, covert failures have the greatest impact on SIF availability because they can go long periods of time in an unrevealed inoperative state.

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Availability calculations

Whichever method is chosen to perform the SIF availability calculations, a common set of failure rate data should be used. This data should be agreed upon by a team of facility personnel who have much experience with the equipment used in implementing SIF. All SIF calculations should use only the agreed upon database.

What is considered a system failure?

In simplest terms, a system should be considered to have failed if it cannot perform the safety function for which it has been designed. First, it presumes that you know safety function the system was designed to perform. There should be a clear description in the unit Process Hazards Analysis of the scenario or hazardous event the SIF was designed to prevent. Next, system component failures should not be considered system failures if they are not in the chain of devices and logic that perform the safety function. Failures of alarms, system resets and diagnostic components usually do not prevent the system from providing the safety function when needed. Increasing system availability may require the use of redundant components. A failure of a single transmitter in a two out of three voting triad should not be considered a system failure since the other transmitters are still available to perform the safety function.

Transmitter or switch drift should be considered a source of system failures if the drift is beyond the acceptable safety tolerance for that system. The tolerance will vary from system to system based on the process hazard and how close the trip point is to the point of hazard. The tolerance on the hazardous side of the trip point may be different than the tolerance on the nuisance side of the trip point. A general guideline might be to set the acceptable tolerance no more than (+) or (-) 10% of the process trip point and at least 5% on the safe side of the point of hazard.

Trip valves which fail to fully stroke when tripped should be considered system failures. Trip valves which leak through when fully closed may or may not be considered failures depending on the process. Many processes can tolerate some amount of leakage through the trip valve and still mitigate the hazardous event. Some processes require tight shut off to prevent the hazardous event. A leak tolerance should be designated for each trip valve. Valve leak testing may be required to ensure process leakage is within tolerance for tight shut off valves.

Plugged impulse lines on transmitters should be considered failures.

Any logic device or switch which fails and prevents any SIF output from tripping when a SIF trip initiator trips should be considered a system failure.

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Annex DD — Model technique for testing SIF valves on-line

How can functional tests of SIF valves be conducted in a long run-time plant?

1. Install manual Bypass Valve. Prove stroke and inspect internals. Operate plant on Bypass Valve while doing test and inspection.

2. Exercise valve for one stroke with plant operating. Use Valve Diagnostic tool to determine valve health.

   - May or may not require Bypass Valve.

   - Portable Diagnostic tool able to detect actuator and mechanical linkage problems plus detect if leakage is significant.

   - Tool available for purchase or as a service from valve vendors.

3. Install redundant valves for a SIL 1 application and extend TI to match plant turnaround schedule.

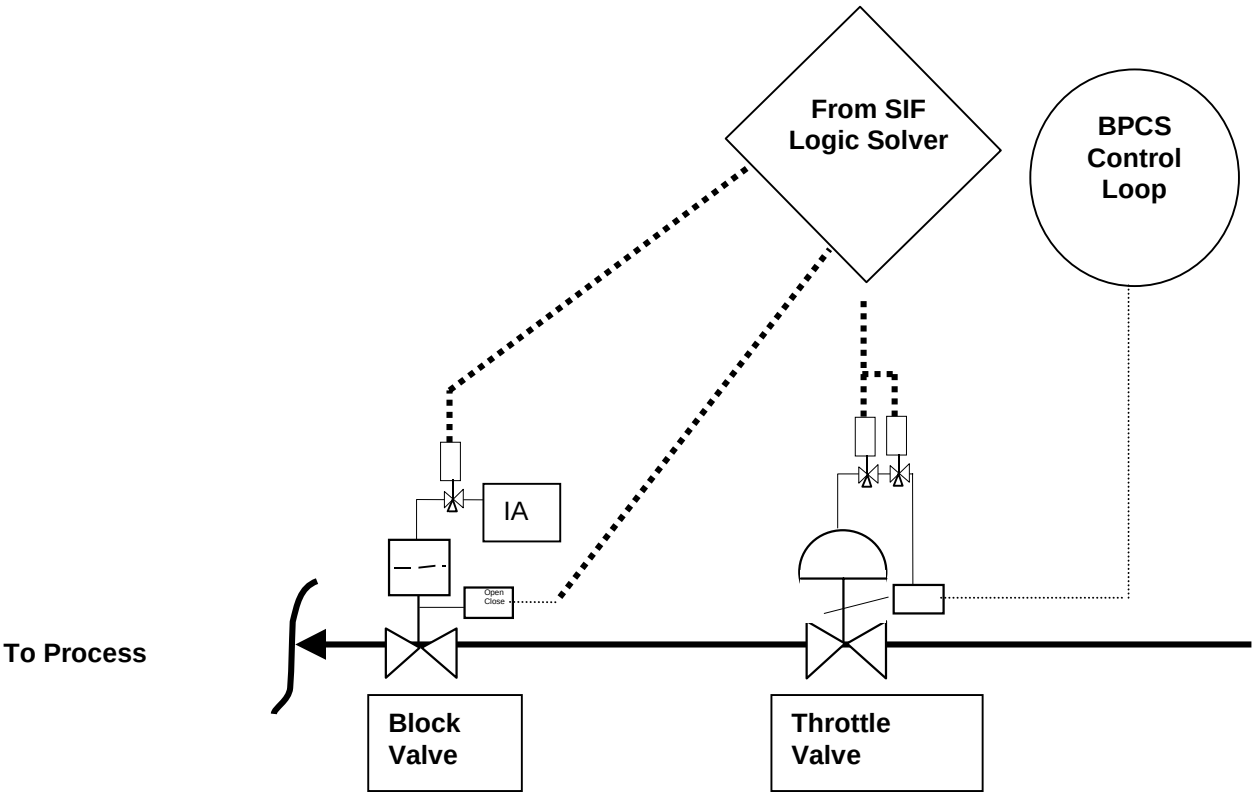
An SIF BV and a shared BPCS throttle valve with redundant SIF solenoid valves provides the maximum SIF Test Intervals. This results from the effect of operator-provided diagnostics for the throttle valve. The valve configuration is shown below.

Procedure No.
Revision Date
Page _ of _

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Annex EE — Automated testing of SIF valves on-line

AutoTest (AT): Requirements

- Hardware
  - ESD Full Flow Bypass Valves for Normally Open Valves
  - ESD Block Valves for Normally Closed Valves
  - ESD Valve Limit Switches
  - SOV Limits Switches
- Software
  - SIF Vendor Auto Test Code
  - DCS Interface Read / Write Points to Start, Abort & End AT.
  - DCS Interface Read Only Points to Report Results & Time Stamp
  - DCS Graphics for AT
- Two Types of AutoTest
  - Logic Auto Test: Logic Test Only w/o Tripping Final Control Elements
  - Trip AutoTest: Tests the Final Control Element Action

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Logic AutoTest (AT): Steps

Furnace Low Pressure Transmitters (2oo2)

1. Operator Calls “Logic Test” Display for the Transmitter Pair on the Appropriate DCS Graphic.
2. Operator Selects “Logic Test” Target if Visible and then “OK”.
3. Target Turns Green.
4. Process Pre-trip & Trip Setpoints are Replaced with Auto Test Trip Setpoints (a fixed percentage (3%) higher than current process value)
5. SIS Sets Alarm Flags in DCS (I.e. Pre-Trip, Trip, First-Out, Marks for Associated Effects on Cause & Effect Matrix).
6. SIS resets Logic Quick Test.

Notes:

a. No Final Control Element is Tripped.

b. Test only validates ESD Logic Functions.

Trip AutoTest (AT): Steps

SETUP STEPS: Furnace Fuel Gas ESD Valve

1. Operator Manually Opens ESD Bypass Valve.
2. SIF Checks: Final Control Element Status (Open / Close), SOV Status on ESD Valve, Bypass Valve & SOV’s.
3. Trip Test Permissive Target is Visible if Permissives Met.
4. Operator Initiates Auto Test for each SIF Final Control Element via DCS Graphic (Trip Test Target).
5. Pop Up Window: “Press OK to Test” - “OK” or “Cancel”
6. “OK” Selection Instructs SIF to Initiate Auto Test.
7. If Setup OK in Field - “Trip Test” Target turns Green - Test Executed.

AT EXECUTION STEPS

1. SOV A is de-energized.
2. SOV A is re-energized & SOV B is de-energized.
3. SOV A & SOV B are Simultaneously De-energized.
4. ESD Valve Trips
5. SIF Checks States of the ESD Valve & SOV’s.

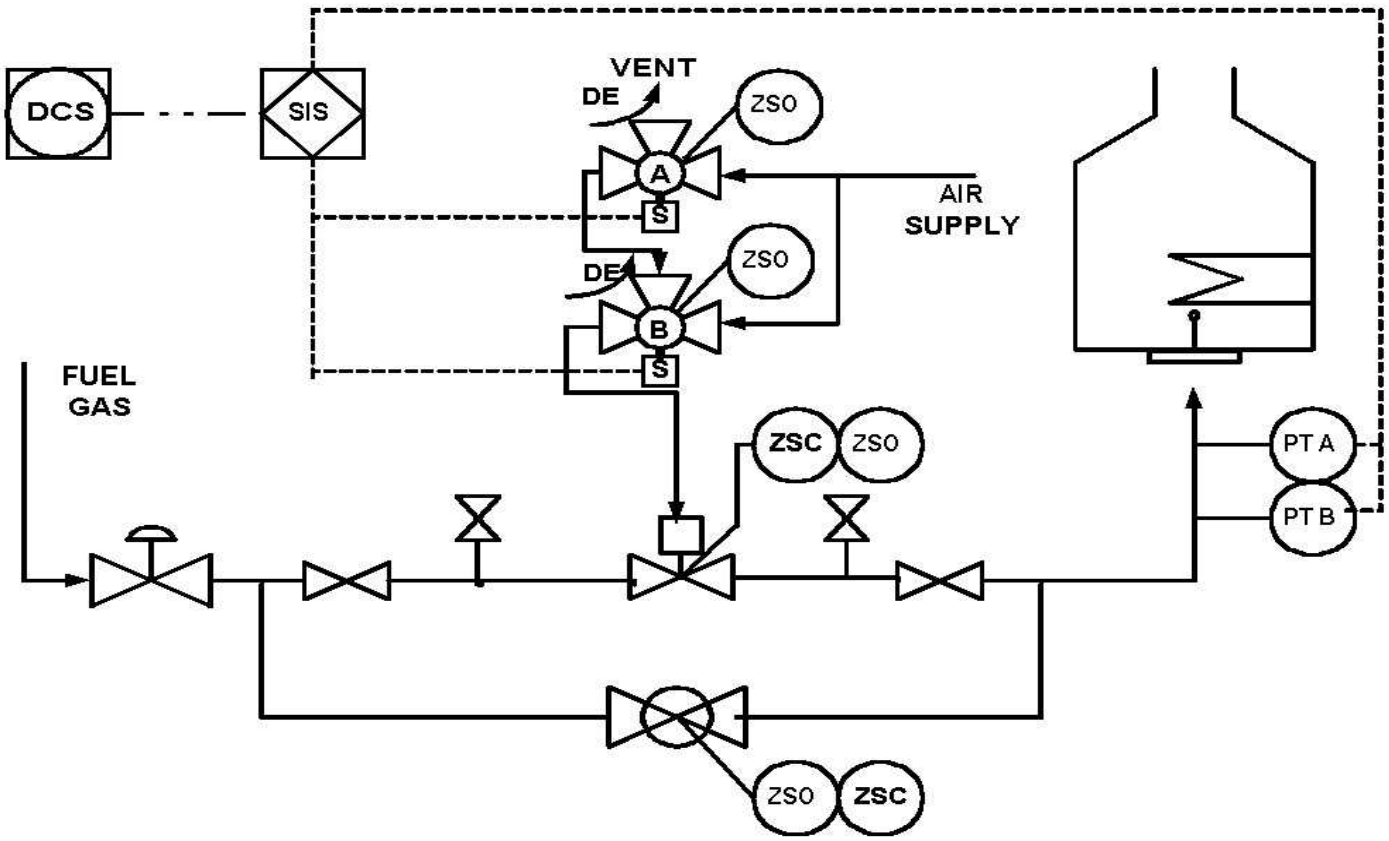
Procedure No.
Revision Date
Page _ of _

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Auto Test Example



Procedure No.
Revision Date
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Annex FF — Possible audit protocol for safety instrumented functions

The following documentation shall be available for the Audit Team at time of audit:

- Copies of SIF Manual for system being audited
- Copies of all plant policies related to SIF
- Copies of all SOPs related to SIF being audited
- List of key personnel responsible for SIF being audited
- Key plant contact during audit _____
- Copy of change logs and history logs of system being audited if not contained in SIF manual

SIF to be audited _____

Audit Team Members: _____ **Location:** _____

_____ _____

_____ _____

_____ _____

_____ _____

Scope of Audit: This audit of the SIF specified above covers the following:

- SIF Documentation
- SIF Procedures
- Adherence to General Design Requirements for SIF
- Validation of SIF Function both before system startup for the first time and maintaining the system's capability

Procedure No.
Revision Date
Page _ of _

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I. Review documentation for SIF

Issue	Standard Reference	Finding	Auditor
A. SIF Manual			
1. All copies are the same			
2. Contents of manual NOTE All of the following documents do not have to be in the same manual (binder), but they must be readily available for use if required.			
a. TOC or Index			
b. Drawings describing shutdown system (list available)			
c. Narrative description of shutdown system			
d. Simple block schematic of shutdown system (optional)			
e. List of Pre-Alarm and S/D set points			
f. Copies of change authorizations with approvals			
g. Copy of change procedure			
h. Copy of Functional Test Procedure			
i. Indication of required manual test frequency			
j. Copies of any bypass procedures required			
k. Bypass procedure approvals			
l. System audit records			
m. Copies of system availability calculations, if appropriate			

Procedure No.
Revision Date
Page _ of _

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I. Review documentation for SIF (con't)

Issue	Standard Reference	Finding	Auditor
B. Other Documentation			
1. Copy of history register (log) of events associated with system, i.e., trips, equipment failures, etc.			
2. Copy of system configuration, i.e., equipment arrangements with Rev. numbers, Serial Numbers, etc.			
3. Copy of Functional Requirements Specifications (may be several documents)			
a. Description of each SIF system initiators purpose and function in system			
b. Description of logic requirements			
c. Description of actions system must take and how this is accomplished			
d. Describe requirements related to operator interface			
e. Description of other requirements as appropriate			
C.Documentation Control Procedures			
a. Identification of responsibility for maintenance of documentation			
b. Number of copies of documentation controlled			

Criteria to consider in audit: Appropriateness of documents, number of copies of documents maintained, completeness of documentation, clarity of documentation, accessibility of documentation, and identification of documents as being a part of a SIF.

Procedure No.
Revision Date
Page _ of _

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II. Review of Procedures Associated with SIF

Issue	Standard Reference	Finding	Auditor
A. Personnel responsibility			
1. Process familiarity			
2. System familiarity			
3. Design standards familiarity			
4. Peer review of design			
B. Design, Review and Approval			
1. Design Criteria Followed			
a. WDT, if appropriate			
b. Independent Trip Switch			
c. No Automatic Reset			
d. No Blind Initiators			
e. Failure alarms (opposite direction to trip)			
f. Power separation			
2. Initial design review			
C. Management of Change Procedures			
1. Set Point changes			
2. Logic changes			
3. Vendor software changes			
4. Valve action changes			
5. Hardware changes			
6. Wiring changes			
7. Testing frequency changes			
8. Process changes			

Procedure No.
Revision Date
Page _ of _

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II. Review of Procedures Associated with SIF (con't)

Issue	Standard Reference	Finding	Auditor
D. By-pass Procedures			
1. No master bypasses			
2. Number of bypasses minimized			
3. Permissives controlled			
4. Bypassing only during stable operation			
5. Acceptable bypass methods			
6. Evidence of training on bypassing			
E. Operating SOPs Available			
1. Readily Accessible			
2. Understood by operators			
F. Maintenance SOPs Available			
1. Readily Accessible			
2. Understood by technicians			
3. Appropriate for components being maintained			
4. Cautions about working on and around Safety System equipment			
G. Availability of system spare parts			
H. Records of any internal audits performed			

Criteria to consider in audit: Appropriateness of procedures, appropriate levels of experience involved in design, evidence of adherence to procedures, frequency of audits, understanding of procedures by operations, maintenance and engineering personnel, qualifications of those approving changes, and evidence of enforcement of procedures by management.

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III. Use of Approved Equipment for SIF

Issue	Standard Reference	Finding	Auditor
A. Field Components			
1. Sensors			
2. Valves			
B. Logic Solvers			
C. Software			
1. Configuration software			
2. Vendor software Version			

Criteria to consider in audit: Conformance to approved vendor list for components, use of approved vendor revision levels for internal software, use of approved configuration software, and appropriate approvals for any deviations.

IV. Separation between BPCS and SIF

Issue	Standard Reference	Finding	Auditor
A. Sensors either separate or redundant			
B. Logic separation			
C. Software separation			
D. I/O conversion separation			
E. Final control element separation			
F. Logic Solver programming station separation			
G. Operator Interface separation			

Procedure No.
Revision Date
Page _ of _

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V. Validation of SIF Functions

Issue	Standard Reference	Finding	Auditor
A. Field I/O Verification			
1. Proper installation			
2. Wiring connections			
3. Valves			
a. PM schedule in place			
b. Record of maintenance			
4. Visual inspection of field devices			
B. Functional Test Procedure			
1. Written Procedure			
2. Specific to this system			
3. Manual frequency specified			
4. Forms for recording data			
a. All components included in test			
b. As found condition			
c. As left condition			
5. Test techniques identified and followed			
6. Copy of last functional test performed available			
7. Tests of approved changes included			
8. Identification of who is authorized to perform test			
9. Test equipment appropriate			

Procedure No.
Revision Date
Page _ of _

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Annex GG — Example of checklist for auditing an SIF

(a) Is there a register, schedule, or listing of all Safety Functions included in the SIS? Is it up to date?

(b) Do written test procedures exist for SIF?

(c) Are the tests regularly reviewed to ensure that they meet the current standards and operational requirements?

(d) Do the tests check that the whole system operates correctly?

(e) Is the purpose of each system recorded and is this reflected in the system Integrity Level?

(f) Are settings and the rational for them recorded?

(g) Has consideration been given to the behavior of systems outside their normal operating boundaries?

(h) Are changes to equipment, settings, test procedures, and test intervals made via an established management of change procedure?

(i) Is the test schedule up to date? Do you inspect it and take action on reports of overdue tests?

(j) Is there a formal SOP, which takes full technical consideration of the consequences, for the bypass or defeat of safety systems?

(k) Are defects in safety systems repaired quickly?

(l) Are all safety systems tested before being returned to service after repair or modification?

(m) Have process and maintenance personnel received the training necessary to operate, test, and repair the SIF so as to maintain their design intent and performance?

(n) Do operators and supervisors understand the correct operation of the systems is a part of their responsibilities?

(o) Have any operational difficulties or incompatibilities between the plant operation and safety system performance been reported and acted upon?

(p) Are audits carried out which establish if the questions on this list are answered?

(q) Is there documentary evidence to support the answers?

Procedure No.
Revision Date
Page _ of _

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Annex HH — Partial instrument trip test (PITT)

INTRODUCTION

In process plants, valves employed for shut off applications remain open while the process is in safe and controlled state. These valves close only upon a plant trip arising from an out of control process or during a normal maintenance outage. The performance of such valves is tested only during the shutdown condition of the process. Economic considerations have driven plant operators to reduce the frequency of maintenance outages extending continuous operation of plants for many years. State of the art SIF have extensive features to detect incipient failures within them and redundancy to offer a high degree of availability. However, the shutoff valve, which is one of the critical elements in the SIF loop, typically does not have any means of ensuring availability when a demand arises. The availability of the shutoff valve can be enhanced by periodic partial stroking of the valves on-line without causing process upset. Almost all SIF valves have pneumatic cylinder actuators driving the valve to a closed state quickly on the loss of the pneumatic supply. A combination of 3-way solenoid valve and quick exhaust valve controls the pneumatic drive. On a trip signal the solenoid valve de-energizes cutting off air supply to the cylinders. The quick exhaust valve vents the cylinder driving to close the valve.

Partial Stroking Of Shut-Off Valves

Partial Instrument Trip Testing applied to shutoff valve is a scheme of partial stroking of the valve to ensure its functionality without causing process upset or shutdown in the process plant. The scheme as indicated in the figure was designed, developed, and tested for on-line implementation of Partial Instrument Trip Test on shutoff valves.

Under normal operating condition the main trip solenoid valve remains energized passing air supply through quick exhaust valve to the cylinder of the actuator keeping the valve open.

The PITT solenoid valve, which remains de-energized normally, is energized to initiate a partial stroke test. Energisation of PITT solenoid valve causes partial bleeding of the air supply to the shutoff valve actuator causing the valve to move from its open state. The PITT will be terminated either on travel of the valve about 10% sensed by 10% limit switch or after a predetermined time.

In case of a trip during the test the main solenoid valve will cutoff the air supply and the cylinder will be vented through both the quick exhaust valve as well as the PITT valve.

The travel time during the 10% limit during PITT can be used for monitoring the stroke performance of shutoff valve.

The 10% travel limit actuation during PITT is an indication of the success of the test.

The logic for conducting the PITT is implemented in the SIF system and all information related to PITT is transmitted to BPCS for report generation and archiving purpose.

Procedure No.
Revision Date
Page _ of _

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Salient Features of the Scheme

1. PITT is independent.
2. PITT action will not hamper the trip action.
3. Action of PITT solenoid valve improves travel time of shut off valve on a trip.
4. Any failures in PITT solenoid valve will not effect trip action.
5. In the event of failure of main trip solenoid valve, the PITT solenoid valve will act as a backup to close the valve.
6. Adjustable travel time during PITT.
7. Automated hardcopy report generation as a proof of successful valve test.
8. Facilitates on-line maintenance of PITT solenoid valve.
9. Increase in the frequency of valve test leading to early detection of incipient failures.

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GLOSSARY

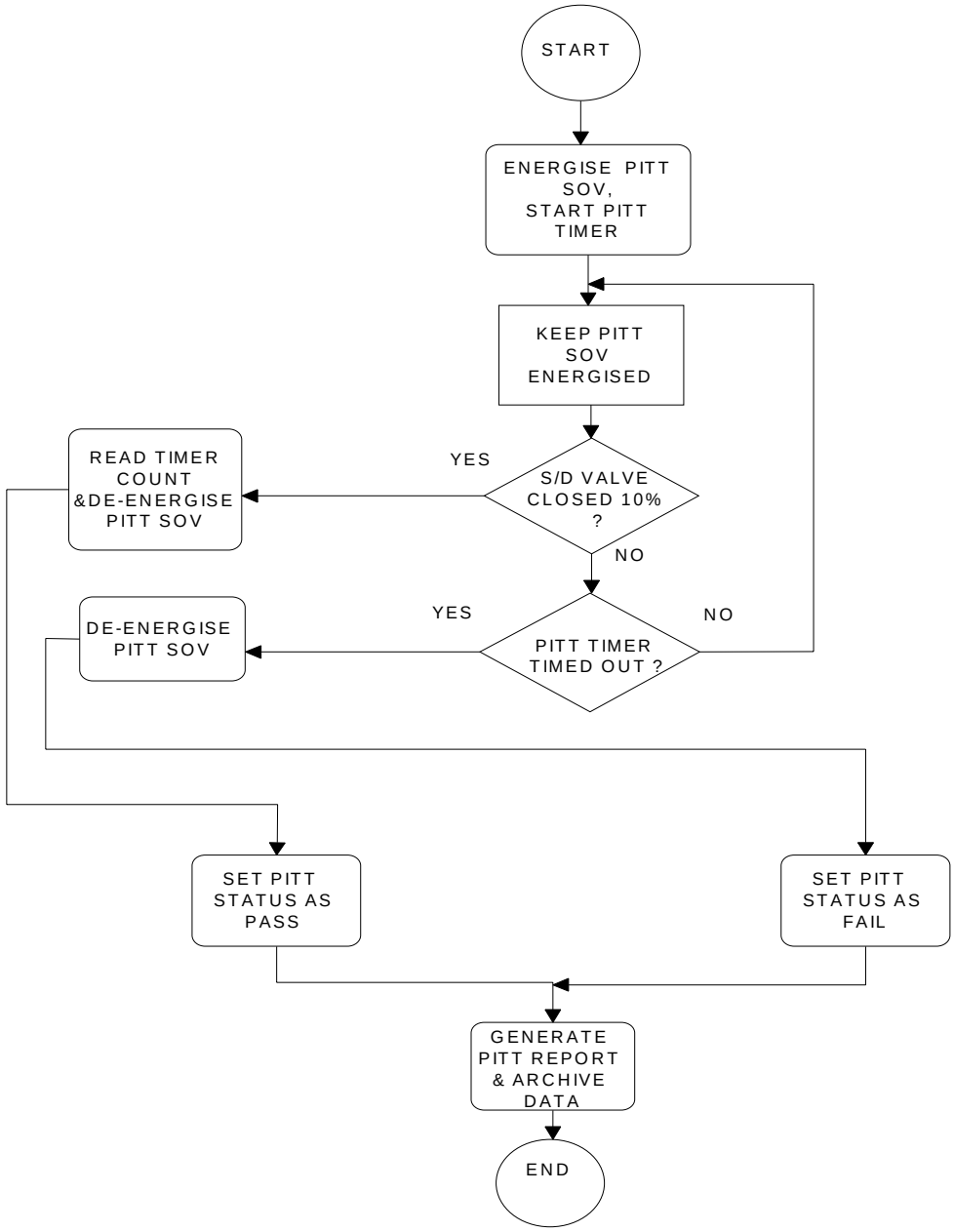
PITT	**Partial Instrument Trip Test**
ESD System (Emergency Shutdown System)	Emergency Shutdown system, which shuts down the plant to a safe state in the event of any out of control processes. The system is also used for PITT of shutoff valves periodically.
Shutdown Valve	Shutdown valve is a safety device which remains open and will close (fail-safe position) in case of trip/shutdown. PITT is performed on this valve.
Main Solenoid Valve	Main Solenoid valve is the safety device on the SHUTDOWN VALVE which is normally energized. De-energizes to vent air through exhaust port to close Shutdown valve on trip/ shutdown.
PITT Solenoid Valve	PITT Solenoid valve is the test solenoid valve to perform PITT. It is independent of main ESD solenoid valve. The partial closing is achieved by energizing the PITT solenoid valve for partially bleeding the air supply to achieve predetermined valve closing of approximately 10%. PITT solenoid valve energizes on trip signal complementing the exhaust valve to improve the speed of shutoff valve closure. Since the PITT solenoid valve is programmed to energize on a trip it acts as a backup to the main solenoid valve.
Quick Exhaust Valve	It is a pneumatic actuated valve. It allows the SHUTDOWN valve to close very quickly (<1 sec) by bleeding the actuator pressure through its exhaust port.
Isolation Valve	It isolates the PITT Solenoid for any maintenance. It is also useful to control test travel time/stroke by throttling (adjusting the bleed rate).
100% open limit switch	Valve open status
Close limit Switch	Valve close status
10% close limit Switch	10% Valve close status when PITT is on.
PC with Printer	To monitor/ record the program and timings.

Procedure No.
Revision Date
Page _ of _

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PITT ROUTINE LOGIC FLOW



NOTE.: PART OF THE ESD APPLICATION SOFTWARE. TO BE EXECUTED ON INITIATION OF PITT REQUEST. DOCUMENT N0. 4571-00-16-51-4091B .

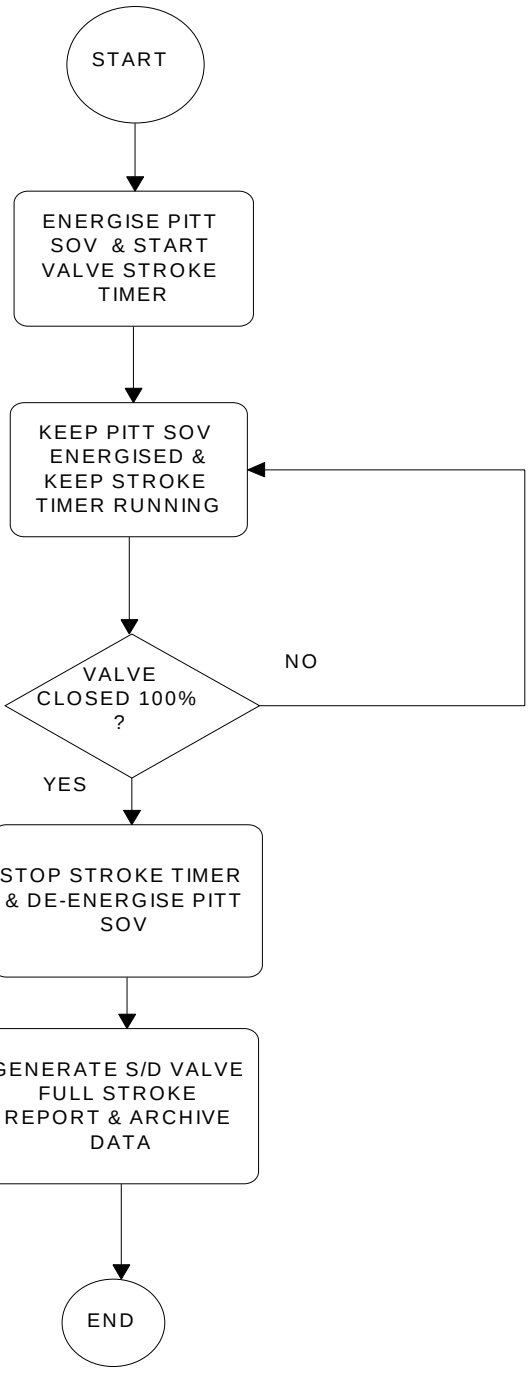
Procedure No.
Revision Date
Page _ of _

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ROUTINE TO ENHANCE ACTUATOR BLEED ON A TRIP
LOGIC FLOW



NOTE. PART OF THE ESD APPLICATION PROGRAM. INITIATED IN THE EVENT OF A TRIP SIGNAL

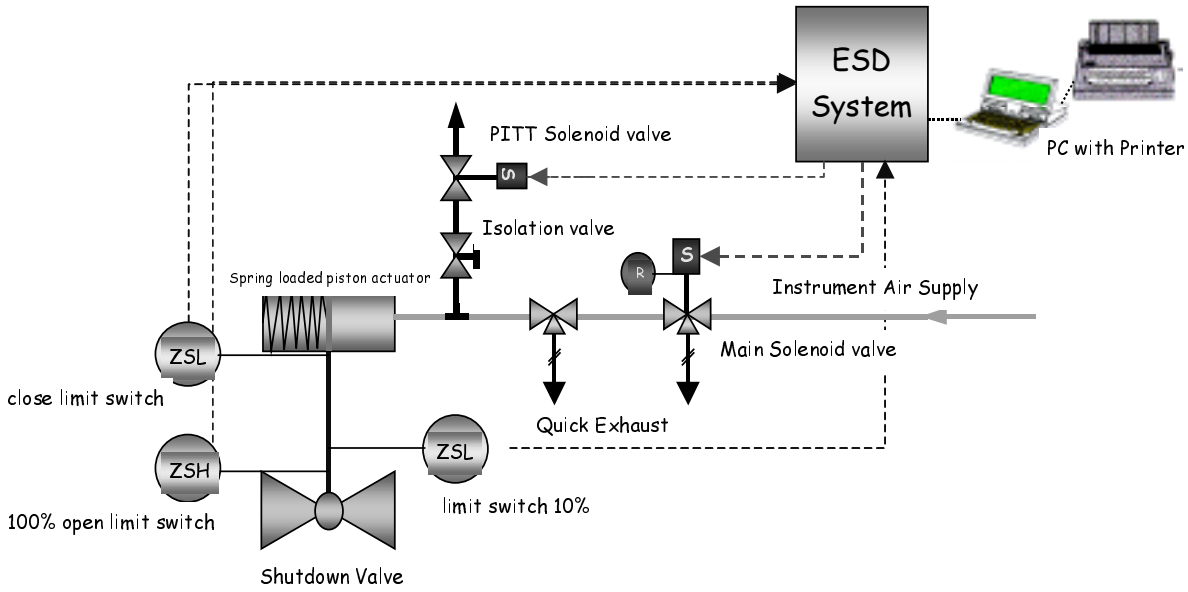
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Procedure No.
Revision Date
Page _ of _

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Partial Instrument Trip Test (PITT) - Schematic



Procedure No.
Revision Date
Page _ of _

--`,,`-`-`,,`,,`,`,,`---

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Annex JJ — Vendor packages to perform partial stroke testing of SIF valves

There are a number of valve manufacturers who now provide a package system for performing diagnostics and partial stroke testing of both sliding stem and 90º turn valves that may be used in SIF applications. The listing, which follows, does not claim to be the only manufacturers available to do this. It is just the listing of companies who submitted information related to testing to the committee developing this document. A brief description of what each system provides is included with the vendor information for clarification.

Neles Automation

Neles offers a package called the ValvGuard System, which provides automated testing of a valve by performing a partial stroke of the valve, and measuring valve position as related to air pressure in the actuator. A “fingerprint” of the valve can be obtained and compared with original condition of the valve for analysis of any potential problems. The vendor claims third party certification of their product and estimates that > 85% of the time the valve will perform the function required of it by the SIF if appropriate maintenance is performed.

Contact the North American subsidiary at 42 Bowditch Drive, Shrewsbury, MA 01545-8004, telephone number 1-508-852-3567.

Tyco Valves & Controls

Tyco offers a package called K-MOVE™ (Manually Operated Verification Equipment), which allows testing valves without shutting them down. The system works only with rotary action valves at the present time. The package moves the valve about 20º to minimize the impact on flow through the valve. It is possible to have the SIF initiate the test and information can be fed back that the test has been performed.

Tyco can be contacted at 9700 West Gulf Bank Road, Houston, TX 77040, and telephone number 713-466-1176.

DRALLIM Controls

Drallim offers a non Contact Real Time Testing and Monitoring system for emergency isolation valves and associated control devices called VALVEWATCH. They claim that due to the speed of the test action that in some cases full closure of the valve may be possible.

Drallim can be contacted at Drallim Industries Inc., Grogans Mill Rd, Suite 125, The Woodlands, TX 77380, telephone number 261-296-1665.

Siemens

Siemens offers a smart valve positioner that provides diagnostic capabilities with the information readily available using the HART protocol.

Siemens can be contacted at Siemens Energy & Automation, Inc., Process Industries Division, Mail Stop 510, 1201 Sumneytown Pike, Spring House, PA 19477-0900, telephone number 215-646-7400.

Emerson Controls

Emerson Controls, formerly Fisher-Rosemount, offers a valve diagnostic package called FIELDVUE DVC6000 for Safety Instrumented Systems.

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For information contact Emerson Process Management - Fisher Controls Division, 205 South Center Street, Marshalltown, IA 50158, telephone number 641-754-3011.

Industrial Control Specialists

Industrial Control Specialists has developed a technique called “Shurshut” for testing a control valve used in a SIF application while the process is in operation.

Industrial Control Specialists may be contacted at 1320 Gauthier Road in Lake Charles, LA and telephone number 337-474-3163.

Note that additional vendors will be added when information is received.

Procedure No.
Revision Date
Page _ of _

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Annex KK — Possible technique for evaluating benefit of partial stroke testing of SIS valves in PFD_{avg} calculations

The following presents the procedure that one recognized consultant in the safety arena uses to evaluate the benefit of partial stroke testing of SIS valves in determining the PFD_{avg} for the SIF. Users are cautioned to fully understand this procedure in light of the requirements for the SIF being installed.

Partial-stroke testing can be used to supplement full-stroke testing to reduce the block valve PFD_{avg}. The amount of the reduction is dependent on the valve and its application environment. The partial-stroke test involves moving the valve a minimum of 10-20 percent, which tests a portion of the valve failure modes. The remainder of the failure modes is tested using a full-stroke test. The main purpose of the partial-stroke test is to reduce the required full-stroke testing frequency.

Partial-stroke testing may not eliminate the need for a full flow bypass. If the valve is partial-stroke tested and determined to be non-functional, maintenance will need a bypass or the process will have to be shutdown for valve repair.

How does partial-stroke testing affect the PFD_{avg}? A complete functional test of the valve can be viewed as consisting of two parts: the partial-stroke (PS) and the full-stroke (FS). For the calculation, the dangerous failure rate, λ^D, must be divided into what can be tested at the partial-stroke (λ^D_{PS}) and what can only be tested with a full-stroke (λ^D_{FS}). The resulting equation for the PFD is as follows:

$$PFD_{avg} = \lambda^D_{PS} * TI_{PS}/2 + \lambda^D_{FS} * TI_{FS}/2 \qquad (1)$$

The division of λ^D into parts requires an evaluation of the failure modes of the valve. Table KK.1 provides a listing of typical dangerous failure modes for block valves and the corresponding effect of these failure modes. The test strategy indicates whether the failure mode can be detected by partial-stroke testing or only by full-stroke testing. The equation (1) can then be shown as follows:

$$PFD_{avg} = PD * \lambda^D * TI_{PS}/2 + (1\text{-}PD)*\lambda^D * TI_{FS}/2 \qquad (2)$$

Where the percentage detected (PD) represents the percentage of the total failures detected by the partial stroke test.

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Table KK.1 — Dangerous failure modes and effects with associated test strategy

Failure Modes	Effects	Test Strategy
Actuator sizing is insufficient to actuate valve in emergency conditions	Valve fails to close (or open)	Typically not tested
Valve packing is seized	Valve fails to close (or open)	Test valve – Partial or full-stroke
Valve packing is tight	Valve is slow to move to closed or open position	Not tested unless speed of closure is monitored.
Air line to actuator crimped or plugged vent port	Valve is slow to move to closed or open position	Not tested unless speed of closure is monitored. Physical inspection
Air line to actuator blocked	Valve fails to move to closed or open position	Test valve – Partial or full-stroke
Valve stem sticks	Valve fails to close (or open)	Test valve – Partial or full-stroke
Valve seat is scarred	Valve fails to seal off	Full-stroke test with leak test
Valve seat contains debris	Valve fails to seal off	Full-stroke test
Valve seat plugged due to deposition or polymerization	Valve fails to seal off	Full-stroke test

The failure modes listed in Table KK.1 can be compared to the failure mode distributions presented in the Offshore Reliability Data Handbook (OREDA) for various valve types and sizes. Based on the OREDA data, the percentage of the failures that can be detected by a partial-stroke test is approximately 70%. The remaining 30% of the failures can only be detected using a full-stroke test.

Users are cautioned that this breakdown is based on average valve performance in offshore installations and may not represent the breakdown for the User's application. This evaluation should be done for each valve type, based on the application environment and the shutoff requirements. If the service is erosive, corrosive, or plugging, the failure rate and failure mode breakdown will be different from that shown in this Annex. If the valve is specified as tight-shutoff, the contribution of minor seat deformation or scarring will be more significant than shown in this Annex. For these reasons, it is recommended that partial-stroke testing not used as a substitute for full-stroke testing for a *single* block valve application when:

a) the valve has been shown to fail in the service due to process deposition or plugging,

b) the valve is specified as tight-shutoff for safety reasons, and

c) valve leakage can generate a hazardous incident.

Some analysts choose to neglect the PFD_{avg} associated with the failures detected at the partial stroke test by using the diagnostic coverage (DC) model.

$$PFD_{avg} = (1\text{-}DC)\ \lambda^{D} * TI_{FS}/2 \qquad (3)$$

Procedure No.
Revision Date
Page _ of _

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However, the diagnostic coverage (DC) model is usually reserved for on-line fault detection where the "testing interval" is within or very near the process time constant. For example, comparison of analog transmitter signals is performed each scan and can be alarmed on deviation. This means that the transmitter "test" is performed at least every 150 to 300 ms with a programmable logic controller operating with a reasonable scan rate. When the transmitter PFD_{avg} is calculated, the appropriate diagnostic coverage is selected and used with the failure rate and off-line testing frequency for the calculation. In the case of the transmitters, it is common to neglect the diagnosed portion in the PFD_{avg} calculation, assuming that the operator will be notified immediately that the SIS is degraded (due to failed transmitter), has operating procedures to address safe operation during degraded SIS performance, and has the means and authority to shutdown the operation if necessary.

In contrast to the transmitter, partial stroke tests are typically only performed monthly, quarterly, or annually. This means that there is a substantial time window in which the valve could be in a dangerous, undetected state. Neglecting the partial stroke portion of the valve failure rate can yield substantial error in the calculation. The following is a comparison of the two calculations, assuming 1-year partial stroke testing, 3-year full stroke test, and MTBF of 35 years.

Using DC model:

$$(1-0.70)*(1/35yr)*3yr/2 = 0.0129$$

Using partial test model:

$$(1-0.70)*(1/35yr)*3yr/2 + (0.70)*(1/35yr)*1yr/2 = 0.0229$$

The DC model under predicts the PFD_{avg} of the valve by a factor of 2 at the annual partial stroke test. As the partial stroke test frequency is increased, the error is, of course, reduced. However, even at monthly partial stroke test, the contribution of the PFD_{avg} associated with the partial stroke test is still within the SIL 3 PFD_{avg} range. For the DC model assumption to be correct, the testing must be frequent enough that the PFD_{avg} for partial stroke test is at *least an order of magnitude* lower than SIL 3 (less than 10^{-5}).

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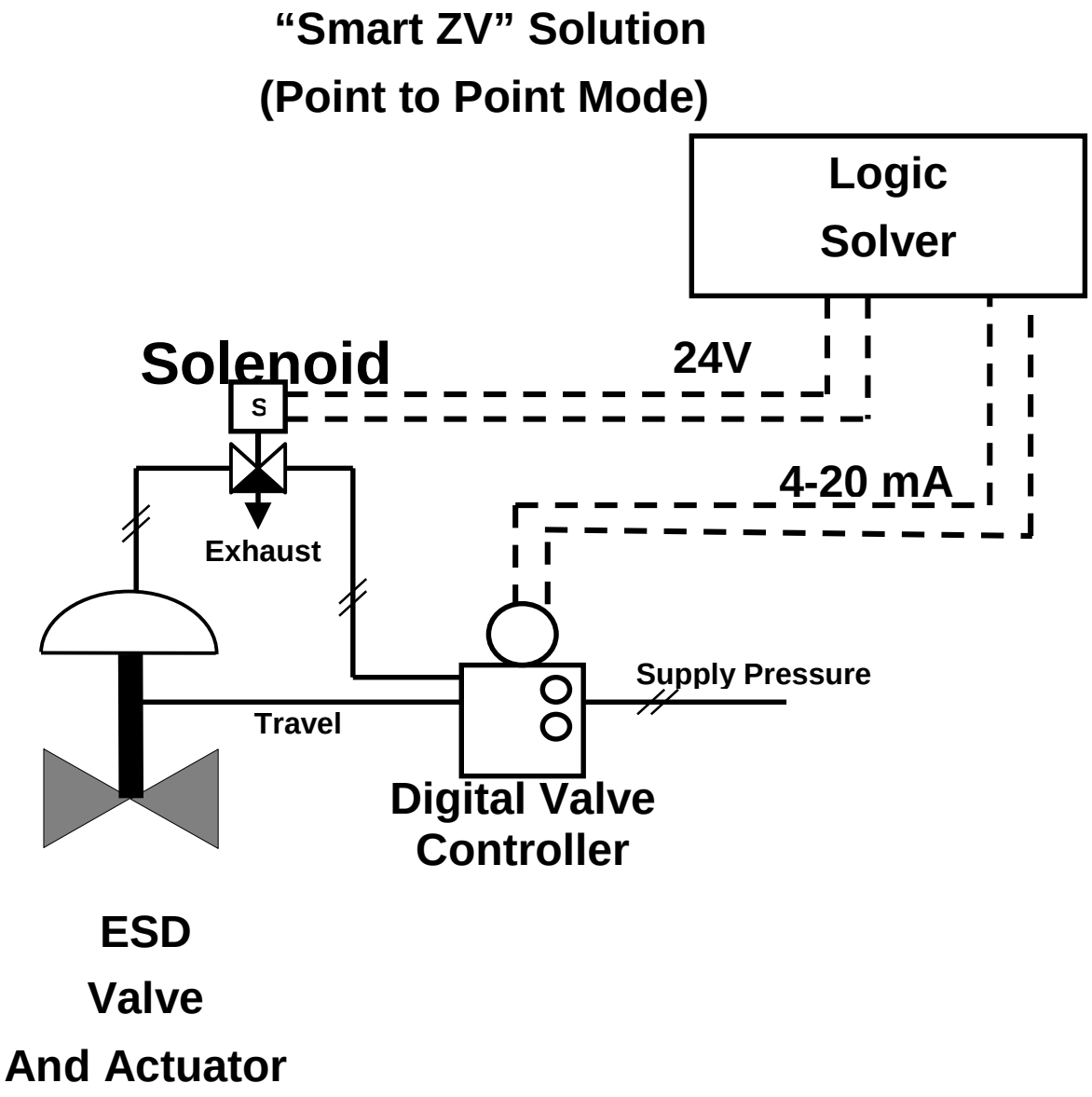
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Annex LL —Example method for partial stroke testing of SIS valves

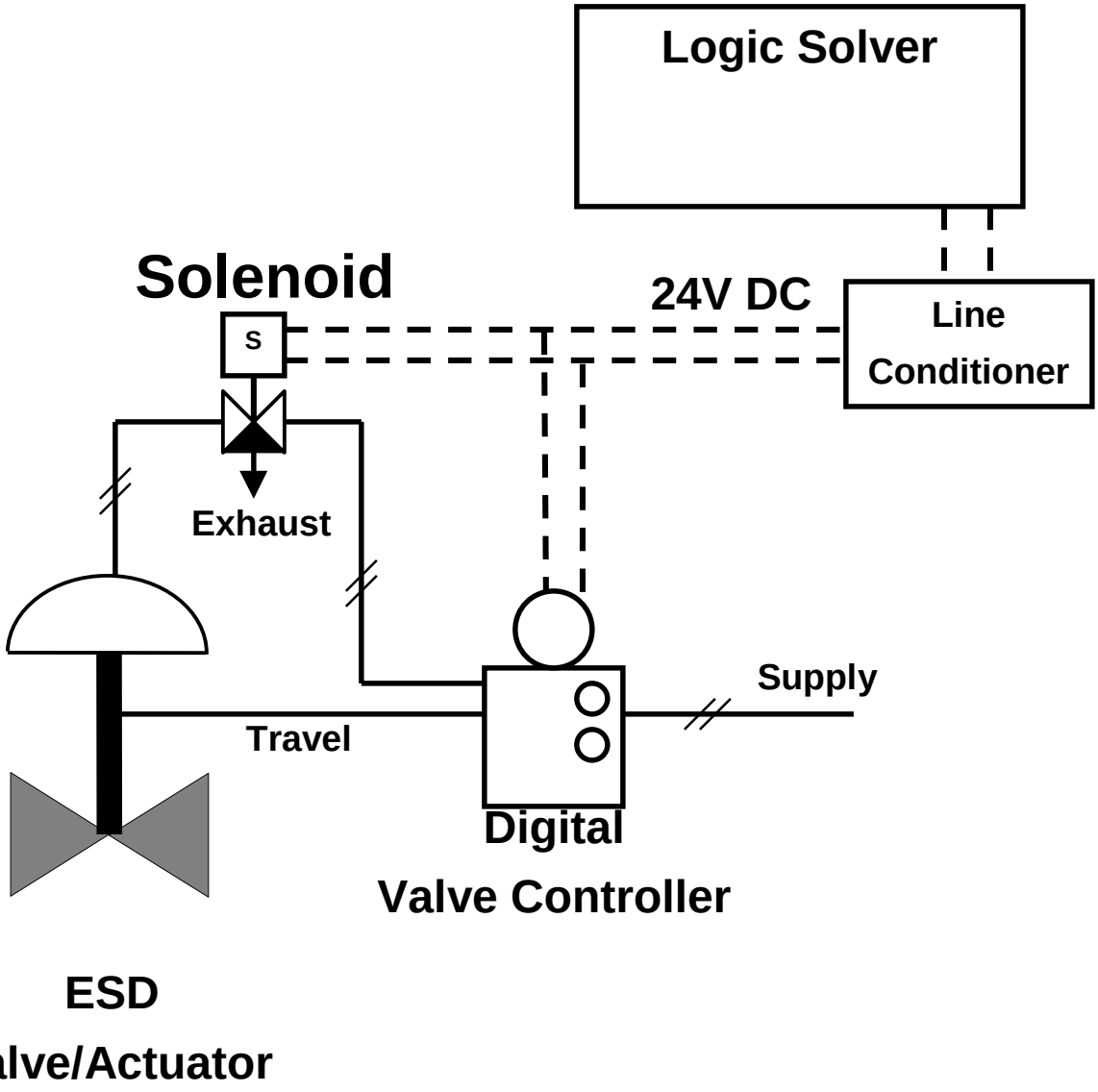


Procedure No.
Revision Date
Page _ of _

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"Smart ZV" Solution
(Multi-drop Mode)



Procedure No.
Revision Date
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Smart ZV Approach

How it works

- **Configuration** – Using the HART handheld communicator or laptop running vendor specific software (Valvelink with Fisher Rosemount DVC 6000), the test parameters are downloaded onto the positioner.

- **Local Test Push Button** – when pressed in the field, the positioner performs the predefined limited travel “partial stroke” test of the ZV. The results of last test are saved in memory on the positioner.

- **ESD Override** – A separate ESD output to the SOV overrides the positioner and drives the valve to the fail safe position.

Best Application

In pneumatic applications single acting or double acting ZV actuators (normally energized or normally de-energized). Ideal where on-line testing is not possible between scheduled T&I’s.

Features

- **Versatile, modular, design** — can handle any ESD signal to the SOV (normally energized or normally de-energized).

- **Continuously monitored** — with the 4-20 mA option, ZV’s are monitored, even after a trip.

Proven performance — installed base in Saudi Aramco has demonstrated reliability.

The smart valve positioner (Fisher Rosemount) is used to perform "limited travel" testing while the valve is in service on a quarterly basis and full stroke the valve annually.

The smart valve positioner collects valve signature data that can be used to compare with previous test results to identify changes in valve performance. In addition, the valve signatures collected during functional testing, provide an audit trail of past functional test results.

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Annex MM — Examples of techniques to perform on-line testing of solenoid valves

There are a variety of methods that can be implemented for on-line testing of solenoids. Each method requires the installation of test facilities and the development of test procedures. Any functional test of a solenoid must determine that the solenoid can vent the air (or other fluids) from the valve actuator. Consequently, the test must determine that the solenoid valve can change states and that the air can be vented through the solenoid vent port to the atmosphere.

The following discussion provides some examples of on-line solenoid testing methods, including brief descriptions of the equipment and procedures. Users are cautioned to fully understand how the field design and test procedures will work in concert to prevent nuisance trips or hazardous situations during testing.

Solenoid in Bypass

A manual test station can be built that uses hand operated valves to bypass the solenoid valve and place air directly on the valve actuator, holding the valve in position. Since this results in the bypass of the final control element, the board operator and field operator must be have a procedure for implementing a safe shutdown in the event of a process demand during the test.

Limit switches are often incorporated on the hand operated valves to allow bypass alarming to the operator HMI. Once the instrument air is in bypass, the solenoid is de-energized and pressure indication is used to determine that the solenoid has properly vented. If 2oo2 solenoid voting is used, no instrument air bypass is required. For 2oo2 voting, each solenoid is de-energized one-at-a-time and pressure is monitored to determine that each solenoid has successfully vented.

Solenoid is Pulsed

In this method, the solenoid is tested by pulsing the power to the solenoid. The operator activates a pushbutton or switch to initiate the test to de-energize the solenoid for as long as the field operator holds the switch. The field operator monitors the valve position and releases the button when the operator confirms valve movement. When the valve moves, it can be inferred that the solenoid successfully vented. Also, if the partial movement of the valve is sufficiently large (10-20%), this test can provide partial stroke testing of the final control element. The main risk is that the operator may hold the switch too long or the switch may fail to return to the normal state, allowing the valve to close all the way. However, most operators quickly learn how long they can press the switch without causing a nuisance trip.

This method of testing was mandated by the MMS (Government Agency that oversees safety for Oil/Gas Operations in US Offshore waters). MMS requires that an operator initiate and monitor the test. This method has worked very well in a number of offshore installations.

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Shuttle Valve

Another method uses dual solenoids mounted in parallel with a shuttle valve in the middle. During the test, pressure indication (e.g. switches or gauges) is used to monitor the discharge pressure of the solenoids. The test is performed by de-energizing each solenoid separately and verifying that the solenoid has vented. The reliability of this technique depends on successful operation of the shuttle valve during the test of each solenoid valve. Improper operation may result in the air being vented from the actuator.

Integrated Test Package

A fully integrated solenoid package is available from ASCO (*2oo2D-SOV, patent pending*) that provides on-line diagnostics of solenoid coil failure and facilitates on-line solenoid testing. During normal operation, the air signal passes through the package from the signal source to the valve actuator. When a trip occurs, the solenoids vent the air from the valve actuator and allow the valve to move to its fail-safe position. The ASCO package can be used in two operational modes:

- A normal 2oo2 configuration where both solenoids must de-energize for shutdown. The pressure switches are used to individually alarm if either solenoid goes to the vent state when not commanded, reducing the potential for spurious trips. The pressure switches also facilitate automatic on-line testing, where each solenoid is de-energized individually with pressure switch confirmation of venting.

- A 1oo1 configuration where one solenoid is on-line for the shutdown action. The PLC is programmed so that if the primary solenoid goes to the vent state without being commanded (as detected by the pressure switch), the secondary solenoid is energized, preventing the spurious trip. Solenoid testing is performed by cycling the solenoids and verifying vent state. This configuration provides the safety availability of a 1oo1 configuration with the spurious trip rate of a 2oo2 configuration.

Either configuration can be used for partial stroke testing by pulsing the power to the solenoids for just long enough to achieve the partial stroke. To verify the movement of the valve, a position transmitter or limit switch is used. The position indication is also used to prevent over stroking of the block valve, i.e., if the valve moves too far during the timed stroke, the solenoids are re-energized. Due to solenoid valve redundancy, this method for pulsing the solenoids has a reduced potential for spurious trips during the partial stroke test (i.e., both solenoids must fail to return to position to incur a spurious trip.)

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Annex NN — Model procedure for testing mA pressure transmitters

Using a 4-20 mA signal simulator verify the transmitter fault logic by performing the following steps:

1. The root valve is closed and the system is safely vented prior to connecting the calibrated pressure source.

2. Connect the simulator to the instrument loop being tested.

3. Drive the output current to 21.2 mA (a different value may be selected by the user with assurance that upscale overdrive has taken place) and verify readout device indicates bad measurement.

4. Drive the output current to 3.5 mA (a different value may be selected by the user with assurance that downscale overdrive has taken place) and verify readout device indicates bad measurement.

5. Disconnect the simulator from the loop being tested.

Perform the following steps for verification of transmitter input processing and trip check:

1. Connect the calibrated pressure source to the input side of the transmitter downstream of the process root valve.

2. Set the calibrated pressure source to allow simulation of the input pressure over the calibrated range of the transmitter.

3. Increase the simulated pressure until a High pressure pre-alarm and trip occurs as indicated by the loop documentation (if applicable). Verify and document that pre-alarm and trip occur at correct set point.

4. Decrease the simulated pressure until the High pressure trip and pre-alarm clears as indicated by loop documentation (if applicable). Verify and document that trip and pre-alarm clear at correct set point. Also verify that the SIF does not automatically reset after the trip condition has cleared.

5. Decrease the simulated pressure until a Low pressure pre-alarm and trip occurs as indicated by loop documentation (if applicable). Verify and document that pre-alarm and trip occurs at correct set point.

6. Increase the simulated pressure until the Low pressure trip and pre-alarm clears as indicated by loop documentation (if applicable). Verify and document that pre-alarm and trip clear at correct set point. Also verify that the SIF does not automatically reset after the trip condition has cleared.

7. Document as found and as left alarm and trip settings on appropriate place in test procedure. Table NN.1 is an example of a way to document this data.

Procedure No.
Revision Date
Page _ of _

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8. Verify that process root valve is open.

Table NN.1 — Sample documentation for high alarm and trip settings

Pressure Input	Input Range P1234 (0-xxx psi) (0-yyy " H2O)	High Pre-Alarm Setpoint P1234 (xxx psi) (yyy "H_2O) (zzz mA)	High Trip Setpoint P1234 (xxx psi) (yyy " H_2O) (zzz mA)	Pre-Alarm Setpoint (As Found)	Pre-Alarm Setpoint (As Left)	Trip Setpoint (As Found)	Trip Setpoint (As Left)
PT1234							

Procedure No.
Revision Date
Page _ of _

--`,,,`,-`-`,,`,,`,`,,`---

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Annex PP — Model procedure for testing mA temperature transmitters

Verify the thermocouple (T/C) fault protection by disconnecting the thermocouple and verifying that the Open T/C tag alarms in control center. The user should be aware that this might be alarmed high, low or last depending on the Safety Requirements Specifications (SRS) and the application.

Using a 4-20 mA signal simulator verify the transmitter fault logic by performing the following steps:

1. Connect the simulator to the instrument loop being tested.

2. Drive the output current to 21.2 mA (a different value may be selected by the user with assurance that upscale overdrive has taken place) and verify readout device indicates bad measurement.

3. Drive the output current to 3.5 mA (a different value may be selected by the user with assurance that downscale overdrive has taken place) and verify readout device indicates bad measurement.

4. Disconnect the simulator from the loop being tested.

Perform the following steps for verification of transmitter input processing and trip check:

1. Connect the calibrated temperature source to input side of transmitter.

2. Set the calibrated temperature source to allow simulation of the input temperature over the calibrated range of the transmitter.

3. Increase the simulated temperature until a High temperature pre-alarm and trip occurs as indicated by the loop documentation (if applicable). Verify and document that pre-alarm and trip occur at correct set point.

4. Decrease the simulated temperature until the High temperature trip and pre-alarm clears as indicated by loop documentation (if applicable). Verify and document that trip and pre-alarm clear at correct set point. Also verify that the SIF does not reset automatically.

5. Decrease the simulated temperature until a Low temperature pre-alarm and trip occurs as indicated by loop documentation (if applicable). Verify and document that pre-alarm and trip occurs at correct set point.

6. Increase the simulated temperature until the Low temperature trip and pre-alarm clears as indicated by loop documentation (if applicable). Verify and document that pre-alarm and trip clear at correct set point. Also verify that the SIF does not reset automatically.

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a) Thermocouples

Verify the thermocouple type by physical examination of tag or color code on thermocouple.

Using a calibrated temperature simulator and a portable ice bath, measure the thermocouple output or temperature with the thermocouple inserted into the ice bath. Verify correct reading for type of thermocouple used.

Repeat above for ambient temperature measurement and verify that thermocouple output indicated correct ambient temperature.

If the process temperature measurement must meet a SIL 3 application, use of a certified thermocouple should be considered.

b) Resistance Temperature Detectors

Verify the resistance temperature detector (RTD) type by physical examination of tag or color code on sensor.

Using a calibrated temperature simulator and a portable ice bath, measure the RTD output or temperature with the RTD inserted into the ice bath. Verify correct reading for type of RTD used.

Repeat above for ambient temperature measurement and verify that RTD output indicated correct ambient temperature.

If the process temperature measurement must meet a SIL 3 application, use of a 4-wire certified RTD element should be considered.

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Annex QQ — Model procedure for testing mV temperature transmitters

Thermocouple Input Validation and Trip Check

Perform the following steps using Table 5 for verification of thermocouple input processing validation and trip check.

1. Verify the T/C fault by disconnecting the thermocouple and verifying that the Open T/C tag alarms in control center.

2. Connect the mV simulator to the thermocouple wiring at the sensor end and simulate the T/C input over the operating range indicated in the table.

3. Increase the simulated T/C temperature until a high temperature trip occurs as indicated by readout device in control center.

4. Decrease the simulated T/C temperature until the high temperature trip clears as indicated by readout device in control center. Also verify that SIF does not automatically reset.

5. Remove the mV signal generator and re-connect the thermocouple.

6. Verify that the readout device in control center High Temp Trip Alarm is Normal.

Repeat the above procedure for low temperature pre-alarm and trip settings as appropriate.

Procedure No.
Revision Date
Page _ of _

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Annex RR — Model procedure for testing pressure switches

Perform the following steps for verification of switch input processing validation and trip check:

1. Connect the calibrated pressure source to the input of the pressure switch downstream of process root valve.

2. Set the calibrated pressure source to allow simulation of the input pressure over the calibrated range of the pressure switch.

3. Increase the simulated pressure until a High pressure pre-alarm and trip occurs as indicated by the loop documentation (if applicable). Verify and document that pre-alarm and trip occur at correct set point.

4. Decrease the simulated pressure until the High pressure trip and pre-alarm clears as indicated by loop documentation (if applicable). Verify and document that trip and pre-alarm clear at correct set point. Also verify that the SIF does not automatically reset.

5. Decrease the simulated pressure until a Low pressure pre-alarm and trip occurs as indicated by loop documentation (if applicable). Verify and document that pre-alarm and trip occurs at correct set point.

6. Increase the simulated pressure until the Low pressure trip and pre-alarm clears as indicated by loop documentation (if applicable). Verify and document that pre-alarm and trip clear at correct set point. Also verify that the SIF does not automatically reset.

7. Disconnect pressure source and reconnect switch to process tap and open process root valve.

Procedure No.
Revision Date
Page _ of _

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